Statistical Mechanics

A Set of Lectures

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Editor's Foreword

Addison-Wesley's *Frontiers in Physics* series has, since 1961, made it possible for leading physicists to communicate in coherent fashion their views of recent developments in the most exciting and active fields of physics—without having to devote the time and energy required to prepare a formal review or monograph. Indeed, throughout its nearly forty-year existence, the series has emphasized informality in both style and content, as well as pedagogical clarity. Over time, it was expected that these informal accounts would be replaced by more formal counterparts—textbooks or monographs—as the cutting-edge topics they treated gradually became integrated into the body of physics knowledge and reader interest dwindled. However, this has not proven to be the case for a number of the volumes in the series: Many works have remained in print on an on-demand basis, while others have such intrinsic value that the physics community has urged us to extend their life span.

The *Advanced Book Classics* series has been designed to meet this demand. It will keep in print those volumes in *Frontiers in Physics* or its sister series, *Lecture Notes and Supplements in Physics*, that continue to provide a unique account of a topic of lasting interest. And through a sizable printing, these classics will be made available at a comparatively modest cost to the reader.

These notes on Richard Feynman's lectures on Statistical Mechanics were first published some twenty-five years ago. As is the case with all of Feynman's lectures, the presentation in this work reflects his deep physical insight, the freshness and originality of his approach to understanding physics, and the overall pedagogical wizardry of Richard Feynman. This volume will be of interest to everyone concerned with teaching and learning statistical mechanics. In

addition to providing an elegant and concise introduction to the basic concepts of statistical physics, the notes contain a description of some of the many original and profound contributions—ranging from polaron theory to the theory of liquid helium—that Feynman made to the field of condensed matter physics.

David Pines
Urbana, Illinois
December 1997

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CHAPTER 1

INTRODUCTION TO STATISTICAL MECHANICS

1.1 THE PARTITION FUNCTION

The key principle of statistical mechanics is as follows:

If a system in equilibrium can be in one of N states, then the probability of the system having energy E_n is $(1/Q)e^{-E_n/kT}$, where

$$Q = \sum_{n=1}^{N} e^{-E_n/kT},$$

k = Boltzmann's constant, and T = temperature. Q is called the *partition function.*

If we take $|i\rangle$ as a state with energy E_i and A as a quantum-mechanical operator for a physical observable, then the expected value of the observable is

$$\langle A \rangle = \frac{1}{Q} \sum_{|i\rangle} \langle i|A|i\rangle e^{-E_i/kT}.$$

This fundamental law is the summit of statistical mechanics, and the entire subject is either the slide-down from this summit, as the principle is applied to various cases, or the climb-up to where the fundamental law is derived and the concepts of thermal equilibrium and temperature T clarified. We will begin by embarking on the climb.

If a system is very weakly coupled to a heat bath at a given "temperature," if the coupling is indefinite or not known precisely, if the coupling has been on for a long time, and if all the "fast" things have happened and all the "slow" things not, the system is said to be in *thermal equilibrium.*

For instance, an enclosed gas placed in a heat bath will eventually erode its enclosure; but this erosion is a comparatively slow process, and sometime before the enclosure is appreciably eroded, the gas will be in thermal equilibrium.

Consider two different states of the system that have the same energy, $E_r = E_s$. The probabilities of the system being in states r and s are then equal. For if the system is in state r, any extremely small perturbation will cause the system to go into a different state of essentially the same energy, such as s. The

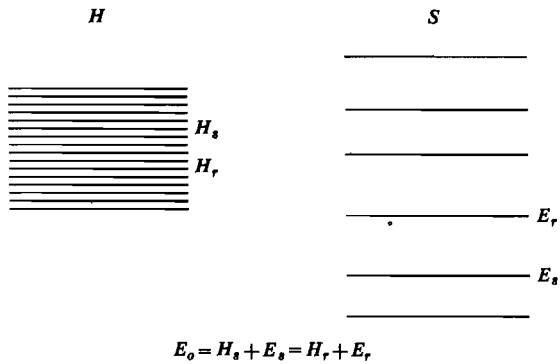

Fig. 1.1 Energy levels in a system S and a heat bath H.

same is true if the system is in state s. Since the system remains in contact with the heat bath for a long time, one would expect states of equal energy to be equally likely. Also, states of different energies would be expected to have different probabilities.

Because two states of the same energy are equally probable, the probability of a state having energy E is a function only of the energy; $P = P(E)$.

Now consider a system, S, in equilibrium with a large heat bath, H (see Fig. 1.1). Since experience shows that the behavior of a system in equilibrium is independent of the nature of the heat bath, the bath may be assumed extremely large and its total energy E very great. Also, the possible energy levels of the heat bath may be assumed quasi-continuous.

Let the energy levels of the heat bath be denoted by H_i. These levels are distributed quasi-continuously. Let the energy levels of S be denoted by E_j. Then $H_i \gg E_j$ for all i, j. The bath plus the system can be thought of as a new system, T, which is also in thermal equilibrium.

T has a definite energy, but as that energy is not fixed exactly (the bath is in contact with the outside world), we can assume that the energy may be anywhere in the range $E_0 \pm \Delta$. If Δ is sufficiently small, we can assume that the states of the heat bath are equally likely in the range $H_i \pm \Delta$. Let $\eta(H_r)$ be the number of states per unit energy range in the heat bath H around energy H_r.

The probability, $P(E_r)$, that S is in a state with energy E_r is proportional to the number of ways S can have that energy. In other words, it is proportional to $\eta(E_0 - E_r)2\,\Delta$ = the number of states of H that allow T to have energy in the range $E_0 \pm \Delta$. Then

$$\frac{P(E_r)}{P(E_{r'})} = \frac{\eta(E_0 - E_r)}{\eta(E_0 - E_{r'})} = e^{\ln \eta(E_0 - E_r) - \ln \eta(E_0 - E_{r'})}.$$

t. & S.D energy.

Remember that $E_r \ll E_0$. If it is true that $(d/dE) \ln \eta(E) = \beta(E)$ is almost constant for E in the range under consideration,* then we can say

$$\frac{P(E_r)}{P(E_{r'})} = e^{-\beta(E_r - E_{r'})};$$

so

$$P(E_{r'}) \propto e^{-\beta E_{r'}}.$$

Normalization requires that $P(E_i) = (1/Q)e^{-\beta E_i}$ where $Q = \sum_i e^{-\beta E_i}$.

The fundamental law has just been shown to be quite plausible. But for those who doubt the constancy of β(E), let us consider some examples.

First, assume that the heat bath consists of N independent harmonic oscillators. The energy of the bath is

F: energy of the bath

$$F = \sum_{i=1}^{N} n_i \hbar w_i,$$

where we assume that n_i is very large, and we neglect the zero-point energy. How many states are there with energy less than F? If $N = 2$, we have the situation shown in Fig. 1.2.

For $N = 2$, the number of states is proportional to the area of the triangle. Clearly the number of states with energy less than F for large F is proportional to F^N; so the number of states per unit energy range is $\eta(F) \propto (d/dF)F^N \propto F^{N-1}$.

$$\ln \eta(F) = \text{constant} + (N - 1) \ln F,$$

$$\frac{d \ln \eta(F)}{dF} = \frac{N - 1}{F} \approx \frac{N - 1}{E_0} \qquad \text{since} \qquad E_r \ll E_0.$$

---

* The basic assumption made here is, that the system governing the probabilities has a quasi-continuous spectrum in the region considered and for which there is no particular characteristic energy. For such a system, if ε_1 and ε_2 are two energy values, then, since energy is only defined up to an additive constant ε, we should have

$$\frac{f(\varepsilon_1)}{f(\varepsilon_2)} = \frac{f(\varepsilon_1 + \varepsilon)}{f(\varepsilon_2 + \varepsilon)},$$

where $f(\varepsilon)$ is the probability. Defining

$$f(\varepsilon) = g(\varepsilon - \varepsilon_2),$$

we obtain

$$g(\varepsilon)g(\varepsilon_1 - \varepsilon_2) = g(0)g(\varepsilon_1 - \varepsilon_2 + \varepsilon),$$

which is (uniquely) solved by

$$g(\varepsilon) = g(0)e^{-\beta\varepsilon} \qquad (\beta \text{ constant}),$$

that is,

$$\frac{f(\varepsilon_1)}{f(\varepsilon_2)} = e^{-\beta(\varepsilon_1 - \varepsilon_2)}$$

$f(\varepsilon_1) \cdot g(\varepsilon_1 - \varepsilon_2$

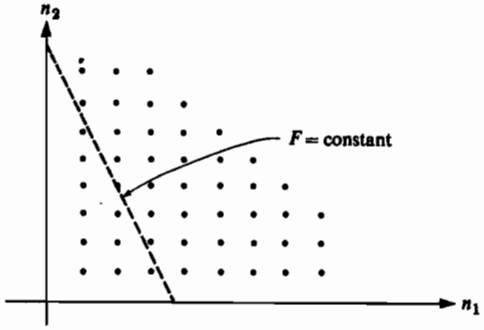

Fig. 1.2 States with energy less than F for two independent harmonic oscillators.

For N large,

$W \approx \frac{t_0}{\boxed{N}}$

$$\beta = \frac{1}{\text{energy per oscillator}} = \frac{1}{W},$$

where W = energy per oscillator.

Alternatively, we can go back to the equation

$$\frac{P(E_r)}{P(E_{r'})} = \frac{\eta(E_0 - E_r)}{\eta(E_0 - E_{r'})}$$

and plug in N directly. We get

$$\frac{\eta(E_0 - E_r)}{\eta(E_0 - E_{r'})} = \frac{(E_0 - E_r)^{N-1}}{(E_0 - E_{r'})^{N-1}}$$

$$\cancel{\simeq} \frac{(1 - E_r/NW)^{N-1}}{(1 - E_{r'}/NW)^{N-1}} \to \frac{e^{-E_r/W}}{e^{-E_{r'}/W}} \quad \text{as} \quad N \to \infty.^*$$

We then get the same value for β as by our previous method.

As a second example, consider the case of a heat bath consisting of N particles in a box.

$$F = \frac{P_{x_1}^2 + P_{y_1}^2 + P_{z_1}^2}{2m} + \frac{P_{x_2}^2 + P_{y_2}^2 + P_{z_2}^2}{2m} + \cdots + \frac{P_{x_N}^2 + P_{y_N}^2 + P_{z_N}^2}{2m}.$$

Assuming periodic boundary conditions,

$$P_x = \frac{2\pi\hbar(\text{integer})}{L} = \frac{2n_x\pi\hbar}{L},$$

* *Problem*: Why is W independent of N?

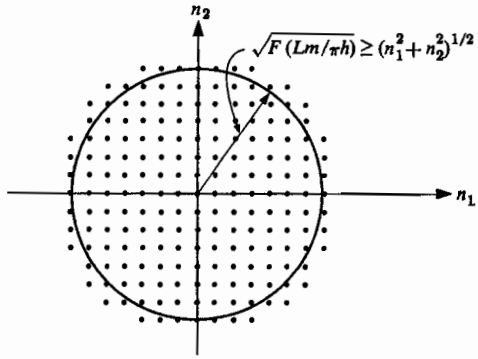

Fig. 1.3 Number of states with energy less than F is roughly equal to the area of the circle shown.

where L is the length of the sides of the box. Then

$$F = \frac{2\pi\hbar}{2Lm}\,[n_{x_1}^2 + n_{y_1}^2 + n_{z_1}^2 + n_{x_2}^2 + \cdots + n_{z_N}^2].$$

If there were just two η's, we would calculate β with the help of Fig. 1.3. The number of states with energy less than F is roughly equal to the area of the circle with radius $\sqrt{F(Lm/\pi\hbar)}$. For N particles, we must use a hypersphere in $3N$ dimensions, and the number of states with energy less than F is proportional to $(\sqrt{F})^{3N}$. It follows that β is roughly constant, and equals $1/(\frac{2}{3}W)$, where $W =$ energy per particle.

Consider two independent systems S_A and S_B with energy levels A_i and B_j. The probability of system S_A having energy A_i is

$$P_A(A_i) = \frac{e^{-\beta A_i}}{\sum e^{-\beta A_j}}.$$

Now place S_A and S_B in loose contact with each other and consider the combined system $S_T = S_A + S_B$, with combined energy $T_k = A_i + B_j$.

$$P_T(T_k) = P_T(A_i + B_j) = \frac{e^{-\beta_T(A_i + B_j)}}{\sum_i e^{-\beta_T A_i} \sum_j e^{-\beta_T B_j}}$$
$$= \frac{e^{-\beta_T A_i}}{\sum_i e^{-\beta_T A_i}} \frac{e^{-\beta_T B_j}}{\sum_j e^{-\beta_T B_j}}.$$

The probability of system S_T being such that system S_A has energy A_i is

$$P_T(A_i) = \sum_j \left[\frac{e^{-\beta_T A_i}}{\sum_i e^{-\beta_T A_i}}\right] \frac{e^{-\beta_T B_j}}{\sum_j e^{-\beta_T B_j}} = \frac{e^{-\beta_T A_i}}{\sum_i e^{-\beta_T A_i}} = P_A(A_i).$$

Similarly,

$$P_B(B_j) = \frac{e^{-\beta_T B_j}}{\sum_j e^{-\beta_T B_j}}.$$

We see that if two systems are placed in loose contact with each other, in equilibrium they have the same β. Temperature has a similar property, and in fact, by the way temperature is conventionally defined, $\beta = 1/kT$ where k is Boltzmann's constant.*

From the basic principle of statistical mechanics, once

$$Q = \sum_n e^{-E_n/kT}$$

is known, all thermodynamic properties can be found. We define F, the Helmholtz free energy, so that

$$Q = \sum_n e^{-E_n/kT} \equiv e^{-F/kT}, \tag{1.1}$$

$$F = -kT \ln Q = -kT \ln\left(\sum_n e^{-E_n/kT}\right), \tag{1.2}$$

$$S = \text{entropy, is defined as } -k \sum_n P_n \ln P_n, \tag{1.3}$$

---

* Furthermore, suppose our spectrum has a density of states $n(\varepsilon)$ which is relatively constant when $\varepsilon \to \beta\varepsilon$, that is

$$n(\beta\varepsilon) = n(\varepsilon)q(\beta) \tag{*}$$

Then $U(\beta)$ (see 1.7) $= C/\beta$ and we recognize in C the usual (total) heat capacity, so that

$$\frac{C_1}{\beta_1} + \frac{C_2}{\beta_2} = \frac{C_1 + C_2}{\beta},$$

which is the well-known (experimental) formula.

Equation (*) has the unique solution

$$n(\varepsilon) = A\varepsilon^n$$

which is, indeed, the form of $n(\varepsilon)$ for microscopic bodies (see pp. 3, 5).

Also, it can easily be seen (from 1.7 below) that

$$\frac{dU}{d\beta} = -\frac{1}{2Q^2} \sum_{n,m} (\varepsilon_n - \varepsilon_m)^2 e^{-\beta(\varepsilon_n + \varepsilon_m)} < 0,$$

so that U is a decreasing function of β. Suppose we mix two systems having different β's, $\beta_1 < \beta_2$. After mixing there will be a common β, according to

$$U_1(\beta_1) + U_2(\beta_2) = U_1(\beta) + U_2(\beta),$$

whence, since the U's decrease, $\beta_1 < \beta < \beta_2$, and energy flows from the low-β body to the high-β body. This also fits into our intuitive perception of temperature.

where

$$P_n = \frac{1}{Q} e^{-E_n/kT} \tag{1.4}$$

From Eq. (1.2) it can be seen that

$$-\left(\frac{\partial F}{\partial T}\right)_V = -k \sum_n \left[\frac{1}{Q} e^{-E_n/kT} \left\{\frac{-E_n}{kT} - \ln Q\right\}\right] = S \tag{1.5}$$

$$P = \text{pressure} = \sum_n - P_n \frac{\partial E_n}{\partial \text{Vol}} = -\left(\frac{\partial F}{\partial V}\right)_T \tag{1.6}$$

$$U = \text{average energy} = \frac{1}{Q} \sum_n E_n e^{-E_n/kT} \tag{1.7}$$

But

$$\frac{\partial Q}{\partial T} = \frac{1}{kT^2} \sum_n E_n e^{-E_n/kT},$$

$$U = \frac{kT^2}{Q} \frac{\partial Q}{\partial T} = -T^2 \frac{\partial}{\partial T}\left(\frac{F}{T}\right) = F - T\frac{\partial F}{\partial T} = \frac{\partial}{\partial(1/T)} \frac{F}{T}, \tag{1.8}$$

since $S = -\partial F/\partial T$, $U = F + TS$.

$$C_v = \left(\frac{\partial U}{\partial T}\right)_V = \frac{\partial F}{\partial T} - \frac{\partial}{\partial T}\left(T\frac{\partial F}{\partial T}\right) = \frac{\partial F}{\partial T} - \frac{\partial F}{\partial T} - T\frac{\partial^2 F}{\partial T^2}$$

$$= -T\left(\frac{\partial^2 F}{\partial T^2}\right)_V \tag{1.9}$$

To get a clearer idea of the nature of pressure, consider a possible alternative definition. The Hamiltonian operator for the system is dependent on the volume. Set H = Hamiltonian = $H(V)$. We can take as a pressure operator $P_{OP} = -\partial H/\partial V$.

$$P = \frac{1}{Q} \sum_i \langle i|P_{OP}|i\rangle e^{-E_i/kT}.$$

Our alternative definition is equivalent to the first one if

$$\langle i|\frac{\partial H}{\partial V}|i\rangle = \frac{\partial E_i}{\partial V}.$$

But

$$\frac{\partial E_i}{\partial V} = \lim_{V' \to V} \frac{E_{i'}(V') - E_i(V)}{V' - V} = \lim_{V' \to V} \frac{\langle i'|H'|i'\rangle - \langle i|H|i\rangle}{V' - V},$$

where $|i'\rangle$ is the eigenvector of H' corresponding to $|i\rangle$, the ith eigenvector of H. $H' = H(V')$.

$H' = H + (H' - H)$; so we can apply first-order perturbation theory to say that $\langle i'|\, H'\, |i'\rangle \approx \langle i|\, H\, |i\rangle + \langle i|\, H' - H\, |i\rangle$ for $H' - H \approx 0$. Then

$$\frac{\partial E_i}{\partial V} = \lim_{V' \to V} \frac{\langle i|H' - H|i\rangle}{V' - V} = \langle i|\, \frac{\partial H}{\partial V}\, |i\rangle.$$

Our two definitions are equivalent.

Because H may be a function of the shape of the system, as well as its volume, our definition of pressure might depend on how the volume is changed. In general, for any parameter α, there is a force that can be computed by

$$(\text{Force})_\alpha = \frac{1}{Q} \sum_i \left(- \frac{\partial E_i}{\partial \alpha} \right) e^{-E_i(\alpha)/kT},$$

and we can write

$$\frac{\partial E_i}{\partial \alpha} = \langle i|\, \frac{\partial H}{\partial \alpha}\, |i\rangle.$$

A third definition of pressure is

$$P = - \frac{\partial U}{\partial V} (V, S).$$

That this definition is equivalent to the other two may be verified without difficulty from the equation $U = F + TS$.

The equation $S = -(\partial F/\partial T)_V$ holds only at equilibrium, where F is defined. Away from equilibrium, S always increases with time. To see this, note that time-dependent perturbation theory gives

$$\frac{dP_m}{dt} = \sum_n (|V_{mn}|^2 P_n - |V_{mn}|^2 P_m),$$

where $|V_{nm}|^2$ is the probability per unit time of transition from state n to state m, and $|V_{nm}|^2 = |V_{mn}|^2$. Then

$$\frac{dS}{dt} = -k \sum_i \left(\frac{dP_i}{dt} + \ln P_i \frac{dP_i}{dt} \right) = -k \sum_i \frac{dP_i}{dt} \ln P_i,$$

since

$$\sum_i \frac{dP_i}{dt} = \frac{d}{dt} \sum_i P_i = \frac{dP}{dt} = 0.$$

Then

$$\begin{aligned} \frac{dS}{dt} &= -k \sum_{mn} |V_{nm}|^2 (P_n - P_m) \ln P_m \\ &= -\frac{k}{2} \sum_{mn} |V_{nm}|^2 (P_n - P_m)(\ln P_m - \ln P_n). \end{aligned}$$

But each term in the sum is negative, for the sign of $P_n - P_m$ is opposite to the sign of $(\ln P_m - \ln P_n)$. So $dS/dt > 0$.

Before we start to make calculations, note that if we have a system that is a combination of several independent subsystems, with $E_{\text{total}} = E_n = \sum_i E_{n_i} = \sum_i$ energies of the subsystems,

$$F = -kT \ln Q = -kT \ln \sum_n e^{-\beta E_n} = -kT \ln \sum_{n_1, n_2, \ldots} e^{-\beta \sum_i E_{n_i}}$$

$$= -kT \ln \prod_i \left(\sum_{n_i} e^{-\beta E_{n_i}} \right) = -kT \sum_i \left[\ln \sum_{n_i} e^{-\beta E_{n_i}} \right] = \sum_i F_i.$$

The free energy of the whole system is the sum of the free energies of its non-interacting parts.

1.2 LINEAR HARMONIC OSCILLATORS

Consider a system of harmonic oscillators in thermal equilibrium. The partition function Q, free energy F, and average energy of the system of oscillators can be found as follows: The oscillators do not interact with each other, but only with the heat bath. Since each oscillator is independent, one can find F_i of the ith oscillator and then

$$F = \sum_{i=1}^{M} F_i$$

(M oscillators).

$$Q_i = \sum_n e^{-E_n^i/kT} \tag{1.10}$$

$$E_n^i = \hbar_i \omega_i (n + \tfrac{1}{2}) \qquad \text{from quantum mechanics } (n = 0, 1, 2, \ldots) \tag{1.11}$$

$$Q_i = \sum_n e^{-\hbar_i \omega_i (n + 1/2)/kT}$$

$$Q_i = \frac{e^{-\hbar\omega_i/2kT}}{1 - e^{-\hbar\omega_i/kT}} \left[\frac{1}{1 - e^{-\hbar\omega_i/kT}} = 1 + e^{-\hbar\omega_i/kT} + e^{-2\hbar\omega_i/kT} + \cdots \right] \tag{1.12}$$

$$F_i = -kT \ln Q_i = \frac{\hbar\omega_i}{2} + kT \ln (1 - e^{-\hbar\omega_i/kT}) \tag{1.13}$$

$U_i =$ average energy of a single oscillator in thermal equilibrium

$$= \frac{1}{Q_i} \sum_n E_n^i \, e^{-E_n^i/kT} = \frac{\partial}{\partial(1/T)} \frac{F_i}{T} \tag{1.14}$$

$$= \frac{\hbar\omega_i}{2} + \frac{\hbar\omega_i e^{-\hbar\omega_i/kT}}{1 - e^{-\hbar\omega_i/kT}} = \frac{\hbar\omega_i}{2} + \frac{\hbar\omega_i}{e^{\hbar\omega_i/kT} - 1}. \tag{1.15}$$

$$F = \sum_i F_i = \sum_i \left[\frac{\hbar\omega_i}{2} + kT \ln (1 - e^{-\hbar\omega_i/kT}) \right] \tag{1.16}$$

$$U = \sum_i U_i = \sum_i \left[\frac{\hbar\omega_i}{2} + \frac{\hbar\omega_i}{e^{\hbar\omega_i/kT} - 1} \right].$$

It is customary to define an average n_i, $\boldsymbol{n}_i$, according to

$$U_i = (\boldsymbol{n}_i + \tfrac{1}{2})\hbar\omega_i.$$

Thus

$$\boldsymbol{n}_i = \frac{1}{e^{\hbar\omega_i/kT} - 1}.$$

Returning to the example on p. 3 we see that for very high temperature (and U) we indeed have

$$W \approx U_i \approx kT = 1/\beta,$$

which is independent of U.

Note that the contribution to F of the ith oscillator is negligible if $\hbar\omega_i \gg kT$ (except for the $\hbar\omega_i/2$ term). At low temperatures the high-frequency modes are "frozen out" and do not contribute to the specific heat.

1.3 BLACKBODY RADIATION

In dealing with blackbody radiation, our point of view will be as follows: In a cavity (blackbody), there are a great number of modes of oscillation. The number of modes per unit volume per frequency bandwidth is given by classical considerations. Each mode, however, behaves as an independent quantum harmonic oscillator, except that the $\hbar\omega/2$ term is neglected. That is, $E_n = n\hbar\omega$.

We want to get rid of the $h\omega/2$ because it leads to infinite energy when there are an infinite number of modes. A Hamiltonian that eliminates the $h\omega/2$ is

$$H_i = \tfrac{1}{2}(P_i^2 + \omega_i^2 q_i^2) - \frac{\hbar\omega_i}{2}.$$

With the above assumptions, we can obtain an expression for the energy per unit volume per unit frequency.

First we will find the number of modes per unit volume per frequency (or wave number).

Assume a gigantic box of dimensions a, b, and c. The demand is made that the waves be periodic at the walls of the box.

$$\frac{1}{\lambda_x} = \text{number of waves/cm in } x\text{-direction}$$

$$\frac{a}{\lambda_x} = \text{number of waves in box } (x\text{-direction}).$$

Because of the periodic boundary condition,

$$\frac{a}{\lambda_x} = n_x = \text{an integer.}$$

Let $k_x = 2\pi/\lambda_x$. Then

$$\frac{ak_x}{2\pi} = n_x,$$

and similarly

$$\frac{bk_y}{2\pi} = n_y, \qquad \frac{ck_z}{2\pi} = n_z.$$

$$\frac{a\,dk_x}{2\pi} = dn_x, \qquad \frac{b\,dk_y}{2\pi} = dn_y, \qquad \frac{c\,dk_z}{2\pi} = dn_z$$

$$d^3n = dn_x\,dn_y\,dn_z = abc\,\frac{d^3k}{(2\pi)^3} = (abc)\,\frac{dk_x\,dk_y\,dk_z}{(2\pi)^3}.$$

For each k, there are two possible polarizations. Thus, the number of modes per unit volume with wave number between $\boldsymbol{k}$ and $\boldsymbol{k} + d\boldsymbol{k}$ is

$$\frac{2\,d^3n}{(abc)} = 2\,\frac{d^3k}{(2\pi)^3}.$$

Now $\omega = kc$ where c is the velocity of light. Also, the number of modes is so large that the sum over the modes can be replaced by an integral.

$$\frac{F}{V} = \iiint k_bT \ln\left(1 - \exp\left[\frac{-\hbar\omega(k)}{k_bT}\right]\right)\frac{2\,d^3k}{(2\pi)^3}. \tag{1.17}$$

(Remember that the $h\omega/2$ term has been omitted.)

From symmetry,

$$\frac{F}{V} = 2\int k_bT \ln\left(1 - \exp\left[\frac{-\hbar\omega(k)}{k_bT}\right]\right)\frac{4\pi k^2\,dk}{(2\pi)^3}.$$

$$\frac{U}{V} = 2\int \frac{\hbar\omega(k)\exp\left[-\hbar\omega(k)/k_bT\right]}{1 - \exp\left[-\hbar\omega(k)/k_bT\right]}\,\frac{4\pi k^2\,dk}{(2\pi)^3}. \tag{1.18}$$

Let $x = \hbar\omega/k_bT = \hbar kc/k_bT$. Then

$$\frac{U}{V} = \frac{8\pi}{(2\pi)^3}\,\frac{(k_bT)^4}{\hbar^3c^3}\int_0^\infty \frac{e^{-x}}{1 - e^{-x}}\,x^3\,dx$$

$$= \frac{\pi^2(k_bT)^4}{15\hbar^3c^3} = \sigma T^4;$$

$$\sigma = \frac{\pi^2k_b^4}{15\hbar^3c^3}. \tag{1.19}$$

The preceding results can be summarized and put in a more familiar form by replacing $\hbar\omega$ with $h\nu$.

The number of modes/unit volume between k and $k + dk$ is

$$2 \cdot \frac{4\pi k^2\, dk}{(2\pi)^3} = \frac{k^2\, dk}{\pi^2} = \frac{8\pi\nu^2\, d\nu}{c^3},$$

since $k = 2\pi\nu/c$. The average energy of an oscillator of frequency ν is $h\nu/e^{h\nu/kT}$—1. Thus, the energy per unit volume between ν and $\nu + d\nu$ is

$$U_\nu\, d\nu = \frac{dU}{V} = \frac{8\pi\nu^2}{c^3} \frac{h\nu\, d\nu}{e^{h\nu/kT} - 1}. \tag{1.20}$$

This is the *Planck Radiation Law*, and is the same as Eq. (1.18).

$$\int U_\nu\, d\nu = \sigma T^4$$

is the *Stefan Boltzmann Law*.

For T very large ($kT \gg h\nu$),

$$U_\nu\, d\nu = \frac{8\pi\nu^2}{c^3} \frac{h\nu\, d\nu}{1 + (h\nu/kT) - 1} = \frac{8\pi\nu^2}{c^3} kT\, d\nu. \tag{1.21}$$

This is the *Rayleigh-Jeans Law*.

Let $F^* = F/V$ and $U^* = U/V$.

$$C_v = \frac{\partial U^*}{\partial T} = 4\sigma T^3$$

is the heat capacity of a gas of photons in equilibrium with the container.

If an oscillator (or mode) is excited to the Nth level, that is, if $E = \hbar\omega_i N$, one says that there are N photons with energy $\hbar\omega_i$. Since photons are defined as the degree of excitation of a mode, one cannot consider a permutation of photons as a new state. That is, photons are indistinguishable. This point is the basis of quantum statistics and will be dealt with in a later section.

1.4 VIBRATIONS IN A SOLID

We want to find the specific heat of a solid. However, en route to the specific heats, we will derive important results that will prove useful in many other cases; chief among these results is the calculation of the normal modes of a crystal.

The program will be as follows:

1. Consider the solid to be a crystal lattice of atoms, each atom behaving as an harmonic oscillator. These oscillators are, of course, coupled.

2. Find the normal modes of the system. There are as many normal modes as there are degrees of freedom; namely, $3(AN)$, where (AN) is the number of atoms in the crystal. The modes behave as independent quantum oscillators.
3. Given the modes, calculate F, the free energy.
4. From F, calculate C_v and any other thermodynamic quantities of interest. Once F is found, we can quickly calculate U, the total energy of the system. In the following derivation, we will calculate C_v directly from U without always writing down F first.

Method of Labeling

Consider a crystal with A atoms per unit cell. For convenience, assume that the unit cell is a rectangular solid of dimensions a, b, c, along three mutually perpendicular axes, x, y, z. Let the origin be at the "center" of a cell. This cell can be denoted by the triplet (0, 0, 0). The cell to its right along the x-axis is denoted by (1, 0, 0), and so on. Thus, any cell can be denoted by a vector $\boldsymbol{N} = n_x\boldsymbol{a} + n_y\boldsymbol{b} + n_z\boldsymbol{c}$ where $\boldsymbol{a} = a\hat{\imath}, \boldsymbol{b} = b\hat{\jmath}, \boldsymbol{c} = c\hat{k}$. If there are A atoms/cell, $3A$ additional coordinates must be given to locate each atom. Let α denote one of these $3A$ coordinates.

Call the displacement from equilibrium of the coordinate in the Nth cell $Z_{\alpha,N}$. (We either consider m = mass equal to 1 or absorb it in Z.)

$Z_{\alpha,N+M}$ is the displacement of an atom in a cell close to N; if, for example, there are two atoms A_1, A_2 per cell, the displacement of A_1 in the x direction is denoted by $Z_{1,N}$, that of A_2 (in same cell) by $Z_{4,N}$, and that of A_1 in an adjacent cell by $Z_{1,N+1}$. $\mathbf{1} = (1, 0, 0)$ or $(0, 1, 0)$, or $(0, 0, 1)$.

Normal Modes

$$T = (1/2) \sum_{\substack{\text{particles} \\ \text{and directions}}} \dot{Z}^2_{\alpha,N} = (1/2) \sum_{\alpha,N} \dot{Z}^2_{\alpha,N} \tag{1.22}$$

$$V = V(0) + \sum_{\alpha,N} \left(\frac{\partial V}{\partial Z_{\alpha,N}}\right)_{Z=0} Z_{\alpha,N} + \frac{1}{2} \sum_{\alpha,\beta,N,M} \left(\frac{\partial^2 V}{\partial Z_{\alpha,N}\, \partial Z_{\beta,N+M}}\right)_{Z=0} Z_{\alpha,N} Z_{\beta,N+M} + \cdots \tag{1.23}$$

Assume that the electrons in the crystal always have time to adjust themselves to the configuration with lowest energy, even when the crystal is vibrating. In this configuration, there is no net force on the nuclei when the $Z_{\alpha,N}$ are zero, so

$$\left(\frac{\partial V}{\partial Z_{\alpha,N}}\right)_{Z=0} = 0.$$

The additive constant, $V(0)$, will not affect our answers, so we might as well drop it. Let

$$\left(\frac{\partial^2 V}{\partial Z_{\alpha,N}\,\partial Z_{\beta,N+M}}\right)_{Z=0} = C^M_{\alpha\beta},$$

a number that depends on the relative positions of the cells of the two atoms, and not on their absolute positions. Note that $C^M_{\alpha\beta} = C^{-M}_{\beta\alpha}$.

Neglecting higher orders, take

$$V = \tfrac{1}{2} \sum_{\alpha,\beta,N,M} C^M_{\alpha\beta} Z_{\alpha,N} Z_{\beta,N+M}. \tag{1.24}$$

For low temperatures it is not too unreasonable to neglect higher orders, because the separation between atoms is of the order of 1 Å and at room temperature the vibrations have amplitude of the order of 0.1 Å. But we should not be too surprised if experiment shows our idealization to be false. In matters such as the one under consideration, the general approach is to make idealizations and then try to find corrections to our assumptions that will give better results.

In order to motivate the procedure that we will use for finding the vibrations of a solid, let us consider the classical problem of vibrations of coupled oscillators. Let the Hamiltonian be

$$H = \sum_i \frac{P_i'^2}{2M_i} + \sum_{ij} \tfrac{1}{2} C'_{ij} q'_i q'_j,$$

where the q'_i are the coordinates of the amount of displacement from equilibrium, $P'_i = M_i \dot{q}'_i$ is the momentum, and $C'_{ij} = C'_{ji}$ are constants. To eliminate the constants M_i, let

$$q_i = q'_i \sqrt{M_i} \qquad \text{and} \qquad C_{ij} = \frac{C'_{ij}}{\sqrt{M_i M_j}}.$$

$$P_i = \frac{\partial \text{Lagrangian}}{\partial \dot{q}_i} = \frac{1}{\sqrt{M_i}} P'_i.$$

Then we get

$$H = \sum_i \frac{P_i^2}{2} + \frac{1}{2} \sum_{ij} C_{ij} q_i q_j. \tag{1.25}$$

The equations of motion are

$$\dot{q}_i = \frac{\partial H}{\partial P_i} = P_i, \qquad \dot{P}_i = \frac{-\partial H}{\partial q_i} = -\sum_j C_{ij} q_j.$$

We now break the motion of the system into modes, each of which has its own frequency. The total motion of the system is a sum of the motions of the modes.

Let the αth mode have frequency ω_α so that

$$q_i^{(\alpha)} = e^{-i\omega_\alpha t}\, a_i^{(\alpha)}$$

for the motion of the αth mode, with $a_i^{(\alpha)}$ independent of time. Then

$$\omega_\alpha^2 a_i^{(\alpha)} = \sum_j C_{ij} a_j^{(\alpha)}.$$

The classical problem of vibrations of coupled oscillators has just been reduced to the problem of finding the eigenvalues and eigenvectors of the real, symmetric matrix, $\|C_{ij}\|$. In order to get the ω_α we must solve the equation

$$\det \|C_{ij} - \omega^2 \delta_{ij}\| = 0. \tag{1.26}$$

Then the $a_i^{(\alpha)}$ (the eigenvectors) can be found. It is possible to choose the $a_i^{(\alpha)}$ so that

$$\sum_i a_i^{(\alpha)} a_i^{(\beta)} = \delta_{\alpha\beta}.$$

The general solution for q_i is

$$q_i = \sum_\alpha C_\alpha q_i^{(\alpha)},$$

where the C_α are arbitrary constants. If we take $Q_\alpha = C_\alpha e^{-i\omega_\alpha t}$, we get $q_i = \sum_\alpha a_i^{(\alpha)} Q_\alpha$. From this it follows that

$$\sum_i a_i^{(j)} q_i = \sum_{\alpha,i} a_i^{(j)} a_i^{(\alpha)} Q_\alpha = \sum_\alpha \delta_{\alpha j} Q_\alpha = Q_j.$$

Making the change of variables, $Q_j = \sum_i a_i^{(j)} q_i$, we get $H = \sum_\alpha H_\alpha$, where

$$H_\alpha = \tfrac{1}{2} P_\alpha^2 + \tfrac{1}{2}\omega_\alpha^2 Q_\alpha^2.$$

This has the expected solutions $Q_\alpha = C_\alpha e^{-i\omega_\alpha t}$.

Now suppose we wish to solve the quantum-mechanical problem of coupled oscillators. Again we have

$$H = \sum_i \frac{P_i'^2}{2M_i} + \sum_{ij} C_{ij}' q_i' q_j',$$

where this time

$$P_i' = \frac{\hbar}{i}\frac{\partial}{\partial q_i'}.$$

Making the same change of variables as before, we get

$$Q_\alpha = \sum_i a_i^{(\alpha)} q_i = \sum_i a_i^{(\alpha)} \sqrt{M_i}\, q_i'$$

$$H = \sum_\alpha H_\alpha,$$

where

$$H_\alpha = \frac{-\hbar^2}{2}\frac{\partial^2}{\partial Q_\alpha^2} + \tfrac{1}{2}\omega_\alpha^2 Q_\alpha^2.$$

It follows immediately that the eigenvalues of our original Hamiltonian are $E = \sum_\alpha (N_\alpha + \frac{1}{2})\hbar\omega_\alpha$. The solution of a quantum-mechanical system of coupled oscillators is trivial once we have solved the equation

$$0 = \det \|C_{ij} - \omega^2\delta_{ij}\| = \det \|C'_{ij}/\sqrt{M_i M_j} - \omega^2\delta_{ij}\|.$$

If we have a solid with $\frac{1}{3}(10^{23})$ atoms we must apparently find the eigenvalues of a 10^{23} by 10^{23} matrix. But if the solid is a crystal, the problem is enormously simplified. The classical Hamiltonian for a crystal is

$$H = \tfrac{1}{2}\sum_{\alpha,N} \dot{Z}^2_{\alpha,N} + \tfrac{1}{2}\sum_{\alpha,\beta,N,M} C^M_{\alpha\beta} Z_{\alpha,N} Z_{\beta,N+M},$$

and the classical equation of motion is (using $C^M_{\alpha\beta} = C^{-M}_{\beta\alpha}$)

$$\ddot{Z}_{\alpha,N} = -\sum_{M,\beta} C^M_{\alpha,\beta} Z_{\beta,N+M}.$$

In a given mode, if one cell of the crystal is vibrating in a certain manner, it is reasonable to expect all cells to vibrate the same way, but with different phases. So we try

$$Z_{\alpha,N} = a_\alpha(\boldsymbol{K}) e^{-i\omega t} e^{i\boldsymbol{K}\cdot\boldsymbol{N}},$$

where $\boldsymbol{K}$ expresses the relative phase between cells. The $e^{i\boldsymbol{K}\cdot\boldsymbol{N}}$ factor allows for wave motion. We now want to find the dispersion relations, or $\omega = \omega(\boldsymbol{K})$.

$$\omega^2 a_\alpha e^{i\boldsymbol{K}\cdot\boldsymbol{N}} = \sum_{M,\beta} (C^M_{\alpha,\beta} a_\beta e^{i\boldsymbol{K}\cdot\boldsymbol{M}}) e^{i\boldsymbol{K}\cdot\boldsymbol{N}}.$$

Let

$$\gamma_{\alpha\beta}(\boldsymbol{K}) = \sum_M C^M_{\alpha,\beta} e^{i\boldsymbol{K}\cdot\boldsymbol{M}}.$$

(Note that $\gamma_{\alpha\beta}(\boldsymbol{K})$ is Hermitian. See end of p. 18).

Then $\omega^2 a_\alpha = \sum_\beta \gamma_{\alpha\beta} a_\beta$, and we must solve the characteristic equation of a $3A$-by-$3A$ matrix:

$$\det \|\gamma_{\alpha\beta} - \omega^2\delta_{\alpha\beta}\| = 0. \tag{1.27}$$

The solutions of the characteristic equation are

$$\omega^{(r)}(\boldsymbol{K}) \equiv \omega^{(r)}_{\boldsymbol{K}},$$

where r runs from 1 to $3A$. The motion for a particular mode can be written

$$Z^{(r)}_{\alpha,N}(\boldsymbol{K}) = a^r_\alpha(\boldsymbol{K}) e^{-i\omega^{(r)}(\boldsymbol{K})^r} e^{i\boldsymbol{K}\cdot\boldsymbol{N}}$$

where

$$\sum_\alpha a^r_\alpha a^{*r'}_\alpha = \delta_{rr'}.$$

Then the general motion can be described by

$$Z_{\alpha,N} = \sum_{K,r} \frac{C_r(K)}{\sqrt{\eta}} a_\alpha^r(K) e^{-i\omega^{(r)}(K)^r} e^{iK\cdot N}$$

where the $C_r(K)/\sqrt{\eta}$ are arbitrary constants, and η is the total number of unit cells.

The factor $1/\sqrt{\eta}$ is inserted to make things look nicer later, but it is not strictly necessary.

Let $Q_r(K) = C_r(K)e^{-i\omega^{(r)}(K)^r}$. $Q_r(K)$ describes the motion of a particular mode.

$$Z_{\alpha,N} = \sum_{K,r} Q_r(K) a_\alpha^r(K) e^{iK\cdot N}(1/\sqrt{\eta}). \tag{1.28}$$

It follows that

$$Q_r(K) \propto \sum_{\alpha,N} Z_{\alpha,N} a_\alpha^{*r}(K) e^{-iK\cdot N}, \tag{1.29}$$

and the Hamiltonian for the system is

$$\begin{aligned} H &= \frac{1}{2}\sum_{\alpha,N}\left[\dot{Z}_{\alpha,N}^2 + \sum_{\beta,M} C_{\alpha\beta}^M Z_{\alpha,N} Z_{\beta,N+M}\right] \\ &= \frac{1}{2}\sum_{K,r}[|\dot{Q}_r(K)|^2 + \omega^{2(r)}(K)|Q_r(K)|^2]. \end{aligned} \tag{1.30}$$

If we consider $Q_r(K)$ and its complex conjugate to be independent variables, we get the same equations of motion from

$$H = \sum_{K,r}[|\dot{Q}_r(K)|^2 + \omega^{2(r)}(K)|Q_r(K)|^2]. \tag{1.31}$$

1.5 SPECIFIC HEAT OF A CRYSTAL

We now want to find F, so we must sum over all possible modes.

For one mode, $F = kT \ln(1 - e^{-\hbar\omega_K(p)/kT})$. There is a quasi-continuum of K values, and for each K value there are $3A$ $\omega_K^{(p)}$'s. Assuming the crystal has volume V, and assuming periodic boundary conditions, we have in the range K to $K + dK$ approximately $d3K/(2\pi)^3 V$ modes with a given p. Thus,

$$\begin{aligned} \frac{F}{V} &= kT\int_K \sum_p \frac{\hbar\omega_K(p)}{2} + kT\ln[1 + e^{-\hbar\omega_K(p)/kT}]\frac{d^3K}{(2\pi)^3} \\ d^3K &= dK_x\, dK_y\, dK_z. \end{aligned} \tag{1.32}$$

$\int_K$ is shorthand for

$$\int_{-\pi/a}^{\pi/a}\int_{-\pi/b}^{\pi/b}\int_{-\pi/c}^{\pi/c}.$$

The reason for the limits $-\pi/a$, π/a is as follows: the factor $\boldsymbol{K}$ was introduced in $e^{i\boldsymbol{K}\cdot\boldsymbol{N}}$. Then $e^{i\boldsymbol{K}\cdot\boldsymbol{N}} = e^{iK_x n_x a} e^{iK_y n_y b} e^{iK_z n_z c}$; n_x, n_y, n_z are integers. But,

$$e^{iK_x n_x a} = e^{i(K_x + 2\pi/a)n_x a} = e^{iK_x n_x a}\, e^{2\pi i}.$$

No new modes are introduced if $\boldsymbol{K}$ ranges beyond the prescribed limits.

We now demonstrate that the specific heat of a crystal determined by our present method agrees with experiment; that is, at high T, $C_v = 3R$. At low T, $C_v \to 0$ as T^3.

We choose to look at U rather than at F. First we note that there is a maximum frequency, ω_M. Two adjacent atoms can be no more than 180° out of phase, and thus the minimum wavelength must be of the order of twice the atomic spacing (call it $2a$). Thus $K_{\max} \approx 2\pi/2a = \pi/a$, and $\omega_M = \omega_M^{(p)}(K_M)$.

First, we consider the high-temperature limit. For $kT \gg \hbar\omega_M$, all modes are excited to approximately the same energy; that is, if there are N atoms,

$$\begin{aligned} U &= \sum_i \frac{\hbar\omega_i}{2} + \sum_i \frac{\hbar\omega_i}{\exp(\hbar\omega_i/kT) - 1} \approx \frac{\hbar}{2}\sum_i \omega_i + \sum_i kT \\ &= U_0 + 3NkT = U_0 + 3RT, \end{aligned} \tag{1.33}$$

if N is Avogadro's number. $C_v = 3R$ for large T. Note that we might as well neglect the zero-point energy, U_0.

For very small values of T ($kT \ll \hbar\omega_M$), the behavior of C_v can also be approximately determined.

Ignoring the zero-point energy, we see that the contribution to U of the high-frequency modes is very small, because

$$\frac{\hbar\omega}{\exp(\hbar\omega/kT) - 1} \approx 0$$

when $h\omega \gg kT$. We also know that when $\boldsymbol{K}$ is zero, there are three zero-frequency modes arising from translation of the entire crystal. For small $\boldsymbol{K}$, there should be three very low-frequency modes. To see how those modes vary with $\boldsymbol{K}$, consider $\gamma_{\alpha\beta}(\boldsymbol{K})$ for low $\boldsymbol{K}$. Since

$$\gamma_{\alpha\beta}(\boldsymbol{K}) = \sum_{\boldsymbol{M}} C_{\alpha,\beta}^{\boldsymbol{M}} e^{i\boldsymbol{K}\cdot\boldsymbol{M}},$$

where the $C_{\alpha\beta}^{\boldsymbol{M}}$ are real,

$$\gamma_{\beta\alpha}^*(\boldsymbol{K}) = \sum_{\boldsymbol{M}} C_{\beta,\alpha}^{\boldsymbol{M}} e^{-i\boldsymbol{K}\cdot\boldsymbol{M}} = \sum_{\boldsymbol{M}} C_{\beta,\alpha}^{-\boldsymbol{M}} e^{i\boldsymbol{K}\cdot\boldsymbol{M}} = \gamma_{\alpha\beta}(\boldsymbol{K}).$$

Also, for real $(\boldsymbol{K})$, $\gamma_{\beta\alpha}^*(\boldsymbol{K}) = \gamma_{\beta\alpha}(-\boldsymbol{K})$, so $\gamma_{\alpha\beta}(\boldsymbol{K}) = \gamma_{\beta\alpha}(-\boldsymbol{K})$, and

$$\begin{aligned} f(\omega, \boldsymbol{K}) &= \det \|\gamma_{\alpha\beta}(\boldsymbol{K}) - \omega^2\delta_{\alpha\beta}\| = \det \|\gamma_{\beta\alpha}(\boldsymbol{K}) - \omega^2\delta_{\beta\alpha}\| \\ &= \det \|\gamma_{\alpha\beta}(-\boldsymbol{K}) - \omega^2\delta_{\alpha\beta}\| = f(\omega^2, -\boldsymbol{K}). \end{aligned}$$

Then, because f is an even function of $\boldsymbol{K}$,

$$\frac{\partial f}{\partial \boldsymbol{K}}(\omega^2, 0) = 0$$

Since the characteristic equation must have a solution with $\omega = 0$ for $K = 0$, $f(0, 0) = 0$. The characteristic equation becomes

$$0 = \omega^2 \frac{\partial f}{\partial \omega^2} - \frac{1}{2!} \sum_{i,j} K_i K_j \frac{\partial^2 f}{\partial K_i \, \partial K_j}\bigg|_0 + \text{higher order terms,}$$

so that, for low ω and $|\boldsymbol{K}|$ we have

$$\omega = |\boldsymbol{K}| \left(\frac{\cos \alpha_i \cos \alpha_j \dfrac{\partial^2 f}{\partial K_i \, \partial K_j}\bigg|_0}{2\partial f/\partial \omega^2} \right)^{1/2},$$

where $\cos \alpha_i$ are the direction cosines of $\boldsymbol{K}$.

In other words, $\omega = |\boldsymbol{K}|V$, where V is the velocity of sound, which may depend on the direction of $\boldsymbol{K}$ when the eigenvalues of $\partial^2 f/\partial K_i dK_j|_0$ are not all equal. But, for sufficiently low frequency, V does not depend on the frequency. This is the *Debye approximation.*

For convenience, assume that the three sound velocities are equal; that is, $\omega = V_0 K$.

$$U = \sum_i \left[\frac{\hbar\omega_i}{2} + \frac{\hbar\omega_i}{e^{\hbar\omega_i/kT} - 1} \right]$$

$$= \sum_i \frac{\hbar\omega_i}{2} + 3 \int_0^{K_M} \frac{\hbar\omega(K)}{e^{\hbar\omega/kT} - 1} \frac{4\pi K^2}{(2\pi)^3} \, dK; \qquad K = \frac{\omega}{V_0}.$$

The factor 3 before the integral is there because for each K there are three modes with low ω.

$$U = U_0 + \frac{3\hbar}{2\pi^2 V_0^3} \int_0^{\omega_M} \frac{\omega^3}{e^{\hbar\omega/kT} - 1} \, d\omega, \qquad \text{where} \qquad \omega_M = V_0 K_{\max}. \tag{1.34}$$

$K_{\max}$ can be found by setting the total number of modes equal to $3(AN)$, where (AN) is the number of atoms in the crystal.

$$3(AN) = \int_0^{K_{\max}} \frac{3 \, d^3K V}{(2\pi)^3} = \int_0^{K_{\max}} \frac{3V}{(2\pi)^3} 4\pi K^2 \, dK = \frac{K_M^3 V}{2\pi^2}.$$

Then

$$K_M = (6\pi^2\rho)^{1/3} \text{ where } \rho \text{ is the number of atoms per unit volume.}$$

Let

$$x = \frac{\hbar\omega}{kT} = \frac{\Theta}{T}; \qquad \omega = \frac{kTx}{\hbar}; \qquad \omega^3\, d\omega = \frac{k^3T^3x^3}{\hbar^3}\frac{kT\, dx}{\hbar}; \qquad \Theta = \frac{\hbar KV}{k}.$$

$$\frac{U}{V} = \frac{U_0}{V} + \frac{3k^4T^4}{2\pi^2\hbar^3V_0^3}\int_0^{\Theta_M/T}\frac{x^3\, dx}{e^x - 1} = U_0 + AT^4. \tag{1.35}$$

Here Θ is the *Debye temperature.*

$$C_v = 4AT^3 \tag{1.36}$$

where $A \to \pi^2k^4/10h^3V_0^3$ as $T/\Theta \to 0$.
So C_v is proportional to T^3 at low temperatures.

The T^3 dependence of the specific heat at low T can be made more plausible as follows: We assume that for $\hbar\omega_i > kT$, the contribution to the energy of the ω_i mode is negligible and for $\hbar\omega_i < kT$, the contribution is kT.

The number of modes with wave number less than K is

$$3V\int_0^K \frac{4\pi k^2}{(2\pi)^3}\, dk = \frac{K^3V}{2\pi^2}.$$

At low T, $K = \omega/V_0$. Therefore the number of modes with frequency less than ω_c is $\omega_c^3V/2\pi^2V_0^3$. But $\hbar\omega_c = kT$ or $\omega_c = kT/\hbar$ and

$$n = (\text{number of modes with frequency less than } \omega_c) = k^3T^3V/2\pi^2V_0^3h^3.$$

The energy varies as

$$\frac{U}{V} = \frac{k^3T^3V}{2\pi^2\hbar^3V_0^3}(kT) = \frac{k^4T^4V}{2\pi^2\hbar^3V_0^3}.$$

And thus C_v is proportional to T^3. The numerical factor is, of course, not correct.

If the mode with frequency ω_i is excited to the nth level, $E_i = \hbar\omega_i(n + \frac{1}{2})$, we say that there are n phonons of frequency ω_i and energy $h\omega_i$ in the crystal.

Figure 1.4 shows the general form of the specific heat of a solid as a function of temperature.

Example: Assume the unit cell to be a cubic lattice with one atom per cell (Fig. 1.5). Each atom behaves as an harmonic oscillator, with spring constants k_A (nearest neighbors), and k_B (next-nearest neighbors). This case is fairly simple, and we can simplify the notation: $\alpha = 1, 2, 3$.

$$Z_{1,N} = X_N, \qquad Z_{2,N} = Y_N, \qquad Z_{3,N} = Z_N.$$

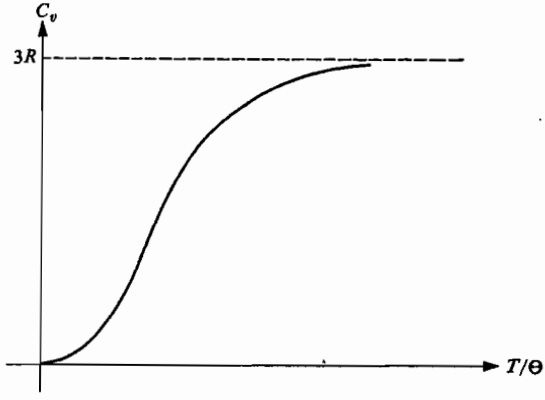

Fig. 1.4 Specific Heat of a Solid.

We wish to find the three natural frequencies associated with each K of the crystal. To do this, we must find $C^M_{\alpha,\beta}$ and then $\gamma_{\alpha,\beta}$. In complex coordinates

$$V = \sum_{\alpha,\beta} V_{\alpha\beta}$$

where

$$V_{\alpha\beta} = \sum_{N,M} C^M_{\alpha\beta} Z^*_{\alpha,N} Z_{\beta,N+M}. \tag{1.37}$$

For example,

$$V_{11} = \sum_{N,M} C^M_{11} X^*_N X_{N+M}. \tag{1.37'}$$

If we express the displacement of atom N from its normal position as X_N, then the potential energy from the distortion of the spring between atoms N and M is

$$\tfrac{1}{2}K_M \left[(X_N - X_{N+M}) \cdot \frac{M}{|M|} \right]^2;$$

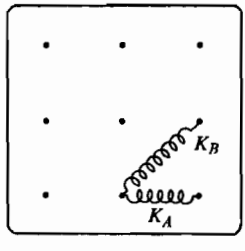

Fig. 1.5 Cubic lattice with one atom per cell.

$K_M = K_A$ for $N + M$ a nearest neighbor to N, $K_M = K_B$ for $N + M$ a next-nearest neighbor.

In summing over N and M to get the total potential energy we must divide by two, for we count each spring twice. If we use complex coordinates, however, we multiply V by two to get the correct equations of motion.

$$V = \frac{1}{2} \sum_{N,M} K_M \left| (X_N - X_{N+M}) \cdot \frac{M}{|M|} \right|^2, \tag{1.38}$$

$$\begin{aligned} V_{11} &= \frac{1}{2} \sum_{N,M} K_M \left| \frac{M_X}{|M|} \right|^2 (X_N^* - X_{N+M})(X_N - X_{N+M}) \\ &= \frac{1}{2} \sum_{N,M} K_M \left(\frac{M_X}{|M|} \right)^2 [(X_N^* X_N + X_{N+M}^* X_{N+M}) - (X_N^* X_{N+M} + X_{N+M}^* X_N)] \\ &= \sum_{N,M} K_M \left(\frac{M_X}{|M|} \right)^2 [X_N^* X_N - X_N^* X_{N+M}]. \end{aligned} \tag{1.38'}$$

Comparing Eqs. (1.37′) and (1.38′), we see that

$$C_{11}^0 = \sum_M K_M \left(\frac{M_X}{|M|} \right)^2; \qquad C_{11}^{M \neq 0} = -K_M \left(\frac{M_X}{|M|} \right)^2. \tag{1.39}$$

Here $(M_X/|M|)^2 = 1$ for $M = (\pm 1, 0, 0)$, $\frac{1}{2}$ for $M = (\pm 1, \pm 1, 0)$ and $M = (\pm 1, 0, \pm 1)$, and zero for the other nearest and next nearest neighbors. So

$$C_{11}^0 = 2K_A + 4K_B, \qquad C_{11}^{\pm(1,0,0)} = -K_A, \quad \text{and so on.}$$

In this way, all the $C_{\alpha,\beta}^M$ can be found. We can then calculate

$$\gamma_{\alpha,\beta}(K) = \sum_M C_{\alpha,\beta}^M [e^{iK \cdot M}].$$

We wish to solve $\det |\gamma_{\alpha,\beta} - \omega^2 \delta_{\alpha,\beta}| = 0$.

For each K, there are three solutions for ω. Thus we obtain $3N$ values of ω, $\omega^{(p)}(K)$.

1.6 THE MÖSSBAUER EFFECT

If a free, excited nucleus goes to its ground state by emission of a photon, the energy of the photon will be less than the excitation energy because the nucleus recoils. But if the excited nucleus is in a crystal, there is a finite probability that it will emit a photon with the full excitation energy. In other words, there is a finite probability that the state of the crystal after radiation will be the same as the initial state. Similarly, there is a finite probability that the crystal state will be unchanged by absorption of a photon. This effect, called the *Mössbauer effect*, can be discussed in terms of modes of oscillation of a crystal. For example,

suppose the temperature is absolute zero. Then the crystal must be in its vibrational ground state before emission of a photon. We will find an expression for the probability that the crystal is in its ground state after emission.

Let $\boldsymbol{R}$ be the position of the excited atom, and let $\boldsymbol{P}$ be the momentum of the emitted photon. We will assume, without proof, that the amplitude for the crystal being in a given final state is

$$a = \langle \text{final } |Ae^{i\boldsymbol{P}\cdot\boldsymbol{R}/\hbar}| \text{ initial}\rangle.$$

Furthermore, we will consider only the vibrational states of the crystal, and will neglect any effect due to the fact that the nucleus changes state.

Let the Mth atom be the one that emits the photon; let $\boldsymbol{R}_{0,M}$ be its mean position, and let $\boldsymbol{Z}_M$ be its displacement from that position.

$$\boldsymbol{R} = \boldsymbol{R}_{0,M} + \boldsymbol{Z}_M.$$

We can take the origin of our coordinate system at the mean position of the excited atom. Then $\boldsymbol{R}_{0,M} = \boldsymbol{0}$.

A single one-dimensional harmonic oscillator in the ground state is described by the wave function

$$\psi = \left(\frac{\omega}{\pi h}\right)^{1/4} e^{-\omega(Q^2/2\hbar)},$$

where Q is the position measured in units such that the mass can be set equal to unity. The wave function of a crystal in its ground state is a product of the wave functions for each mode of vibration.

$$\psi_{\text{crystal}} = \prod_{\boldsymbol{K},r} \left(\frac{\omega^{(r)}(\boldsymbol{K})}{\pi\hbar}\right)^{1/4} e^{-\omega^{(r)}(\boldsymbol{K})|Q_r^2(\boldsymbol{K})|/2\hbar}$$

We wish to calculate:

$$F = C\left|\iiint \psi^*_{\text{crystal}} e^{i\boldsymbol{P}\cdot\boldsymbol{Z}_M/\hbar}\psi_{\text{crystal}}\right|^2 \tag{1.40}$$

where F is the probability that the crystal will remain in its ground state and C is a constant independent of P. As $\boldsymbol{P} \to 0$, F must tend to unity. So, if we can get a formula for F without the correct constant in front of it, the constant can be determined easily. We will ignore such factors as "A" and "$(\omega/\pi\hbar)^{1/4}$."

We take $Z_{\alpha,M} = Z_{\alpha,0}$ as the displacement of the nucleus that emits the photon. For simplicity, assume the momentum of the photon is in the α direction. Then we wish to compute

$$F^{1/2} \propto \iiint \left(\prod \exp\left[\frac{-\omega^{(r)}(\boldsymbol{K})Q_r^2(\boldsymbol{K})}{2\hbar}\right]\right)(\exp\left[iPZ_{\alpha,0}/\hbar\right])$$
$$\times \left(\prod \exp\left[\frac{-\omega^{(r')}(\boldsymbol{K}')Q_{r'}^2(\boldsymbol{K}')}{2\hbar}\right]\right).$$

But

$$Z_{\alpha,0} = \sum_{K,r} Q_r(K) a_\alpha^r(K) \frac{1}{\sqrt{\eta}},$$

so

$$F^{1/2} \propto \prod_{K,r} \left[\int \exp\left[\frac{-\omega^{(r)}(K)|Q_r^2(K)|}{\hbar} \right] \exp\left[i \left(\frac{P}{\hbar\sqrt{\eta}} \right) Q_r(K) a_\alpha^r(K) \right] dQ_r(K) \right]$$

$$\propto \prod_{K,r} \exp\left[\frac{-P^2|a_\alpha^r(K)|^2}{4\hbar\omega^{(r)}(K)\eta} \right] = \exp\left[- \sum_{K,r} \frac{P^2|a_\alpha^r(K)|^2}{4\hbar\omega^{(r)}(K)\eta} \right].$$

Notice that

$$\langle (Z_{\alpha N})^2 \rangle = \left\langle \sum_{\substack{K,r \\ K',r'}} a_\alpha^{(r)} [a_\alpha^{(r')}]^* Q_r(K) Q_{r'}^*(K') e^{i(K-K')\cdot N} \frac{1}{\eta} \right\rangle$$

$$= \sum_{K,r} |a_\alpha^r(K)|^2 \langle Q_r^2(K) \rangle \frac{1}{\eta} = \sum_{K,r} \frac{\hbar |a_\alpha^r(K)|^2}{2\eta\omega^{(r)}(K)}.$$

So we can write

$$F \propto \exp\left[- \left(\frac{P^2}{\hbar^2} \right) \langle Z_{\alpha,M}^2 \rangle \right].$$

Because $F = 1$ at $P = 0$, we can write

$$F = \exp\left[- \left(\frac{P^2}{\hbar^2} \right) \langle Z_{\alpha,M}^2 \rangle \right]. \tag{1.41}$$

The value of F can also be found for nonzero temperature. If "i" is the number describing the state of the crystal, we must compute

$$F_i = |\langle i | e^{iPZ_{\alpha,0}/\hbar} | i \rangle|^2$$

and then

$$F = \frac{1}{Q} \sum_i F_i e^{-E_i/kT},$$

$$F_i e^{-E_i/kt} = \prod_{K,r} \left| \langle n | \exp\left[\frac{iP}{\hbar\sqrt{\eta}} Q_r(K) a_\alpha^r(K) \right] | n \rangle \right|^2 \exp\left[\frac{-(n + \frac{1}{2})\hbar\omega^{(r)}(K)}{kT} \right],$$

where $|n\rangle$ denotes the nth state of a single one-dimensional harmonic oscillator with frequency $\omega^{(r)}(K)$. As η is very large, we can write

$$F_i \exp\left(\frac{-E_i}{kT}\right) = \prod_{K,r} \left| 1 - \frac{1}{2!}\frac{P^2}{\eta\hbar^2} \langle Q_r^2(K)\rangle_n |a_\alpha^r(K)|^2 + \cdots \right|^2$$

$$\times \exp\left[-(n + \tfrac{1}{2})\frac{\hbar\omega^r(K)}{kT}\right]$$

$$\approx \prod_{K,r} \left| 1 - \frac{1}{2}\frac{P^2(2n+1)\hbar}{\eta\hbar^2 2\omega^{(r)}(K)} |a_\alpha^r(K)|^2 \right|^2 \exp\left[\frac{-(2n+1)\hbar\omega^{(r)}(K)}{2kT}\right],$$

$$F \propto \sum \prod \left[1 - \frac{P^2(2n+1)}{2\omega^{(r)}(K)\eta\hbar} |a_\alpha^r(K)|^2\right] \exp\left[\frac{-(2n+1)\hbar\omega^{(r)}(K)}{2kT}\right]$$

$$= \prod_{K,r} \sum_{n=0}^{\infty} \left[1 - \frac{P^2(2n+1)}{2\omega\eta\hbar} |a_\alpha^r|^2\right] \exp\left[\frac{-(2n+1)\hbar\omega}{2kT}\right]$$

$$\propto \prod_{K,r} \left[1 - \frac{P^2}{2\omega\eta\hbar} |a_\alpha^r|^2 \frac{1 + e^{-\hbar\omega/kT}}{1 - e^{-\hbar\omega/kT}}\right]$$

$$\approx - \sum_{K,r} \left[\frac{P^2}{2\omega\eta\hbar} |a_\alpha^r(K)|^2 \frac{1 + e^{-\hbar\omega/kT}}{1 - e^{-\hbar\omega/kT}}\right]$$

As before, we get for our answer

$$F = \exp\left[\frac{-P^2}{\hbar^2} \langle Z_{\alpha,M}^2\rangle\right]. \tag{1.41}$$

1.7 QUANTUM STATISTICS FOR A MANY-PARTICLE SYSTEM

Consider a system of N identical particles, and assume that there is no interaction among them. Any two configurations of the system that differ only by an interchange of two or more identical particles are regarded as one and the same state. Thus the state of the system is determined by giving the number of particles n_a with energy ε_a.

Our problem is to calculate the partition function Q with the condition $\sum n_a =$ constant. The values admitted for every n_a may be

(a) $n_a = 0, 1, 2, 3, 4, \ldots$ Bose-Einstein case

(b) $n_a = 0, 1$ Fermi-Dirac case

Bose-Einstein statistics must be used for particles with integral spin (for example, He^4) and Fermi-Dirac statistics for particles with half-integral spin (such as, electrons). For Bose particles, any number of particles may occupy a given state. For Fermi particles, however, there can only be one particle at most in each state (Pauli exclusion principle).

A state of the system is described by the set of numbers n_a, which can take on any set of values allowed both by the statistics and by the condition $\sum_a n_a = N$.

$$Q = \sum_{n_1, n_2, \ldots} \exp\left(-\beta \sum_a n_a \varepsilon_a\right). \tag{1.42}$$

If there were no restriction on the number of particles, we could write

$$Q = \prod_a \left(\sum_{n_a} e^{-\beta n_a \varepsilon_a}\right), \tag{1.43}$$

and we would have in the Bose-Einstein case

$$Q = \prod_a \left(\frac{1}{1 - e^{-\beta \varepsilon_a}}\right).$$

In the Fermi-Dirac case we would have

$$Q = \prod_a (1 + e^{-\beta \varepsilon_a}).$$

Unfortunately, Eq. (1.43) is incorrect, because we have an auxiliary condition that $\sum_a n_a = N =$ constant. With this auxiliary condition, the problem of finding Q becomes much more difficult. It is possible, however, to get around the restriction by considering the system of particles to be a box connected to a large reservoir of particles (Fig. 1.6), and by assuming that it is possible for particles to pass to and from the reservoir.

We further assume that the statistical mechanics of the total system acts as if it took energy μ to remove a particle from the box to the reservoir. μ can be adjusted by, say, a voltage regulator. In the case of electrons in a metal, for example, μ is the work function of the metal. In general, increasing μ will increase the expected number, $\langle N \rangle$, of particles in the box. If we can find $\langle N \rangle$ as

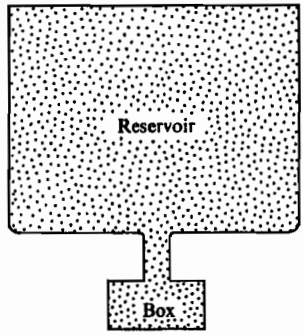

Fig. 1.6 System of particles considered as a box connected to a large reservoir.

a function of μ, we can in principle select μ so that any desired number of particles are in the box.

For a given μ, the energy levels of a particle in the box become $\varepsilon_a - \mu$, rather than ε_a.

By the key principle of statistical mechanics (as applied to the box), the probability of the gas having energy $E = \sum_a n_a(\varepsilon_a - \mu)$ is proportional to $e^{-\beta E}$.

$$Q^{(\mu)} = \sum_{n_1, n_2, \ldots} \exp\ -\beta[\sum n_a(\varepsilon_a - \mu)] \,. \tag{1.44}$$

The summation can be made without restriction. We will now show how μ is determined by N and how thermodynamic quantities depend on $Q^{(\mu)}$ or $g \equiv -1/\beta \ln Q^{(\mu)}$.

$$e^{-\beta g} = Q^{(\mu)} = \sum_{n_1, n_2, \ldots} \exp\left(-\beta\left[\sum_a n_a(\varepsilon_a - \mu)\right]\right). \tag{1.45}$$

Let $N = \sum_a n_a$.

$$\begin{aligned} \frac{\partial Q^{(\mu)}}{\partial \mu} &= \sum \beta \left(\sum_a n_a\right) \exp\left(-\beta\left[\sum_a n_a(\varepsilon_a - \mu)\right]\right) \\ &= \sum \beta N \exp\left(-\beta\left[\sum_a n_a(\varepsilon_a - \mu)\right]\right). \end{aligned} \tag{1.46}$$

But

$$\langle N \rangle = \frac{1}{Q^{(\mu)}} \sum N \exp\left(-\beta\left[\sum_a n_a(\varepsilon_a - \mu)\right]\right). \tag{1.47}$$

$$\langle N \rangle = \frac{1}{\beta Q^{(\mu)}} \frac{\partial Q^{(\mu)}}{\partial \mu} = \frac{1}{\beta} \frac{\partial}{\partial \mu} \ln Q^{(\mu)} = \frac{-\partial g}{\partial \mu}.$$

This equation gives $\langle N \rangle = \langle N(\mu) \rangle$. Inverting it gives μ as a function of $\langle N \rangle$.* Similarly, let $\langle n_a \rangle$ be the average value of n_a for a given μ (or a given $\langle N \rangle$). Then $\langle n_a \rangle = \partial g / \partial f_a$ and $\langle N \rangle = \sum_a \langle n_a \rangle$.

For the purpose of computing the probability of a state, we assumed that the system acts as if it had energy $E = \sum_a n_a(\varepsilon_a - \mu)$. But the μ was inserted only in order to provide for the weighting factor that accounts for different probabilities of different numbers of particles in the system. For such purposes

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* *Problem*: The function

$$f(\mu) = \sum_v \frac{1}{e^{\beta(\varepsilon_v - \mu)} - 1}$$

has many plus-minus infinity points, and therefore the equation $f(\mu) = N$ has many solutions. Why do we still talk about a unique chemical potential for bosons?

as computing the pressure, we do not consider μ to be part of the energy of the system.

$$U = \frac{1}{Q} \sum_{n_1, n_2, \ldots} \left(\sum_a n_a \varepsilon_a \right) \exp \left\{ -\beta \left[\sum_a n_a(\varepsilon_a - \mu) \right] \right\} = \frac{\partial \beta g}{\partial \beta} + \mu \langle N \rangle. \quad (1.48)$$

$$\text{Pressure} = P = \frac{1}{Q} \sum_{n_1, n_2, \ldots} - \left(\frac{\partial}{\partial V} \sum_a n_a \varepsilon_a \right) \exp \left\{ -\beta \left[\sum_a n_a(\varepsilon_a - \mu) \right] \right\}.$$

μ can be treated as a variable or as a function of V and $\langle N \rangle$. If we treat it as an independent variable, so that $g = g(V, \mu)$, we get

$$P = \left. \frac{-\partial g}{\partial V} \right|_{\mu = \text{const.}}.$$

If we treat μ (and g) as functions of V and $\langle N \rangle$, that is, with $g = g[V, \mu(V, \langle N \rangle)]$, then

$$\left. \frac{-\partial g}{\partial V} \right|_{\langle N \rangle = \text{const.}} = P + \langle N \rangle \frac{\partial \mu}{\partial V}$$

is easily shown. Also,

$$\left. \frac{\partial g}{\partial N} \right|_V = -\langle N \rangle \left. \frac{\partial \mu}{\partial \langle N \rangle} \right|_V.$$

If $F = F(V, \langle N \rangle) = g(V, \mu(V, \langle N \rangle)) + \langle N \rangle \mu(V, \langle N \rangle)$, then

$$P = \left. \frac{-\partial F}{\partial V} \right|_{\langle N \rangle = \text{const.}} \quad \text{and} \quad \mu = \left. \frac{\partial F}{\partial \langle N \rangle} \right|_{V = \text{const.}}.$$

$$\text{Entropy} = S = -k \sum_{n_1, n_2, \ldots} P_{n_1, n_2, \ldots} \ln P_{n_1, n_2, \ldots}$$

where

$$P_{n_1, n_2, \ldots} = \frac{1}{Q^{(\mu)}} \exp \left[-\beta \sum_a n_a(\varepsilon_a - \mu) \right].$$

It is easily shown that

$$S = -\left. \frac{\partial g}{\partial T} \right|_{\mu, V = \text{const.}}.$$

In summary:

$$g = \frac{-1}{\beta} \ln Q^{(\mu)} = \frac{-1}{\beta} \ln \sum_{n_1, n_2, \ldots} \exp \left\{ -\beta \left[\sum_a n_a(\varepsilon_a - \mu) \right] \right\}, \quad (1.49)$$

$$\langle n_a \rangle = \frac{\partial g}{\partial \varepsilon_a}, \quad (1.50)$$

$$\langle N \rangle = \frac{-\partial g}{\partial \mu} = \sum_a \langle n_a \rangle, \quad (1.51)$$

$$U = \frac{1}{Q} \sum_{n_1, n_2, \ldots} \left(\sum_a n_a \varepsilon_a \right) \exp\left\{ -\beta \left[\sum_a n_a(\varepsilon_a - \mu) \right] \right\} = \frac{\partial \beta g}{\partial \beta} + \mu \langle N \rangle, \quad (1.52)$$

$$P = \frac{1}{Q} \sum_{n_1, n_2, \ldots} -\left(\frac{\partial}{\partial V} \sum_a n_a \varepsilon_a \right) \exp\left\{ -\beta \left[\sum_a n_a(\varepsilon_a - \mu) \right] \right\}$$

$$= \left. \frac{-\partial g}{\partial V} \right|_{\mu}, \quad (1.53)$$

$$S = \left. \frac{-\partial g}{\partial T} \right|_{\mu, V}. \quad (1.54)$$

Soon we will be able to find g for an ideal Bose gas, and then for an ideal Fermi gas. But first we must evaluate some integrals.

1.8 EVALUATION OF INTEGRALS

We will soon be dealing with integrals of the form $\int_{-\infty}^{\infty} e^{-ax^2}\, dx$ and $\int_{-\infty}^{\infty} x^2 e^{-ax^2}\, dx$. Let us pause for a second to calculate these integrals:

$$\left[\int_{-\infty}^{\infty} e^{-x^2}\, dx \right]^2 = \int_{-\infty}^{\infty} e^{-x^2}\, dx \int_{-\infty}^{\infty} e^{-y^2}\, dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)}\, dx\, dy = \int_0^{2\pi} \int_0^{\infty} e^{-r^2}\, r\, dr\, d\theta$$

$$= -\tfrac{1}{2} \int_0^{2\pi} \int_0^{\infty} e^{-r^2}(-2r\, dr)\, d\theta = [\sqrt{\pi}]^2. \quad (1.55)$$

$$\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}. \quad (1.56)$$

$$\int_{-\infty}^{\infty} e^{-ax^2}\, dx = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} e^{-y^2}\, dy = \frac{\sqrt{\pi}}{\sqrt{a}} \qquad (y = \sqrt{a}\, x). \quad (1.57)$$

$$\int_{-\infty}^{\infty} x^2 e^{-ax^2}\, dx = -\int_{-\infty}^{\infty} \frac{d}{da} e^{-ax^2}\, dx = -\frac{d}{da} \int_{-\infty}^{\infty} e^{-ax^2}\, dx$$

$$= -\frac{d}{da} \frac{\sqrt{\pi}}{\sqrt{a}} = \tfrac{1}{2} a^{-3/2} \sqrt{\pi}. \quad (1.58)$$

$$\int_{-\infty}^{\infty} x^3 e^{-ax^2}\, dx = 0 \quad \text{(odd function)} \quad (1.59)$$

$$\int_{-\infty}^{\infty} x^4 e^{-ax^2}\, dx = \frac{d^2}{da^2} \int_{-\infty}^{\infty} x^2\, e^{-ax^2}\, dx = \tfrac{3}{4} a^{-5/2} \sqrt{\pi}, \quad (1.60)$$

and so forth.

1.9 THE IDEAL BOSE-EINSTEIN GAS

From Eq. (1.45) we have

$$e^{-\beta g} = \sum_{n_1, n_2, \ldots} \exp\{-\beta[n_1(\varepsilon_1 - \mu) + n_2(\varepsilon_2 - \mu) + \cdots]\}$$
$$= \sum_{n_1} \exp[-\beta n_1(\varepsilon_1 - \mu)] \sum_{n_2} \exp[-\beta n_2(\varepsilon_2 - \mu)] \cdots \tag{1.61}$$

For a Bose-Einstein gas, $n_a = 0, 1, 2, \ldots$

$$\sum_{n_i} e^{-\beta n_i(\varepsilon_i - \mu)} = \frac{1}{1 - e^{-\beta(\varepsilon_i - \mu)}} \text{ (Bose Einstein)}, \tag{1.62}$$

$$e^{-\beta g} = \prod_i \frac{1}{1 - e^{-\beta(\varepsilon_i - \mu)}}, \tag{1.63}$$

$$g = \frac{1}{\beta} \sum_i \ln(1 - e^{-\beta(\varepsilon_i - \mu)}). \tag{1.64}$$

Consider a particle contained in a box but otherwise free. The number of modes in the box with momentum in the three-dimensional region d^3p is $s(d^3k/(2\pi)^3)V$, where V = volume and s is the number of possible spin states (e.g., $s = 3$ for spin 1 when the rest mass does not vanish; $s = 2$ for spin 1 when the rest mass vanishes, as for photons). The energy is

$$\varepsilon = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}.$$

Here we have approximated the discrete set of modes by a continuum (which is the case for a completely free particle). The sum for g can then be replaced by an integration:

$$g = s\frac{1}{\beta}\int \ln(1 - e^{-\beta p^2/2m} e^{\beta\mu}) \frac{d^3p}{(2\pi\hbar)^3} V \tag{1.65}$$

$$\rho = \frac{\langle N \rangle}{V} = -\frac{1}{V}\frac{\partial g}{\partial \mu} = s\int \frac{e^{-\beta p^2/2m} e^{\beta\mu}}{1 - e^{-\beta p^2/2m} e^{\beta\mu}} \frac{d^3p}{(2\pi\hbar)^3}. \tag{1.66}$$

For computational purposes, set $\alpha = e^{\beta\mu}$ and $x^2 = \beta p^2/2m$. Then $p^2 = 2mx^2/\beta$ and $dp = (2m/\beta)^{1/2}\, dx$.

$$\frac{1}{(2\pi\hbar)^3} d^3p \to \frac{4\pi p^2\, dp}{(2\pi\hbar)^3} = \frac{x^2}{2\pi^2\hbar^3}\left(\frac{2m}{\beta}\right)^{3/2} dx.$$

$$\rho = s\int_0^\infty \frac{e^{-x^2}\alpha}{1 - e^{-x^2}\alpha}\left[\frac{x^2}{2\pi^2\hbar^3}\left(\frac{2m}{\beta}\right)^{3/2}\right] dx$$
$$= s\frac{1}{4\pi^2\hbar^3}\left(\frac{2m}{\beta}\right)^{3/2}\left[\int x^2\{\alpha e^{-x^2}\, dx + \alpha^2 e^{-2x^2}\, dx + \cdots\}\right]$$
$$= s\left(\frac{mkT}{2\pi h^2}\right)^{3/2}\left[\alpha + \frac{\alpha^2}{2^{3/2}} + \frac{\alpha^3}{3^{3/2}} + \cdots\right], \tag{1.67}$$

where we have used $\int_{-\infty}^{\infty} x^2 e^{-ax^2}\,dx = (\sqrt{\pi}/2a)(1/\sqrt{a})$, from Eq. (1.58). Letting

$$\zeta_r(\alpha) = \sum_{n=1}^{\infty} \frac{\alpha^n}{n^r},$$

we obtain finally

$$\rho = s\left(\frac{mkT}{2\pi h^2}\right)^{3/2} \zeta_{3/2}(\alpha). \tag{1.68}$$

Thus, given ρ, we can find $\alpha = e^{\beta\mu}$ (in principle) by solving the equation

$$\zeta_{3/2}(\alpha) = \frac{1}{s}\left(\frac{2\pi h^2}{mk}\right)^{3/2} \frac{\rho}{T^{3/2}}.$$

For the total energy of the system we have

$$\begin{aligned} U &= \frac{\partial(\beta g)}{\partial\beta} + \mu\langle N\rangle \\ &= s\int \frac{(e^{-\beta p^2/2m}e^{\beta\mu})p^2/2m}{1 - e^{-\beta p^2/2m}e^{\beta\mu}}\, V\, \frac{d^3p}{(2\pi\hbar)^3_3} \\ &= s\tfrac{3}{2}kT\left(\frac{mkT}{2\pi h^2}\right)^{3/2} V\zeta_{5/2}(\alpha). \end{aligned} \tag{1.69}$$

For very low ρ, high T, or both, $\rho/T^{3/2}$ is very small; then $\zeta_{3/2}(\alpha)$ is very small, and thus α is very small. In this case $\zeta_{3/2}(\alpha) \approx \alpha$, $\zeta_{5/2}(\alpha) \approx \alpha$, so that

$$\frac{\zeta_{5/2}(\alpha)}{\zeta_{3/2}(\alpha)} \approx 1$$

and $U/V \approx (3/2)kT\rho$. This condition therefore represents the classical limit.

Now let us look at the other limit. As T becomes lower and lower, with ρ fixed, $\zeta_{3/2}(\alpha)$ becomes larger and α approaches 1. For $\alpha > 1$, ζ diverges.

$$\zeta_{3/2}(1) = 1 + \frac{1}{2^{3/2}} + \frac{1}{3^{3/2}} + \cdots = 2.612,$$

$$\zeta_{5/2}(1) = 1 = \frac{1}{2^{5/2}} + \frac{1}{3^{5/2}} + \cdots = 1.341.$$

The temperature T_c at which $\alpha = 1$ is called the *critical temperature* for Bose-Einstein condensation.

$$\frac{\langle N\rangle}{V} = \rho = s\left(\frac{mkT_c}{2\pi\hbar^2}\right)^{3/2}(2.612), \tag{1.70}$$

$$T_c = \frac{2\pi\hbar^2}{mk}\left(\frac{\rho/s}{2.612}\right)^{2/3}. \tag{1.71}$$

The question arises: Why does our analysis break down at T_c? The answer is that for such low temperatures ($T < T_c$), we cannot replace the sum, Eq. (1.64), by the integral, Eq. (1.65).

Looking back at g and $\langle N \rangle$ before we approximated all summations by integrals, we recall that

$$g = \frac{1}{\beta}\sum_i \ln(1 - e^{-\beta\varepsilon_i}e^{\beta\mu}) = \frac{1}{\beta}\sum_i \ln(1 - e^{-\beta\varepsilon_i}\alpha);$$

$$N = \frac{-\partial g}{\partial \mu} = \sum_i \frac{e^{-\beta\varepsilon_i}\alpha}{1 - e^{-\beta\varepsilon_i}\alpha} = \sum_i \frac{1}{\alpha^{-1}e^{\beta\varepsilon_i} - 1} = \sum_i \frac{1}{e^{-\beta\mu}e^{\beta\varepsilon_i} - 1}.$$

For small α, $e^{-\beta\mu}$ is large; the terms with the lowest ε_i do not contribute much to the sum, and we can replace the sum with an integral. When $e^{-\beta\mu}$ is small, we cannot replace the sum with an integral because the first few discrete terms in the sum are important. Now

$$\langle n_a \rangle = \frac{\partial g}{\partial \varepsilon_a} = \frac{1}{(e^{-\beta\mu}e^{\beta\varepsilon_a}) - 1}.$$

Because $\langle n_a \rangle$ is positive, $(\varepsilon_a - \mu)$ must be greater than zero. For $\langle N \rangle$ to be larger than its value at $T = T_c$, $(\varepsilon_a - \mu)$ must be positive, but very small, in order that the low-energy terms in $\langle N \rangle$ be nonnegligible. Assuming no accidental degeneracy in the lowest level (that is, assuming $\varepsilon_0 \neq \varepsilon_1$) we can conclude that, for sufficiently low temperature,

$$\langle n_1 \rangle = \frac{1}{e^{\beta(\varepsilon_1 - \mu)} - 1} \ll \frac{1}{e^{\beta(\varepsilon_0 - \mu)} - 1} = \langle n_0 \rangle.$$

Without any loss of generality, we may take ε_0 as the zero of the energy. Then

$$\langle n_0 \rangle = \frac{1}{e^{-\beta\mu} - 1} \Rightarrow \mu = -kT \ln\left(1 + \frac{1}{\langle n_0 \rangle}\right) \approx -\frac{kT}{\langle n_0 \rangle} \quad \text{for large } \langle n_0 \rangle.$$

For low temperatures, μ is very close to zero, so for the energy states above ε_0, we can neglect μ. The sum for $\langle N - n_0 \rangle$ can be approximated by an integral:

$$N_{\text{exc}} = \langle N - n_0 \rangle = \sum_{i=1}^{\infty} \frac{1}{e^{\beta\varepsilon_i} - 1} \approx sV \int \frac{d^3p}{(2\pi\hbar)^3} \frac{1}{e^{p^2/2mkT} - 1}$$

$$= 2.612\left(\frac{mkT}{2\pi\hbar^2}\right)^{3/2} sV = \langle N \rangle \left(\frac{T}{T_c}\right)^{3/2}.$$

We see, then, that our definition of the critical temperature is such that when $T < T_c$, there must be a nonnegligible fraction of the particles in the ground state:

$$\langle n_0 \rangle = \langle N \rangle - N_{\text{exc}}$$
$$= \langle N \rangle [1 - (T/T_c)^{3/2}].$$

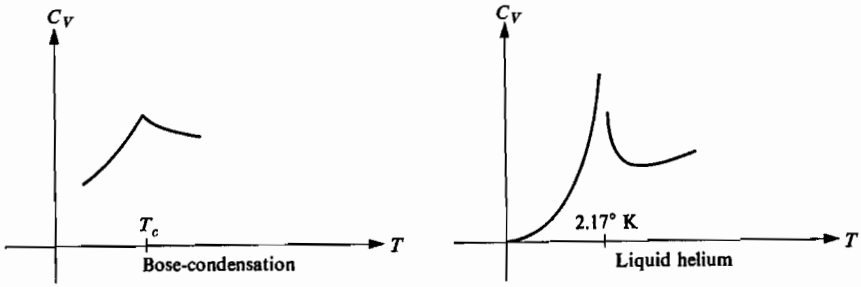

Fig. 1.7 A Bose-Einstein gas around T_c compared with liquid helium around the lambda point.

For $T < T_c$ the total energy is

$$U = \tfrac{3}{2}kT\,s\left(\frac{mkT}{2\pi\hbar^2}\right)^{3/2} V\zeta_{5/2}(1) = \tfrac{3}{2}kTN_{\text{exc}}\frac{\zeta_{5/2}(1)}{\zeta_{3/2}(1)}$$

$$= \tfrac{3}{2}kTN_{\text{exc}}(0.5134)$$

$$= \tfrac{3}{2}kT(0.5134)\left(\frac{T}{T_c}\right)^{3/2}\langle N\rangle. \tag{1.72}$$

Then $C_V = \partial U/\partial T \propto T^{3/2}$ at sufficiently low temperature.

For $T > T_c$, we have $\langle N\rangle = (mkT/2\pi\hbar^2)^{3/2}\zeta_{3/2}(\alpha)sV$. Using the definition of T_c, we have $\zeta_{3/2}(\alpha) = 2.612(T_c/T)^{3/2}$. This equation determines α from $\langle N\rangle$ and T. Then

$$U = \tfrac{3}{2}kT\langle N\rangle\frac{\zeta_{5/2}(\alpha)}{\zeta_{3/2}(\alpha)}. \tag{1.73}$$

The behavior of the specific heat of a Bose-Einstein gas around T_c is rather similar to the behavior of liquid helium near the so-called "lambda point" (Λ point). See Fig. 1.7.* For a given ρ, T_c and Λ are very close to one another (the lambda point is a few degrees K). Because of the mutual forces between particles, liquid helium is certainly not an ideal Bose gas, but perhaps part of the explanation of the lambda transition involves Bose condensation.

* For a calculation of the discontinuity in $(\partial C_v/\partial T)_v$ at the transition point, see L. D. Landau & E. M. Lifshitz, *Statistical Physics*, problem of §59, Pergamon Press, 1959.

A more exact description of the behavior of liquid helium near the lambda point is given by the empirical formula:

$$C_V \approx \begin{cases} a + b \ln |T - T_\Lambda|, & T < T_\Lambda, \\ a' + b \ln |T - T_\Lambda|, & T > T_\Lambda. \end{cases}$$

The explanation of this behavior is left as an exercise for the reader. If successful, publish!

1.10 THE FERMI-DIRAC GAS

For a Fermi gas, we proceed exactly as for the Bose case up to the point

$$e^{-\beta g} = \sum_{n_1} e^{-\beta n_1(\varepsilon_1 - \mu)} \sum_{n_2} e^{-\beta n_2(\varepsilon_2 - \mu)} \cdots . \tag{1.74}$$

Here $n_1 = 0, 1$; $n_2 = 0, 1$; and so on. Thus

$$e^{-\beta g} = \prod_i (1 + e^{-\beta(\varepsilon_i - \mu)}) \tag{1.75}$$

$$g = -\left(\frac{1}{\beta}\right) \sum_i \ln (1 + e^{-\beta(\varepsilon_i - \mu)}). \tag{1.76}$$

Once again, we can approximate the sum by an integral. This time there is no danger that a sizable fraction of the particles will be in the lowest state. Assume that the gas is of electrons, so that $s = 2$.

$$g = -\left(\frac{1}{\beta}\right) \int \ln (1 + e^{-\beta(p^2/2m - \mu)}) \frac{2\, d^3p}{(2\pi\hbar)^3} V, \tag{1.77}$$

$$\rho = \frac{\langle N \rangle}{V} = -\frac{1}{V}\frac{\partial g}{\partial \mu} = \int \frac{e^{-\beta(p^2/2m)} e^{\beta\mu}}{1 + e^{-\beta p^2/2m} e^{\beta\mu}} \frac{2\, d^3p}{(2\pi\hbar)^3}, \tag{1.78}$$

$$\langle n_a \rangle = \frac{\partial g}{\partial \varepsilon_a} = \frac{e^{-\beta(\varepsilon_a - \mu)}}{1 + e^{-\beta(\varepsilon_a - \mu)}} = \frac{1}{e^{\beta(\varepsilon_a - \mu)} + 1}. \tag{1.79}$$

The form of $\langle n_a \rangle$ for $T = 0$ and for any $T > 0$ is shown in Fig. 1.8. Now at $T = 0$, $\langle n_a \rangle = 1$ if $\varepsilon_a < \mu$; and $\langle n_a \rangle = 0$ if $\varepsilon_a > \mu$. Let $\mu = \mu_0$ at $T = 0$. In other words, at $T = 0$ all the states with energy less than μ_0 are occupied, and all with energy greater than μ_0 are empty. μ_0 is called the *Fermi energy* and μ_0/k the *Fermi temperature*.

$$\rho = \int_0^{p_0} 2\left(\frac{4\pi p^2}{(2\pi\hbar)^3}\right) dp = 2\left(\frac{4\pi p_0^3}{3}\right)\frac{1}{(2\pi\hbar)^3}, \tag{1.80}$$

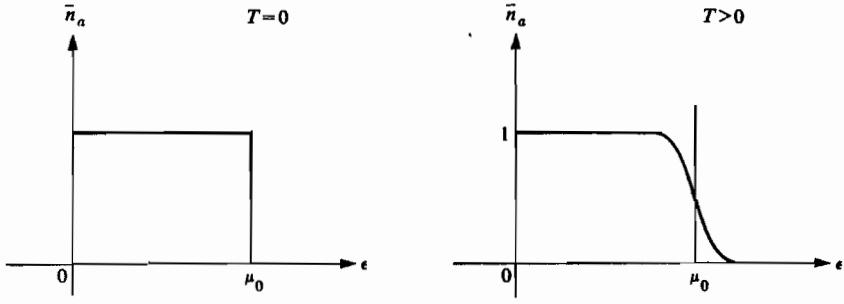

Fig. 1.8 Occupation of states at $T = 0$ and $T > 0$.

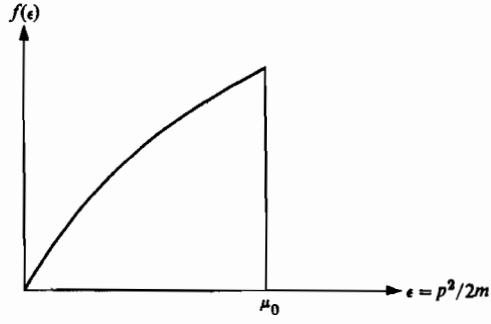

Fig. 1.9 The density of states as a function of energy.

where $\mu_0 = p_0^2/2m$. For most "reasonable" values of μ_0 and T, $\mu_0 \gg kT$ (for example, μ_0/k for copper is 82,000°K*).

Since

$$2\,\frac{4\pi p^2\,dp}{(2\pi\hbar)^3} = \frac{4\pi(2m)^{3/2}}{(2\pi\hbar)^3}\sqrt{\varepsilon}\,d\varepsilon,$$

the density of states with respect to energy is

$$f(\varepsilon) = \frac{4\pi(2m)^{3/2}}{(2\pi\hbar)^3}\sqrt{\varepsilon}.$$

This function is shown in Fig. 1.9.

$$u \equiv \frac{U}{V} = \frac{1}{V}\left(\frac{\partial\beta g}{\partial\beta} + \mu N\right) = \int\left(\frac{p^2}{2m}\right)\frac{e^{-\beta(p^2/2m-\mu)}}{1 + e^{-\beta(p^2/2m-\mu)}}\,\frac{2\,d^3p}{(2\pi\hbar)^3}. \tag{1.81}$$

* Also, in states of very condensed matter, as in a neutron star, μ_0/k of the neutron, proton, and electron liquids are $\sim 10^{10} - 10^{11}$°K whereas $T \sim 10^8$°K.

At $T = 0$,

$$u(0) = \int_0^{p_0} \left(\frac{p^2}{2m}\right) 2 \left(\frac{4\pi p^2\, dp}{(2\pi\hbar)^3}\right) = \frac{4\pi}{(2\pi\hbar)^3} \frac{1}{2m} 2 \left(\int_0^{p_0} p^4\, dp\right)$$

$$= \frac{4\pi}{(2\pi\hbar)^3} 2 \left(\frac{1}{2m}\right) \frac{1}{5} p_0^5, \tag{1.82}$$

$$\frac{u(0)}{\rho} = \frac{\left[\dfrac{4\pi}{(2\pi\hbar)^3}\left(\dfrac{1}{2m}\right)\dfrac{2}{5} p_0^5\right]}{\left[\dfrac{4\pi}{(2\pi\hbar)^3}\dfrac{2}{3} p_0^3\right]} = \frac{3}{5}\left(\frac{p_0^2}{2m}\right) = \tfrac{3}{5}\mu_0. \tag{1.83}$$

We will now tackle the case of finite T. For $T \neq 0$, what is μ and what is the specific heat, etc.?

$$\rho = \frac{\langle N \rangle}{V} = -\frac{1}{V}\frac{\partial g}{\partial \mu} = \int \frac{e^{-\beta(p^2/2m)} e^{\beta\mu}}{1 + e^{-\beta p^2/2m} e^{\beta\mu}} \frac{2\, d^3p}{(2\pi\hbar)^3}, \tag{1.78}$$

$$u = \int \frac{(p^2/2m) e^{-\beta(p^2/2m)} e^{\beta\mu}}{1 + e^{-\beta p^2/2m} e^{\beta\mu}} \frac{2\, d^3p}{(2\pi\hbar)^3}. \tag{1.81}$$

For convenience, we wish to express these integrals in terms of the energy ε:

$$\varepsilon = p^2/2m, \qquad \sqrt{2m\varepsilon} = p, \qquad dp = \sqrt{m/2\varepsilon}\, d\varepsilon.$$

d^3p becomes $4\pi p^2\, dp\, d\varepsilon = 2\pi(2m)^{3/2}\sqrt{\varepsilon}\, d\varepsilon$. Thus,

$$\rho = \int_0^\infty \frac{1}{e^{\beta(\varepsilon-\mu)} + 1} \frac{4\pi(2m)^{3/2}}{(2\pi\hbar)^3} \sqrt{\varepsilon}\, d\varepsilon = a \int_0^\infty \frac{\sqrt{\varepsilon}\, d\varepsilon}{e^{\beta(\varepsilon-\mu)} + 1}, \tag{1.84}$$

and

$$u = \int_0^\infty \frac{1}{e^{\beta(\varepsilon-\mu)} + 1} \frac{4\pi(2m)^{3/2}}{(2\pi\hbar)^3} \varepsilon^{3/2}\, d\varepsilon = a \int_0^\infty \frac{\varepsilon^{3/2}\, d\varepsilon}{e^{\beta(\varepsilon-\mu)} + 1}, \tag{1.85}$$

where

$$a = \frac{4\pi(2m)^{3/2}}{(2\pi\hbar)^3}.$$

In both cases, the integral that must be evaluated is

$$I = \int_0^\infty \frac{g(\varepsilon)\, d\varepsilon}{e^{\beta(\varepsilon-\mu)} + 1},$$

where $g(\varepsilon) = a\sqrt{\varepsilon}$ for ρ, and $g(\varepsilon) = a\varepsilon^{3/2}$ for μ. Then

$$I = \int_0^\infty \frac{g(\varepsilon)\, d\varepsilon}{e^{\beta(\varepsilon-\mu)} + 1} = \int_\mu^\infty \frac{g(\varepsilon)\, d\varepsilon}{e^{\beta(\varepsilon-\mu)} + 1} + \int_0^\mu g(\varepsilon)\, d\varepsilon - \int_0^\mu \frac{g(\varepsilon)\, d\varepsilon}{e^{-\beta(\varepsilon-\mu)} + 1}.$$

In the first term let $x = \beta(\varepsilon - \mu)$, and in the third term let $x = -\beta(\varepsilon - \mu)$.

$$I = \int_0^\mu g(\varepsilon)\, d\varepsilon + \int_0^\infty \frac{g(\mu + x/\beta)}{e^x + 1} \frac{dx}{\beta} - \int_0^{\beta\mu} \frac{g(\mu - x/\beta)}{e^x + 1} \frac{dx}{\beta}. \tag{1.86}$$

For what we have called "reasonable" values of ρ and T, not much error will arise if we replace $g(\mu \pm x/\beta)$ by $g(\mu) \pm (x/\beta)g'(\mu)$ and $\int_0^{\beta\mu}$ by $\int_0^\infty$. Then

$$I \approx \int_0^\mu g(\varepsilon)\, d\varepsilon + \frac{2}{\beta^2} g'(\mu) \int_0^\infty \frac{x\, dx}{e^x + 1} = \int_0^\mu g(\varepsilon)\, d\varepsilon + \frac{\pi^2 g'(\mu)}{6\beta^2}. \tag{1.87}$$

(The integral

$$\int_0^\infty \frac{x\, dx}{e^x + 1}$$

can be evaluated by expanding $xe^{-x}/(1 + e^{-x})$ in a power series, then integrating and summing; the result is $\pi^2/12$.) So,

$$I = \int_0^\mu g(\varepsilon)\, d\varepsilon + \frac{\pi^2}{6} k^2 T^2 g'(\mu). \tag{1.88}$$

Remember that for $I = \rho$, $g(\varepsilon) = a\sqrt{\varepsilon}$, and for $I = u$, $g(\varepsilon) = a\varepsilon^{3/2}$.

$$\rho = a \int_0^\mu \sqrt{\varepsilon}\, d\varepsilon + \frac{\pi^2}{6} (kT)^2 \frac{a}{2\sqrt{\mu}}$$

$$= \left(\frac{2a}{3}\right) \mu^{3/2} + \frac{\pi^2}{6} (kT)^2 \frac{a}{2\sqrt{\mu}}. \tag{1.89}$$

But we have also $\rho = (2a/3)\mu_0^{3/2}$ from Eq. (1.80). Thus

$$\mu \approx \mu_0 \left(1 - \frac{\pi^2}{12} \frac{k^2 T^2}{\mu_0^2}\right) \tag{1.90}$$

$$u \approx a \int_0^\mu e^{3/2}\, d\varepsilon + \frac{\pi^2}{6} (kT)^2 \frac{3a}{2} \sqrt{\mu}$$

$$= \frac{2}{5} a\mu^{5/2} + \frac{\pi^2}{6} (kT)^2 \frac{3a}{2} \sqrt{\mu}$$

$$\approx \frac{2}{5} a\mu_0^{5/2} - a\mu_0^{5/2} \frac{\pi^2}{12} \frac{(kT)^2}{\mu_0^2} + \frac{a\pi^2}{4} (kT)^2 \sqrt{\mu_0}$$

$$= u_0 + \frac{a\pi^2}{6} \sqrt{\mu_0}\, (kT)^2 = u_0 + \gamma T^2; \qquad \gamma = \frac{a\pi^2}{6} \sqrt{\mu_0}\, k^2. \tag{1.91}$$

$$u = u_0 + \gamma T^2; \qquad U = uV = U_0 + \gamma V T^2 = U_0 + \gamma' T^2$$

$$C_V = \frac{\partial U}{\partial T} = 2\gamma' T,$$

$$C_V \text{ of a metal} = \underset{\text{elect.}}{2\gamma' T} + \underset{\text{lattice vibrations}}{\alpha T^3} \text{ at low } T$$

At very low T the $\gamma' T$ term dominates, and the contribution of the electron gas to the specific heat is noticeable.

C_v of a metal $= 2\gamma' T + 3R$ at high T. Since the part of the specific heat due to lattice vibration is constant, the $\gamma' T$ term is again detectable.

Note that at very high T, we have to include higher terms in the expansion of $g(\varepsilon)$, and the part of the specific heat due to the electrons is no longer proportional to T.

We can demonstrate more easily than we have done above that the internal energy is proportional to T^2. At any temperature less than the Fermi temperature almost all the states with energy less than the Fermi energy are filled, and almost all states with energy greater than the Fermi energy are empty. As the electron gas is raised to temperature T from $T = 0$, the average energy than *can* be imparted to an electron is about kT.

Also, only those electrons within about kT of the Fermi level will be excited, because electrons with less energy have no place to go—the states are filled. The number of electrons with energies between μ_0 and $\mu_0 - kT$ is

$$N = \int_{\mu_0 - kT}^{\mu_0} a\varepsilon^{1/2}\, d\varepsilon.$$

For $\mu_0 \gg kT$, N is proportional to kT. Thus the internal energy added is proportional to $(kT)kT = k^2T^2$.

CHAPTER 2

DENSITY MATRICES

2.1 INTRODUCTION TO DENSITY MATRICES

When we solve a quantum-mechanical problem, what we really do is divide the universe into two parts—the system in which we are interested and the rest of the universe. We then usually act as if the system in which we are interested comprised the entire universe. To motivate the use of density matrices, let us see what happens when we include the part of the universe outside the system.

Let x describe the coordinates of the system, and let y describe the rest of the universe. Let $\varphi_i(x)$ be a complete set of wave functions. The most general wave function can be written

$$\psi(x, y) = \sum_i C_i(y)\varphi_i(x). \tag{2.1}$$

At this point we will convert to Dirac notation.

Let $|\varphi_i\rangle$ be a complete set of vectors in the vector space describing the system, and let $|\theta_i\rangle$ be a complete set for the rest of the universe.

$$\varphi_i(x) = \langle x|\varphi_i\rangle \qquad \text{and} \qquad \theta_i(y) = \langle y|\theta_i\rangle.$$

The most general wave function can be written

$$|\psi\rangle = \sum_{ij} C_{ij}|\varphi_i\rangle|\theta_j\rangle \tag{2.2}$$

$$\psi(x, y) = \langle y|\langle x|\psi\rangle = \sum_{ij} C_{ij}\langle x|\varphi_i\rangle\langle y|\theta_j\rangle$$

We can obtain Eq. (2.1) by taking

$$C_i(y) = \sum_j C_{ij}\langle y|\theta_j\rangle.$$

Now let A be an operator that acts only on the system; that is to say, A does not act on the $|\theta_j\rangle$. When A acts on product states (for example, $|\psi\rangle$) we really mean $A|a\rangle|b\rangle \equiv (A|a\rangle)|b\rangle$. In such a case A does not equal

$$\sum_{ii'} A_{ii'}|\varphi_i\rangle\langle\varphi_{i'}|,$$

but equals

$$\sum_{ii'j} A_{ii'}|\varphi_i\rangle|\theta_j\rangle\langle\theta_j|\langle\varphi_{i'}|.$$

Then

$$\begin{aligned}\langle A\rangle = \langle\psi|A|\psi\rangle &= \sum_{\substack{ij\\ i'j'}} C_{ij}^* C_{i'j'}\langle\theta_j|\langle\varphi_i|A|\varphi_{i'}\rangle|\theta_{j'}\rangle \\ &= \sum_{iji'} C_{ij}^* C_{i'j}\langle\varphi_i|A|\varphi_{i'}\rangle \\ &= \sum_{ii'} \langle\varphi_i|A|\varphi_{i'}\rangle\rho_{i'i} \qquad (2.3)\end{aligned}$$

where

$$\rho_{i'i} = \sum_j C_{ij}^* C_{i'j} = \text{density matrix.} \qquad (2.4)$$

We define the operator ρ to be such that $\rho_{i'i} = \langle\varphi_{i'}|\rho|\varphi_i\rangle$. ρ operates only on the system described by x.

$$\begin{aligned}\langle\psi|A|\psi\rangle &= \sum_i \langle\varphi_i|A\sum_{i'}|\varphi_{i'}\rangle\langle\varphi_{i'}|\rho|\varphi_i\rangle \\ &= \sum_i \langle\varphi_i|A\rho|\varphi_i\rangle = \mathrm{Tr}\,\rho A \qquad (2.5)\end{aligned}$$

Where we have used the result

$$\sum_{i'} |\varphi_{i'}\rangle\langle\varphi_{i'}| = 1 \quad \text{(by completeness arguments).}$$

From Eq. (2.4), it is obvious that ρ is hermitian. Therefore it can be diagonalized with a complete orthonormal set of eigenvectors $|i\rangle$ and real eigenvalues w_i,

$$\rho = \sum_i w_i|i\rangle\langle i|. \qquad (2.6)$$

If we let A be 1, we obtain

$$\sum_i w_i = \mathrm{Tr}\,\rho = \langle A\rangle = \langle\psi|\psi\rangle = 1. \qquad (2.7)$$

If we let A be $|i'\rangle\langle i'|$ we have

$$\begin{aligned}w_{i'} = \mathrm{Tr}\,\rho A = \langle A\rangle = \langle\psi|A|\psi\rangle &= \sum_j (\langle\psi|i'\rangle|\theta_j\rangle)(\langle\theta_j|\langle i'|\psi\rangle) \\ &= \sum_j |(\langle i'|\langle\theta_j|)|\psi\rangle|^2. \qquad (2.8)\end{aligned}$$

Therefore,

$$w_i \geq 0 \qquad \text{and} \qquad \sum_i w_i = 1. \qquad (2.9)$$

We now consider the concept of a density matrix independent of the preceding motivation. First let us reformulate quantum mechanics:

Any system is described by a density matrix ρ, where ρ is of the form $\sum_i w_i|i\rangle\langle i|$ and

a) the set $|i\rangle$ is a complete orthonormal set of vectors.

b) $w_i \geq 0$.

c) $\sum_i w_i = 1$.

d) Given an operator A, the expectation of A is given by

$$\langle A\rangle = \mathrm{Tr}\,\rho A.$$

Notice that

$$\begin{aligned}\langle A\rangle = \mathrm{Tr}\,\rho A &= \sum_{i'} \langle i'|\rho A|i'\rangle = \sum_{i'i} w_i\langle i'|i\rangle\langle i|A|i'\rangle \\ &= \sum_i w_i\langle i|A|i\rangle. \end{aligned} \tag{2.10}$$

Since $\langle i|A|i\rangle$ = the expectation value of A in the state $|i\rangle$, it is obvious from (b), (c) and Eq. (2.10) that we can interpret the w_i as the probability that the system is in state i. If all but one of the w_i are zero, we say that the system is in a *pure state*; otherwise it is in a *mixed state*. It is easy to show that a necessary and sufficient condition for a pure state is $\rho = \rho^2$. If a system is in a pure state, $|i_{\text{pure}}\rangle$ we can express the matrix as

$$\begin{aligned}\rho &= |i_{\text{pure}}\rangle\langle i_{\text{pure}}|, \\ \rho_{ij} &= \langle\varphi_i|\rho|\varphi_j\rangle = \langle\varphi_i|i_{\text{pure}}\rangle\langle i_{\text{pure}}|\varphi_j\rangle \\ &= \langle\varphi_i|i_{\text{pure}}\rangle(\langle\varphi_j|i_{\text{pure}}\rangle)^* \end{aligned} \tag{2.11}$$

More generally,

$$\rho_{ij} = \sum_k w_k\langle\varphi_i|k\rangle\langle\varphi_j|k\rangle^*. \tag{2.11a}$$

If it is possible to discuss the system in the x-representation, we can write

$$\begin{aligned}\rho(x', x) &\overset{\text{def.}}{=} \langle x'|\rho|x\rangle = \sum_i w_i\langle x'|i\rangle\langle i|x\rangle \\ &= \sum_i w_i i(x')i^*(x), \end{aligned} \tag{2.12}$$

which, for a pure state $|i\rangle$, becomes

$$\rho(x', x) = i(x')i^*(x) \tag{2.13}$$

In the x-representation we write

$$\langle A\rangle = \mathrm{Tr}\,\rho A = \int dx\langle x|\rho A|x\rangle.$$

But

$$\langle x|\rho A|x\rangle = \langle x|\rho\left(\int dx'|x'\rangle\langle x'|\right)A|x\rangle$$
$$= \int dx'\langle x|\rho|x'\rangle\langle x'|A|x\rangle = \int dx'\rho(x, x')A(x', x).$$
$$\langle A\rangle = \int dx\, dx'\rho(x, x')A(x', x). \tag{2.14}$$

Considering again the problem of a system plus the rest of the universe, we can easily show

$$\langle A\rangle = \int \psi^*(x', y)A(x', x)\psi(x, y)\, dx\, dx'\, dy.$$

We see that in this case

$$\rho(x, x') = \int \psi(x, y)\psi^*(x', y)\, dy. \tag{2.15}$$

From our example in which we split the universe into two parts, we see that pure states are not general enough to describe a quantum mechanical system that does not include the whole universe. It is unknown whether or not the universe is in a pure state. To reformulate quantum mechanics in terms of the more general density matrices, it would be appropriate next to find the equation of motion of ρ. But first, as a simple example of a density matrix, let us try to describe polarized and unpolarized light.

Consider a beam of light travelling in the z-direction. First define

$$\begin{aligned} &\begin{pmatrix}1\\0\end{pmatrix}\text{: Wave function for the } x\text{-polarized state} \\ &\begin{pmatrix}0\\1\end{pmatrix}\text{: Wave function for the } y\text{-polarized state}\end{aligned} \tag{2.16}$$

Any pure state can be written as a linear combination of the two states in Eq. (2.16):

$$\begin{pmatrix}a\\b\end{pmatrix} = a\begin{pmatrix}1\\0\end{pmatrix} + b\begin{pmatrix}0\\1\end{pmatrix}, \tag{2.17}$$

where

$$|a|^2 + |b|^2 = 1.$$

When we use Eq. (2.11), the density matrix for the pure state Eq. (2.17) becomes

$$\rho = \begin{pmatrix}aa^* & ab^*\\ ba^* & bb^*\end{pmatrix}. \tag{2.18}$$

In using Eq. (2.11), we took

$$|\varphi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |\varphi_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Now to examine four pure states and the density matrices corresponding to them:

The x-polarized state: Let $a = 1$ and $b = 0$ in Eq. (2.17). Equation (2.18) gives

$$\rho_{x\text{-pol.}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \tag{2.19}$$

The y-polarized state: Let $a = 0$ and $b = 1$.

$$\rho_{y\text{-pol.}} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \tag{2.20}$$

The 45°-polarized state: $a = 1/\sqrt{2}$ and $b = 1/\sqrt{2}$ gives

$$\rho_{45^\circ\text{-pol.}} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}. \tag{2.21}$$

The 135°-polarized state: $a = -1/\sqrt{2}$ and $b = 1/\sqrt{2}$ gives

$$\rho_{135^\circ\text{-pol.}} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}. \tag{2.22}$$

The ρ's in the last four equations are for pure states. Now consider the following two mixed states:

1. *Mixture of 50% x-polarized and 50% y-polarized states:* From Eq. (2.11a) the density matrix for this mixture is

$$\rho = \tfrac{1}{2}\rho_{x\text{-pol.}} + \tfrac{1}{2}\rho_{y\text{-pol.}} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}. \tag{2.23}$$

2. *Mixture of 50% 45°-polarized and 50% 135°-polarized states:* The density matrix for this mixture is

$$\rho = \tfrac{1}{2}\rho_{45^\circ\text{-pol.}} + \tfrac{1}{2}\rho_{135^\circ\text{-pol.}} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}. \tag{2.24}$$

Thus these two mixtures have the same density matrix and show the same physical effect. Note that a given pure state (for example, x-polarization) determines a state vector (wave function) only up to a phase factor, whereas the density matrix is determined uniquely.

2.2 ADDITIONAL PROPERTIES OF THE DENSITY MATRIX

Recall that the density operator can be written

$$\rho = \sum_i w_i |i\rangle\langle i|, \tag{2.25}$$

where the system is in state $|i\rangle$ with probability w_i. As time changes, the possible states of the system also change; so

$$\rho(t) = \sum_i w_i |i(t)\rangle\langle i(t)|. \tag{2.26}$$

It is easy to find how $|i(t)\rangle$ changes in time, for we can expand $|i(0)\rangle$ in eigenstates of the Hamiltonian, H, and we know how those eigenstates change in time. Let $H|E_n\rangle = E_n|E_n\rangle$.

$$|i(t)\rangle = \sum_n |E_n\rangle\langle E_n|i(0)\rangle. \tag{2.27}$$

It follows that

$$|i(t)\rangle = \sum_n |E_n\rangle e^{-iE_n t}\langle E_n|i(0)\rangle.$$

For convenience we use units in which $\hbar = 1$. If we define $f(H)$ for any function f by the equation $f(H)|E_n\rangle = f(E_n)|E_n\rangle$, we can write

$$|i(t)\rangle = \sum_n e^{-iHt}|E_n\rangle\langle E_n|i(0)\rangle = e^{-iHt}|i(0)\rangle. \tag{2.28}$$

Notice that, according to our definition of $f(H)$, if $f(E_n)$ can be expanded in a power series for any eigenvalue of H, then $f(H)$ can be expanded in the same power series.

Substituting Eq. (2.28) into Eq. (2.26), we get

$$\rho(t) = \sum_i w_i e^{-iHt}|i(0)\rangle\langle i(0)|e^{iHt} = e^{-iHt}\rho(0)e^{iHt}. \tag{2.29}$$

Taking the derivative of ρ with respect to time, we get

$$\dot{\rho} = -i(H\rho - \rho H). \tag{2.30}$$

You may be suspicious about treating H like a number when taking the derivative of terms such as e^{-iHt}, but such a procedure can be justified by expanding e^{-iHt} in a power series. Alternatively, we can obtain Eq. (2.30) from Eq. (2.29) by taking the derivative of $\langle E_n|\rho(t)|E_m\rangle$. In any case, Eq. (2.30) is true, and it plays the same role for density matrices that Schrödinger's equation plays for wave functions. It should be noticed that an observable A in the Heisenberg representation obeys

$$\dot{A} = +i(HA - AH). \tag{2.31}$$

The signs of Eq. (2.30) and Eq. (2.31) are opposite. We could put Planck's constant back into the equations by replacing every H by $H/\hbar$. From Eq. (2.29) we obtain

$$\begin{aligned}\operatorname{Tr} \rho^n(t) &= \operatorname{Tr}\left[e^{-iHt}(\rho^n(0)e^{iHt})\right] = \operatorname{Tr}\left[(\rho^n(0)e^{iHt})e^{-iHt}\right] \\ &= \operatorname{Tr} \rho^n(0) \end{aligned} \tag{2.32}$$

where we use the fact that $\operatorname{Tr} AB = \operatorname{Tr} BA$. It follows that $\operatorname{Tr} \rho^n$ is constant in time, and consequently $\operatorname{Tr} f(\rho)$ for any function f is constant. This result can also be obtained by noticing that

$$\rho^n = \sum_i w_i^n |i\rangle\langle i|,$$

so

$$f(\rho) = \sum_i f(w_i)|i\rangle\langle i|. \tag{2.33}$$

$\operatorname{Tr} f(\rho) = \sum_i f(w_i)$, and the w_i's are constant. [being the eigenvalues of ρ and thus unchanged by a unitary transformation such as Eq. (2.29)].

It is possible to define "entropy" by the equation

$$\text{“}S\text{”} = -\sum_i w_i \ln w_i. \tag{2.34}$$

A pure state has "S" = 0, whereas very impure states have large positive "entropy." The quotation marks are used because, by our definition, the "entropy" cannot increase in time but must remain constant.

Next let us try to get a bit more experience in dealing with density matrices. In ordinary quantum mechanics it was easy to find such things as the expectation value of the position, the expectation value of momentum, and the probability that a system is in a given state.

(a) Expectation value of position: According to Eq. (2.14), if A is an operator,

$$\langle A\rangle = \int dx\, dx' \rho(x, x')A(x', x).$$

In this case $A = x$, and

$$A(x', x) = \langle x'|A|x\rangle = x\langle x'|x\rangle = x\,\delta(x' - x), \tag{2.35}$$

so

$$\langle x\rangle = \int dx\, x\rho(x, x). \tag{2.36}$$

(b) Expectation value of momentum: Now we must take $A = p$. Since

$$\langle x|\, p|\psi\rangle = \frac{\hbar}{i}\frac{\partial}{\partial x}\langle x|\psi\rangle,$$

we have

$$A(x', x) = \frac{\hbar}{i}\frac{\partial}{\partial x}\delta(x' - x). \tag{2.37}$$

Integrate by parts; for any function f we have

$$\int f(x)\frac{\partial}{\partial x}\delta(x' - x)\,dx = -\int \delta(x' - x)\frac{\partial}{\partial x}f(x)\,dx = -f'(x')$$

$$\langle p\rangle = -\int dx\,dx'\,\frac{\hbar}{i}\,\delta(x' - x)\frac{\partial}{\partial x}\rho(x, x')$$

$$= -\frac{\hbar}{i}\int dx\left[\frac{\partial}{\partial x}\rho(x, x')\right]_{x'=x} \tag{2.38}$$

(c) **Probability that a system is in state $|\chi\rangle$:** Finding the probability that a system is found in the state $|\chi\rangle$ is equivalent to finding the expectation value of the operator $|\chi\rangle\langle\chi|$. To see this for the case of a pure state, notice that if the system is in a state $|\psi\rangle$, the probability of experimentally finding it in state $|\chi\rangle$ is

$$|\langle\chi|\psi\rangle|^2 = \langle\psi|\chi\rangle\langle\chi|\psi\rangle = \langle\psi|(|\chi\rangle\langle\chi|)|\psi\rangle.$$

Therefore, the probability that the system is found in state $|\chi\rangle$ can be written

$$P = \text{probability} = \text{Tr}\,\rho|\chi\rangle\langle\chi| = \text{Tr}\sum_i w_i|i\rangle\langle i|\chi\rangle\langle\chi|$$

$$= \sum_i w_i|\langle i|\chi\rangle|^2 = \langle\chi|\rho|\chi\rangle. \tag{2.39}$$

The equation

$$P = \sum_i w_i|\langle i|\chi\rangle|^2$$

is in accord with our interpretation of the w_i as the probability that the system actually is in state $|i\rangle$.

The probability density at value x_0 for a coordinate is the probability that the system is in state $|x_0\rangle$, which by Eq. (2.39) is $\rho(x_0, x_0)$. For a pure state $|\psi\rangle$, $\rho(x_0, x_0) = |\psi(x_0)|^2$.

2.3 DENSITY MATRIX IN STATISTICAL MECHANICS

When $|\varphi_i\rangle$ is an eigenket (eigenfunction), and E_i is the corresponding eigenvalue of the Hamiltonian H of the system, the probability that the system is in the state $|\varphi_i\rangle$ is

$$(1/Q)e^{-\beta E_i}. \tag{2.40}$$

Thus the density matrix is

$$\rho = \sum_n w_n |\varphi_n\rangle\langle\varphi_n|, \qquad \text{where} \qquad w_n = \frac{1}{Q} e^{-\beta E_n}. \tag{2.41}$$

Alternatively, the coordinate representation is

$$\rho(x, x') = \sum_i \frac{1}{Q} e^{-\beta E_i} \varphi_i(x) \varphi_i^*(x'). \tag{2.42}$$

Because $H|\varphi_n\rangle = E_n|\varphi_n\rangle$, we can write Eq. (2.41) as

$$\rho = \frac{1}{Q} \sum_n e^{-\beta H} |\varphi_n\rangle\langle\varphi_n| = \frac{e^{-\beta H}}{Q} \tag{2.43}$$

where

$$e^{-\beta F} = Q = \sum_n e^{-\beta E_n} = \mathrm{Tr}\, e^{-\beta H}, \tag{2.44}$$

so

$$\rho = \frac{e^{-\beta H}}{\mathrm{Tr}\, e^{-\beta H}}. \tag{2.45}$$

The average energy of the system, U, can be written

$$U = \mathrm{Tr}\, \rho H = \frac{\mathrm{Tr}\, [H e^{-\beta H}]}{\mathrm{Tr}\, [e^{-\beta H}]}. \tag{2.46}$$

When we know the free energy and the energy of the system, the entropy is derived from

$$F = U - TS. \tag{2.47}$$

It can be shown that, if H_0 is any other Hamiltonian,

$$F \leq \langle H \rangle_0 - TS_0, \tag{2.48}$$

where

$$\langle H \rangle_0 = \frac{\mathrm{Tr}\, [H e^{-\beta H_0}]}{\mathrm{Tr}\, [e^{-\beta H_0}]}, \tag{2.49}$$

and S_0 is the entropy calculated from H_0. F in Eq. (2.48) is the true free energy corresponding to the true Hamiltonian H, as written in Eq. (2.44). When we write the free energy corresponding to H_0 as

$$F_0 = \langle H_0 \rangle_0 - TS_0 \tag{2.50}$$

and subtract Eq. (2.50) from Eq. (2.48), we may write

$$F \leq F_0 + \frac{\mathrm{Tr}\, [(H - H_0) e^{-\beta H_0}]}{\mathrm{Tr}\, [e^{-\beta H_0}]}. \tag{2.51}$$

This theorem will be proved later, in Section 2.11.

Now, we regard the density matrix as a function of β:

$$\rho(\beta) = e^{-\beta H}/\mathrm{Tr}\, e^{-\beta H}. \tag{2.42}$$

The unnormalized ρ is defined by

$$\rho_U(\beta) = e^{-\beta H}. \tag{2.52}$$

In place of ρ_U we will hereafter write ρ. Next we will show that $\rho(\beta)$ obeys the differential equation

$$-\partial\rho/\partial\beta = H\rho. \tag{2.53}$$

The proof is as follows. In the energy representation, Eq. (2.52) can be written as

$$\rho_{ij} = \delta_{ij} e^{-\beta E_i}. \tag{2.54}$$

Differentiating this equation, we have

$$-\partial\rho_{ij}/\partial\beta = \delta_{ij} E_i e^{-\beta E_i} = E_i \rho_{ij}, \tag{2.55}$$

from which we obtain Eq. (2.53).

The initial condition for Eq. (2.53) is

$$\rho(0) = 1. \tag{2.56}$$

We can write Eq. (2.53) in the position representation as follows:

$$-\partial\rho(xx'; \beta)/\partial\beta = H_x \rho(xx'; \beta). \tag{2.57}$$

Here the subscript x on H_x indicates that H_x operates on x in $\rho(xx'; \beta)$. The initial condition is

$$\rho(xx'; 0) = \delta(x - x'). \tag{2.58}$$

2.4 DENSITY MATRIX FOR A ONE-DIMENSIONAL FREE PARTICLE

We want to solve the differential equation discussed in Section 2.3 for a one-dimensional free particle. The Hamiltonian is

$$H = p^2/2m, \tag{2.59}$$

and the differential equation (from Eq. (3.57)) is

$$\frac{-\partial\rho(x, x', \beta)}{\partial\beta} = \frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\rho(x, x', \beta). \tag{2.60}$$

This is a diffusion-type equation, and we can write down the solution readily:

$$\rho(x, x'; \beta) = \sqrt{\frac{m}{2\pi\hbar^2\beta}} \exp\left[-\left(\frac{m}{2\hbar^2\beta}\right)(x - x')^2\right]. \tag{2.61*}$$

The numerical factor is chosen such that

$$\rho(xx'; 0) = \delta(x - x'). \tag{2.62}$$

For a linear system of length L, the trace of Eq. (2.61) leads to

$$e^{-\beta F} = \int \rho(x, x)\, dx = L\sqrt{\frac{mkT}{2\pi\hbar^2}}. \tag{2.63}$$

This is the partition function for the system.

Problem: Show that the partition function for a three-dimensional system with N particles is

$$e^{-\beta F} = V^N(\sqrt{mkT/2\pi\hbar^2})^{3N}. \tag{2.64}$$

Another way of deriving Eq. (2.61) is as follows:

$$\rho(x, x'; \beta) = \sum_m e^{-\beta E_m}\psi_m(x)\psi_m^*(x'). \tag{2.65}$$

If the particle is in a large box of volume V, we take

$$\sum_m \to \int \frac{V d^3p}{(2\pi\hbar)^3} \quad \text{and} \quad \psi_m(x) \to \psi_p(x) = \frac{1}{\sqrt{V}} e^{ip\cdot x/\hbar}.$$

Then Eq. (3.65) becomes

$$\begin{aligned}\rho(x, x'; \beta) &= \int \frac{d^3p}{(2\pi\hbar)^3} e^{-\beta p^2/2m} e^{ip/\hbar\cdot(x-x')} \\ &= \left(\frac{m}{2\pi\hbar^2\beta}\right)^{3/2} e^{-m|x-x'|^2/2\hbar^2\beta},\end{aligned} \tag{2.66}$$

as in Eq. (2.61).

2.5 LINEAR HARMONIC OSCILLATOR

The Hamiltonian for a linear harmonic oscillator is

$$H = p^2/2m + mw^2x^2/2, \tag{2.67}$$

* Note that for x,x' constant, no expansion in powers of β exists near $\beta = 0$. This is, indeed, an important feature of ρ_{free} and will influence the construction of perturbation series.

so that the differential equation for ρ is

$$\frac{-\partial\rho}{\partial\beta} = \frac{-\hbar^2}{2\,m}\frac{\partial^2}{\partial x^2}\rho + \frac{m\omega^2}{2}x^2\rho. \tag{2.68}$$

Let us write

$$\xi = \sqrt{\frac{m\omega}{\hbar}}\,x, \qquad f = \frac{\hbar\omega}{2}\beta = \frac{\hbar\omega}{2kT}, \tag{2.69}$$

which simplifies Eq. (2.68) to

$$\frac{-\partial\rho}{df} = \frac{-\partial^2\rho}{\partial\xi^2} + \xi^2\rho. \tag{2.70}$$

The initial condition is

$$\rho = \delta(x - x') \qquad \text{for} \quad f = 0, \tag{2.71}$$

or

$$\rho = \sqrt{\frac{m\omega}{\hbar}}\,\delta(\xi - \xi') \qquad \text{for} \quad f = 0. \tag{2.72}$$

The factor comes from the scale change in the δ-function. In general, $\delta(x - x_0) = |f'(x)|\delta[f(x) - f(x_0)]$ for f monotonic.

For low f (high temperature) the particle should act almost like a free particle, as its probable kinetic energy is so high. Therefore we expect that, for low f, the density matrix for a harmonic oscillator will be given approximately by

$$\rho(\xi, \xi', f) \approx \sqrt{m\omega/(4\pi\hbar f)}\,e^{-(\xi-\xi')^2/4f}. \tag{2.73*}$$

Knowing the Gaussian dependence of ρ_{free} on ξ, we try the following form:

$$\rho = \exp\{-[a(f)\xi^2 + b(f)\xi + c(f)]\}. \tag{2.74}$$

Use of Eq. (2.74) in Eq. (2.70) leads us to the equation

$$a'\xi^2 + b'\xi + c' = (1 - 4a^2)\xi^2 - 4ab\xi + 2a - b^2, \tag{2.75}$$

where a prime (′) means a derivative with respect to f. From Eq. (2.75),

$$a' = 1 - 4a^2, \tag{2.76a}$$

$$b' = -4ab, \tag{2.76b}$$

$$c' = 2a - b^2. \tag{2.76c}$$

* This is only true for $\xi \neq \xi'$. When they are equal, $\rho(\xi, \xi', f)$ is equal to ρ_{free} only for $f = 0$. See Section 3.10 for details. Obviously, when T is large and the classical motion has large amplitude, the larger the amplitude the less will the oscillator feel a force between fixed x and x'.

Equation (2.76a) is integrated to give

$$a = (\tfrac{1}{2}) \coth 2(f - f_0),$$

where the initial condition Eq. (2.73) requires that the integration constant f_0 vanish:

$$a = (\tfrac{1}{2}) \coth 2f. \tag{2.77}$$

Equations (2.76b and c) are integrated as

$$b = A/\sinh 2f. \tag{2.78}$$
$$c = (\tfrac{1}{2}) \ln (\sinh 2f) + (A^2/2) \coth 2f - \ln B,$$

where A and B are constants. Use Eqs. (2.77) and (2.78) to get

$$\rho = \frac{B}{\sqrt{\sinh 2f}} \exp\left[-\left(\frac{\xi^2}{2} \coth 2f + \frac{A\xi}{\sinh 2f} + \frac{A^2}{2} \coth 2f\right)\right]. \tag{2.79}$$

When we let $f \to 0$, Eq. (2.79) gives

$$\rho \to \frac{B}{\sqrt{2f}} \exp\left[-\frac{\xi^2 + 2A\xi + A^2}{4f}\right]. \tag{2.80}$$

For this result to agree with Eq. (2.73) we see that

$$A = -\xi', \qquad B = \sqrt{\frac{m\omega}{2\pi\hbar}}. \tag{2.81}$$

Using Eq. (2.81) in Eq. (2.79), we obtain the final form:

$$\rho(x, x'; \beta) = \sqrt{\frac{m\omega}{2\pi\hbar \sinh 2f}} \times \exp\left\{\frac{-m\omega}{2\hbar \sinh 2f}\left[(x^2 + x'^2) \cosh 2f - 2xx'\right]\right\} \tag{2.82}$$

or

$$\rho(x, x'; \beta) = \sqrt{\frac{m\omega}{2\pi\hbar \sinh (\hbar\omega/kT)}} \times \exp\left\{\frac{-m\omega}{2\hbar \sinh (\hbar\omega/kT)}\left[(x^2 + x'^2) \cosh \frac{\hbar\omega}{kT} - 2xx'\right]\right\}. \tag{2.83}$$

The special case when $x = x'$ gives

$$\rho(x, x; \beta) = \sqrt{\frac{m\omega}{2\pi\hbar \sinh 2f}} \exp\left(\frac{-m\omega}{\hbar} x^2 \tanh f\right). \tag{2.84}$$

This is the probability for finding the system at x. Notice that it is of Gaussian form.

Equation (2.84) is used to calculate

$$\langle x^2 \rangle = \frac{\int x^2 \rho(x, x)\, dx}{\int \rho(x, x)\, dx} = \frac{\hbar}{2m\omega} \coth f = \frac{\hbar}{2m\omega} \coth \frac{\hbar\omega}{2kT}. \tag{2.85}$$

The average of the potential energy is thus

$$\langle \text{Potential energy} \rangle = (\tfrac{1}{2}) m\omega^2 \langle x^2 \rangle = (\hbar\omega/4) \coth f. \tag{2.86}$$

We may compare this result with the average of total energy calculated by means of the partition function. We know

$$\langle \text{Energy} \rangle = \frac{\hbar\omega}{2} + \frac{\hbar\omega e^{-\hbar\omega/kT}}{1 - e^{-\hbar\omega/kT}} = \frac{\hbar\omega}{2} \frac{1 + e^{-2f}}{1 - e^{-2f}} = \frac{\hbar\omega}{2} \coth f. \tag{2.87}$$

Equation (2.86) is exactly half of Eq. (2.87). Therefore we know that

$$\langle \text{Kinetic energy} \rangle = \langle \text{Potential energy} \rangle = (\hbar\omega/4) \coth f. \tag{2.88}$$

The partition function is derived from Eq. (2.84):

$$e^{-\beta F} = \int \rho(x, x)\, dx = \sqrt{\frac{m\omega}{2\pi\hbar \sinh 2f}} \sqrt{\frac{\pi\hbar}{m\omega \tanh f}} = \frac{1}{2 \sinh f} \tag{2.89}$$

which leads to

$$\begin{aligned} F &= \frac{1}{\beta} \ln (2 \sinh f) = kT \ln (e^{\hbar\omega/2kT} - e^{-\hbar\omega/2kT}) \\ &= \frac{\hbar\omega}{2} + kT \ln (1 - e^{-\hbar\omega/kT}). \end{aligned} \tag{2.90}$$

This is the free energy, as already derived.

We may examine the limit of high temperature, or small f in Eq. (2.84). In this limit

$$\rho(x, x; \beta) \to \exp\left(\frac{-m\omega^2 x^2/2}{kT}\right) = \exp\left(\frac{-V(x)}{kT}\right), \tag{2.91}$$

except for the factor in front of the exponential. This result, Eq. (2.91), agrees with classical mechanics.

The low-temperature limit is found by inspection of Eq. (2.83):

$$\rho(x, x') \to \sqrt{\frac{m\omega}{\pi\hbar}} \exp\left(\frac{-\hbar\omega}{2kT}\right) \exp\left(\frac{-m\omega x^2}{2\hbar}\right) \exp\left(\frac{-m\omega^2 x'^2}{2\hbar}\right). \tag{2.92}$$

This limit is calculated as

$$\rho(x, x') = \sum_i e^{-\beta E_i} \varphi_i(x) \varphi_i^*(x') \to e^{-\beta E_0} \varphi_0(x) \varphi_0^*(x'), \tag{2.93}$$

because when $\beta \to \infty$, only the ground state has an effective contribution. Equation (2.92) agrees with Eq. (2.93) because we know that for the ground state

$$\varphi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{(-m\omega/2\hbar)x^2}. \tag{2.94}$$

2.6 ANHARMONIC OSCILLATOR

Consider the potential energy shown by the curve in Fig. 2.1, given by the following function:

$$V(x) = \frac{m\omega^2}{2} x^2 + kx^3. \tag{2.95}$$

We will consider the area near $x = 0$ only, so that the region $V(x) < 0$ does not come into the calculation.

Because of the anharmonicity, when the temperature increases the mean position of the oscillation moves out, as shown in Fig. 2.2. Bearing this in mind,

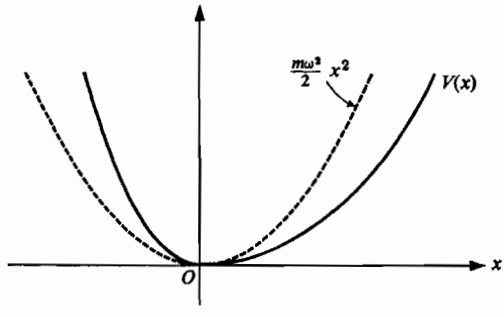

Fig. 2.1 Anharmonic oscillator ($k < 0$).

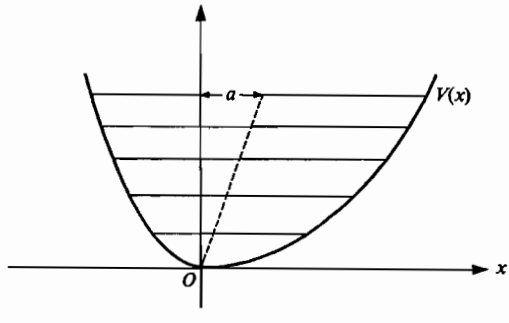

Fig. 2.2 Mean position of the oscillation is displaced.

we will treat the problem using the minimum principle discussed previously. The principle is written as

$$F \leq F_0 + \langle H - H_0 \rangle_0. \tag{2.96}$$

Here F is the true free energy and H is the true Hamiltonian. When H_0 is any Hamiltonian, $\langle\rangle_0$ means the average taken in a system characterized by H_0. F_0 is the free energy of the system of H_0.

In the present problem we take for H_0

$$H_0 = p^2/2m + (m\omega^2/2)(x - a)^2, \tag{2.97}$$

which describes a harmonic oscillator with its center displaced by an amount a. The ω in Eq. (2.97) is the same as that in Eq. (2.95). From Eq. (2.90) we find that

$$F_0 = kT \ln \left(2 \sinh (\hbar\omega/2kT)\right). \tag{2.98}$$

Now we have only to find

$$\langle H - H_0 \rangle_0 = \left\langle \frac{m\omega^2}{2} x^2 + kx^3 - \frac{m\omega^2}{2} (x - a)^2 \right\rangle_0. \tag{2.99}$$

If we make the substitution $y = x - a$, Eq. (2.99) becomes

$$\langle H - H_0 \rangle_0 = \left\langle m\omega^2 ay + \frac{m\omega^2}{2} a^2 + ky^3 + 3ky^2a + 3kya^2 + ka^3 \right\rangle_0. \tag{2.100}$$

Because $H_0 = p^2/2m + (m\omega^2/2)y^2$, we can easily calculate the expectation values of the various powers of y. By symmetry,

$$\langle y \rangle_0 = \langle y^3 \rangle_0 = 0. \tag{2.101}$$

From Eq. (2.85), we obtain

$$\langle y^2 \rangle_0 = (\hbar/2m\omega) \coth (\hbar\omega/2kT). \tag{2.102}$$

Substituting Eq. (2.102) in Eq. (2.100) yields

$$F \leq F_0 + \langle H - H_0 \rangle_0 = F_0 + (m\omega^2/2)a^2 + ka^3 + 3ka \langle y^2 \rangle_0. \tag{2.103}$$

The best estimate of F from Eq. (2.103) is obtained by minimizing the right-hand side. In so doing we neglect the term ka^3 because we are interested only in small oscillations, so that $V(x) < 0$ does not occur. Then the differentiation of Eq. (2.103) with respect to a leads to

$$m\omega^2 a + 3k\langle y^2 \rangle_0 = 0, \tag{2.104}$$

that is,

$$a = \frac{-3k}{m\omega^2} \langle y^2 \rangle_0 = -\frac{3k\hbar \coth (\hbar\omega/2kT)}{2m^2\omega^3}. \tag{2.105}$$

In this case, Eq. (2.103) becomes

$$F \leq F_0 - (m\omega^2/2)a^2. \tag{2.106}$$

In Eq. (2.105), the displacement of the mean position, a, has a simple interpretation. It is the point at which the average force on the oscillator vanishes. This result is seen as follows: from Eq. (2.95),

$$\text{Force} = m\omega^2 x + 3kx^2 = m\omega^2(y + a) + 3ky^2 + \cdots \tag{2.107}$$

$$\langle \text{Force} \rangle = m\omega^2 a + 3k\langle y^2 \rangle + \cdots. \tag{2.108}$$

Therefore, when we neglect the higher powers of a, we arrive at Eq. (2.104).

Problem: Consider a general bounding potential $V(x)$ like that shown in Fig. 2.3. The Hamiltonian is

$$H = p^2/2m + V(x). \tag{2.109}$$

In order to apply the variation principle in Eq. (2.96), try the following H_0:

$$H_0 = \frac{p^2}{2m_0} + \frac{m_0\omega_0^2}{2}(x - a)^2, \tag{2.110}$$

and vary a, m_0, and ω_0. Show that

i) $m_0 = m$; (2.111)

ii) ω_0 and a are determined from

$$\langle V'(x) \rangle_0 = 0, \tag{2.112}$$

$$\langle xV'(x) \rangle_0 = \langle p^2/m \rangle_0. \tag{2.113}$$

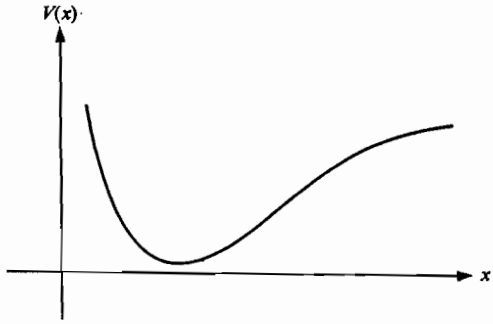

Fig. 2.3 A general bounding potential.

Here $\langle\rangle_0$ is taken for H_0. Equation (2.112) comes from the consideration of the absolute location of the system, and Eq. (2.113) comes from the scale of this variable. The latter corresponds to the virial theorem, which states that

$$\langle xV'(x)\rangle = \langle p^2/m\rangle. \tag{2.114}$$

To show how Eq. (2.114) is related to considerations of the scale of the variable, let us derive the virial theorem in a more general form.

Consider a system inside a cubical box with sides of length L. The Hamiltonian is taken to be

$$H_L = \sum_i \frac{p_i^2}{2m_i} + \sum_{\substack{ij \\ i<j}} V(r_{ij}). \tag{2.115}$$

(We assume that the motion of the system can be described by a potential depending only on the distances separating pairs of the particles making up the system.) $r_{ij} = |\boldsymbol{r}_i - \boldsymbol{r}_j|$.

If the system is bound, then for a sufficiently large box the pressure is essentially zero. But in general,

$$P = \text{Pressure} = -\frac{\partial F}{\partial\,(\text{Vol})} = -\frac{1}{3L^2}\frac{\partial F}{\partial L}. \tag{2.116}$$

Let $F_L = F$ for a box of length L. When we later prove the theorem that $F \leq F_0 + \langle H - H_0\rangle_0$, we will in fact prove that $\langle H - H_0\rangle_0$ is the first-order correction in the perturbation expansion for F. We get an expression for P by writing

$$F_{L(1+\varepsilon)} \approx F_L + \text{correction term}$$

or

$$F_{L(1+\varepsilon)} \approx F_L + \langle H_{L(1+\varepsilon)} - H_L\rangle_{H_L}. \tag{2.117}$$

Now $H_{L(1+\varepsilon)}$ differs from H_L only in that $(\boldsymbol{r}_i)_x$ can run from 0 to $L(1+\varepsilon)$ rather than 0 to L, and similarly for $(\boldsymbol{r}_i)_y$ and $(\boldsymbol{r}_i)_z$. Therefore let us make a change of variables, such that $\boldsymbol{r}_i' = \boldsymbol{r}_i/(1+\varepsilon)$. As $p_i^2 = -\hbar^2\nabla_i^2$, we obtain

$$\begin{aligned} H_{L(1+\varepsilon)} &= \sum_i \frac{p_i'^2}{2m_i(1+\varepsilon)^2} + \sum_{\substack{ij \\ i<j}} V[(1+\varepsilon)r_{ij}'] \\ &\approx \sum_i \frac{p_i^2}{2m_i} + \sum_{\substack{ij \\ i<j}} V(r_{ij}') + \varepsilon\left[-2\sum_i \frac{p_i'^2}{2m} + \sum_{\substack{ij \\ i<j}} r_{ij}'V'(r_{ij})\right] \end{aligned} \tag{2.118}$$

Since the new coordinates (primed) run over the same range as the original (unprimed) ones in Eq. (2.115), we can eliminate the primes over the r_{ij} and P_i and write

$$\langle H_{L(1+\varepsilon)} - H_L\rangle_{H_L} = \varepsilon\left[-2\,\langle\text{kinetic energy}\rangle + \left\langle \sum_{\substack{ij \\ i<j}} r_{ij}V'(r_{ij})\right\rangle\right]. \tag{2.119}$$

From Eq. (2.116) and Eq. (2.117),

$$3PV = -L\frac{\partial F}{\partial L} = -L\frac{F_{L(1+\varepsilon)} - F_L}{L\varepsilon} = -\frac{1}{\varepsilon}\langle H_{L(1+\varepsilon)} - H_L\rangle. \tag{2.120}$$

From Eq. (2.119),

$$3PV = 2\langle \mathrm{KE}\rangle - \left\langle \sum_{\substack{ij\\ i<j}} r_{ij}V'(r_{ij})\right\rangle. \tag{2.121}$$

For the case of a harmonic oscillator, P is zero and we get

$$\langle xV'(x)\rangle = \langle p^2/m\rangle. \tag{2.122}$$

This result should be compared with Eq. (2.113).

For the case of a particle moving in an inverse-square force field, we get the classical result

$$\langle KE\rangle = -(\tfrac{1}{2})\langle \text{potential energy}\rangle. \tag{2.123}$$

This method can be generalized. Let H be the Hamiltonian for the original system and let H_ε be the Hamiltonian with the box changed in an infinitesimal manner. Suppose $F_H = F_{H_\varepsilon}$ and suppose that $H_\varepsilon - H = \varepsilon S$. Then the above method gives $\langle S\rangle_H = 0$. For example, in Eq. (2.118) we have the case for which

$$S = \left[-2\sum_i \frac{p_i^2}{2m} + \sum_{\substack{ij\\ i<j}} r_{ij}V'(r_{ij})\right].$$

As another example, suppose that the box undergoes an infinitesimal translation, ε, in the x direction. We get, by taking $x' = x - \varepsilon$,

$$H_\varepsilon \approx \sum_i \frac{p_i'^2}{2m_i} + \sum_{\substack{ij\\ i<j}} V(r_{ij}') + \varepsilon\sum_{\substack{ij\\ i<j}} \frac{\partial V}{\partial x}(r_{ij}'), \tag{2.124}$$

so that

$$S = \sum_{\substack{ij\\ i<j}} \frac{\partial V}{\partial x}(r_{ij}) \tag{2.125}$$

and

$$\left\langle \sum_{\substack{ij\\ i<j}} \frac{\partial V}{\partial x}(r_{ij})\right\rangle = 0. \tag{2.126}$$

Compare Eq. (2.126) with Eq. (2.112).

2.7 WIGNER'S FUNCTION

The density matrix may be written in the coordinate representation and in the momentum representation as follows:

$$\rho(x, x') = \sum_i e^{-\beta E_i} \varphi_i(x) \varphi_i^*(x'), \tag{2.127}$$

$$\begin{aligned} \rho(p, p') &= \sum_i e^{-\beta E_i} \varphi_i(p) \varphi_i^*(p') \\ &= \int \rho(x, x') e^{(-i/\hbar)(p \cdot x - p' \cdot x')} \, dx \, dx'. \end{aligned} \tag{2.128}$$

The diagonal element

$$\rho(x, x) \equiv P(x) \tag{2.129}$$

represents the probability for finding the particle at x (neglecting the normalization constant $1/\mathrm{Tr}\,\rho$). Similarly

$$\rho(p, p) \equiv P(p) \tag{2.130}$$

is proportional to the probability for finding the particle at p in momentum space.

The expression Eq. (2.130) is used in calculating the average value of the kinetic energy:

$$\left\langle \frac{p^2}{2m} \right\rangle = \frac{\mathrm{Tr}\,[(p^2/2m)\rho]}{\mathrm{Tr}\,[\rho]} = \frac{\int \rho(p, p)(p^2/2m)(dp/2\pi\hbar)}{\int \rho(p, p)(dp/2\pi\hbar)}. \tag{2.131}$$

In classical mechanics, the density function $f(p, x)$ in phase space has the following property:

$$\begin{aligned} P(p) &= \int f(p, x) \, dx, \\ P(x) &= \int f(p, x) \frac{dp}{2\pi\hbar}. \end{aligned} \tag{2.132}$$

We ask ourselves if there is any function $f(p, x)$ in quantum mechanics that satisfies Eq. (2.132). One answer is Wigner's function $f_W(p, x)$:

$$f_W(p, x) = \int \rho\left(x + \frac{\eta}{2}, x - \frac{\eta}{2}\right) e^{-i(p\eta/\hbar)} \, d\eta. \tag{2.133}$$

This function is derived by regarding $\rho(x, x')$ as a function of $(x + x')/2$ and $(x - x')/2$, and then writing $(x + x')/2$ as x and making a Fourier transform with respect to $(x - x')/2$.

Let us check that Eq. (2.133) satisfies Eq. (2.132). First,

$$\int f_W(p, x)\frac{dp}{2\pi\hbar} = \int \rho\left(x + \frac{\eta}{2}, x - \frac{\eta}{2}\right)\left(\int e^{-i(p\eta/\hbar)}\frac{dp}{2\pi\hbar}\right)d\eta$$

$$= \int \rho\left(x + \frac{\eta}{2}, x - \frac{\eta}{2}\right)\delta(\eta)\,d\eta = \rho(x, x) = P(x). \tag{2.134}$$

Thus the second part of Eq. (2.132) is satisfied. Next, using Eq. (2.128) and Eq. (2.130), we have

$$P(p) = \rho(p, p) = \int \rho(x, x')e^{-(i/\hbar)p(x-x')}\,dx\,dx'$$

$$= \int \rho\left(y + \frac{\eta}{2}, y - \frac{\eta}{2}\right)e^{-ip\eta/\hbar}\,dy\,d\eta = \int f_W(p, y)\,dy, \tag{2.135}$$

where we have made the change of variables:

$$\begin{aligned} x &= y + \eta/2 \\ x' &= y - \eta/2. \end{aligned} \tag{2.136}$$

From Eq. (2.132) we see that, if $h(p, x)$ is either a function only of p or a function only of x, then

$$\langle h(p, x)\rangle = \int f_W(p, x)h(p, x)\frac{dp}{2\pi\hbar}\,dx. \tag{2.137}$$

However, Eq. (2.137) is not true for a general function, $h(p, x)$.

Although Wigner's function $f_W(p, x)$ satisfies Eq. (2.132), it cannot be regarded as the probability for finding the particle at the point x and with the momentum p, because $f_W(p, x)$ can become negative for some values of p and x.

Problem: Calculate Wigner's function for a harmonic oscillator. This is a special case where $f_W(p, x)$ is always positive.

In Eq. (2.133), we derived Wigner's function from the density matrix. We can derive the latter from the former:

$$\int f_W\left(p, \frac{x + x'}{2}\right)e^{i[p/\hbar](x-x')}\frac{dp}{2\pi\hbar} = \rho(x, x'). \tag{2.138}$$

Thus $f_W(p, x)$ and $\rho(x, x')$ contain the same information.

Problem: Show that the momentum representation of the free-particle density matrix is

$$\rho(p, p') = 2\pi\hbar\delta(p - p')e^{-(p^2/2m)/kT}. \tag{2.139}$$

2.8 SYMMETRIZED DENSITY MATRIX FOR N PARTICLES

We will now discuss the density matrix for a system of many identical Bose or Fermi particles.

The symbols	will mean
D	Distinguishable
S	Symmetric
A	Antisymmetric

First consider the case of free particles. The Hamiltonian is

$$H = \frac{-\hbar^2}{2m} \sum_k \nabla_k^2. \tag{2.140}$$

The density matrix ρ_D satisfies the equation

$$\partial\rho_D/\partial\beta = -H\rho_D. \tag{2.141}$$

The solution is

$$\rho_D(x_1, x_2, \ldots x_N; x_1', x_2', \ldots x_N'; \beta) = \left(\frac{m}{2\pi\hbar^2\beta}\right)^{3N/2} \exp\left[-\frac{m}{2\hbar^2\beta}\sum_k (x_k - x_k')^2\right]. \tag{2.142}$$

In general when particles are interacting with each other we can write

$$\rho_D(x_1, x_2, \ldots x_N; x_1', x_2', \ldots x_N'; \beta) = \sum_{\text{all states}} e^{-\beta E_i}\psi_i(x_1 \cdots x_N)\psi_i^*(x_1' \cdots x_N'). \tag{2.143}$$

When the particles obey Bose statistics, then

$$\rho_S(x_1 \ldots ; x_1' \ldots x_N'; \beta) = \sum_{\substack{\text{Symmetric}\\ \text{states only}}} e^{-\beta E_i}\psi(x_1 \cdots x_N)\psi_i^*(x_1' \cdots x_N'). \tag{2.144}$$

We introduce the notation P to indicate the permutation of particles. For example, for a permutation

$$P: \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 5 & 4 & 3 & 1 & 2 \end{array}$$

we have

$$Px_1 = x_5.$$

When the state ψ_i is symmetric,

$$\psi_i(Px_k) = \psi_i(x_k).$$

We can construct ρ_S starting from ρ_D. As an example consider the case of two particles:

$$\rho_D(x_1x_2;\, x_1'x_2') = \sum_{\text{all states}} e^{-\beta E_i}\psi_i(x_1, x_2)\psi_i^*(x_1'x_2').$$

Now interchange x_1' and x_2':

$$\rho_D(x_1x_2;\, x_2'x_1') = \sum_{\text{all states}} e^{-\beta E_i}\psi_i(x_1x_2)\psi_i^*(x_2'x_1').$$

Take the average of the two:

$$\begin{aligned}\tfrac{1}{2}[\rho_D(x_1x_2;\, x_1'x_2') + \rho_D(x_1x_2;\, x_2'x_1')] \\ &= \sum_{\text{all states}} e^{-\beta E_i}\psi_i(x_1x_2)\tfrac{1}{2}[\psi_i^*(x_1'x_2') + \psi_i^*(x_2'x_1')] \\ &= \sum_{\substack{\text{symmetric}\\ \text{states only}}} e^{-\beta E_i}\psi_i(x_1x_2)\psi_i^*(x_1'x_2'),\end{aligned}$$

because

$$\tfrac{1}{2}[\psi_i^*(x_1'x_2') + \psi_i^*(x_2'x_1')] = \begin{cases} 0 & \text{antisymmetric } \psi_i^* \\ \psi_i^*(x_1'x_2') & \text{symmetric } \psi_i^*. \end{cases}$$

In general, a group-theoretical argument can show that

$$\frac{1}{N!}\sum_P \psi(Px_1 \cdots Px_k \cdots Px_N) = \begin{cases} \psi(x_1 \cdots x_k \cdots x_N) & \text{symmetric} \\ 0 & \text{for any other symmetry.} \end{cases} \tag{2.145}$$

Here $\sum_P$ means that we sum over all permutations. When we use Eq. (2.145), the case of N particles can be written as

$$\begin{aligned}\rho_S(x_1 \cdots x_k \cdots x_N;\, x_1' \cdots x_k' \cdots x_N') \\ &= \frac{1}{N!}\sum_P \rho_D(x_1 \cdots x_k \cdots;\, Px_1' \cdots Px_k' \cdots Px_N').\end{aligned} \tag{2.146}$$

Thus the partition function becomes

$$\begin{aligned} e^{-\beta F_S} &= \int \rho_S(x_1 \cdots x_k \cdots;\, x_1 \cdots x_N)\, dx_1 \cdots dx_N \\ &= \frac{1}{N!}\sum_P \int \rho_D(x_1 \cdots x_k \cdots;\, Px_1 \cdots Px_k \cdots)\, dx_1 \cdots dx_N. \end{aligned} \tag{2.147}$$

Problem: Derive the partition function for a noninteracting Bose gas. Use Eq. (2.142) in Eq. (2.147) and first write

$$e^{-\beta F_S} = \frac{1}{N!}\left(\frac{m}{2\pi\hbar^2\beta}\right)^{3N/2}$$
$$\times \sum_P \int_V \exp\left\{-\frac{m}{2\hbar^2\beta}[(x_1 - Px_1)^2 + \cdots + (x_k - Px_k)^2 + \cdots + (x_N - Px_N)^2]\right\} dx_1 \cdots dx_N. \tag{2.148}$$

In taking $\sum_P$, we start with the identity permutation that gives V^N from the integral and next go to a pairwise interchange of particles, three-particle cyclic permutation, and so on. The combinatorial factors must be taken into account.

Solution: Any permutation can be broken into cycles. For example,

$$P = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 5 & 3 & 2 & 4 & 6 & 1 \end{pmatrix} = (156)(23)(4) = \begin{pmatrix} 1 & 5 & 6 \\ \downarrow & \downarrow & \downarrow \\ 5 & 6 & 1 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ \downarrow & \downarrow \\ 3 & 2 \end{pmatrix}\begin{pmatrix} 4 \\ \downarrow \\ 4 \end{pmatrix}.$$

Suppose a permutation contains C_ν cycles of length ν, where $\sum_\nu \nu C_\nu = N$. For $\nu > 1$, let

$$h_\nu = \left(\frac{m}{2\pi\hbar^2\beta}\right)^{3\nu/2} \int dx_1\, dx_2 \cdots dx_\nu$$
$$\times \exp\left\{-\frac{m}{2\hbar^2\beta}[(x_1 - x_2)^2 + (x_2 - x_3)^2 + \cdots + (x_{\nu-1} - x_\nu)^2 + (x_\nu - x_1)^2]\right\} \tag{2.149}$$
$$h_1 = \left(\frac{m}{2\pi\hbar^2\beta}\right)^{3/2} \int dx_1 = V\left(\frac{m}{2\pi\hbar^2\beta}\right)^{3/2}.$$

Then Eq. (2.148) can be written

$$e^{-\beta F_S} = \frac{1}{N!}\sum_P \left(\prod_\nu (h_\nu)^{C_\nu}\right). \tag{2.150}$$

It is not difficult to show that

$$\int dy \exp[-a(x - y)^2] \exp[-b(y - z)^2]$$
$$= \left(\frac{\pi}{a + b}\right)^{3/2} \exp\left[-\frac{ab}{a + b}(x - z)^2\right]. \tag{2.151}$$

Using Eq. (2.151) we obtain

$$h_\nu = V\left(\frac{m}{2\pi h^2 \beta \nu}\right)^{3/2}. \tag{2.152}$$

We must now count the possible permutations. Let $M(C_1, \ldots C_q)$ = the number of permutations that have

C_1 cycles of length 1

C_2 cycles of length 2

$\vdots$

C_q cycles of length q

Then

$$e^{-\beta F_S} = \frac{1}{N!} \sum_{C_1, C_2, \ldots, C_q} M(C_1, \ldots, C_q) \prod_\nu (h_\nu)^{C_\nu}. \tag{2.153}$$

As a preliminary to the determination of $M(C_1, \ldots, C_q)$, consider the case of $N = 6$, $C_1 = 3$, $C_3 = 1$, $C_2 = C_4 = C_5 = C_6 = 0$. An example is the permutation $P = (5)(3)(2)(461)$. Notice that the sequence $Q = 5, 3, 2, 4, 6, 1$ corresponds to P. In other words, from the sequence Q, we can uniquely construct the permutation P with $C_1 = 3$, $C_3 = 1$. But from $Q' = 2, 5, 3, 4, 6, 1$, or from $Q'' = 5, 3, 2, 1, 4, 6$, we can also construct in the same way the same P. In fact, it can be easily seen that there are 3! times 3 = 18 different sequences that can be used in the way described above to construct the permutation $P = (5)(3)(2)(461)$.

Generalizing to any set of C's, we see that there are two ways we can get a new Q that leads to the same P:

a) we can interchange cycles of the same length,

b) we can make cyclic permutation within a given cycle.

There are $\prod_\nu C_\nu!$ ways of interchanging cycles of the same length, and $\prod_\nu \nu^{C_\nu}$ cyclic permutations within cycles. Thus for each P there are $\prod_\nu C_\nu! \prod_\nu \nu^{C_\nu}$ different Q's that lead to the same P. Because there are $N!$ possible Q's,

$$M(C_1, \ldots, C_q) = \frac{N!}{\prod_\nu C_\nu!\, \nu^{C_\nu}}.$$

It follows that

$$e^{-\beta F_S} = \sum_{C_1, C_2, \ldots C_q} \prod_\nu \frac{h_\nu^{C_\nu}}{C_\nu!\, \nu^{C_\nu}}, \tag{2.154}$$

where $\sum_\nu \nu C_\nu = N$.

Instead of trying to do the sum in Eq. (2.154), we will find the free energy of the grand canonical ensemble (that is, we will let N vary).

$$e^{-\beta F} = \sum_{N=1}^{\infty} e^{-\beta F_N} e^{+N\mu\beta}. \tag{2.155}$$

As usual, let $\alpha = e^{+\mu\beta}$. Then,

$$e^{-\beta F} = \sum_{C_1,\ldots C_q} \prod_{\nu} \frac{h_\nu^{C_\nu}}{C_\nu!\, \nu^{C_\nu}} \alpha^{\nu C_\nu} = \sum_{C_1,\ldots C_q \ldots} \prod_{\nu} \frac{[h_\nu(\alpha^\nu/\nu)]^{C_\nu}}{C_\nu!},$$

where now each C_q runs from 0 to ∞. Interchanging $\prod$ and $\sum$, we have

$$e^{-\beta F} = \prod_{\nu} \sum_{C_\nu=0}^{\infty} \frac{[h_\nu(\alpha^\nu/\nu)]^{C_\nu}}{C_\nu!} = \prod_{\nu} \exp\left(h_\nu \frac{\alpha^\nu}{\nu}\right) = \exp\left(\sum_{\nu} h_\nu \frac{\alpha^\nu}{\nu}\right).$$

Then,

$$\beta F = -\sum_{\nu=1}^{\infty} \frac{h_\nu \alpha^\nu}{\nu} = -\left(\frac{m}{2\pi\hbar^2\beta}\right)^{3/2} V \sum_{\nu} \frac{\alpha^\nu}{\nu^{5/2}}. \tag{2.156}$$

This result is identical to the one we obtained in a much simpler way earlier.

When the density of the gas is not large, only the identity permutation is important for high temperatures or for the limit of $\hbar \to 0$, since we have such factors as $e^{-[(mkT/2\hbar^2)(x-x')^2]}$. This means that the quantum effect appears only for low temperatures.

For the antisymmetric case,

$$\rho_A(x_1 \cdots x_k \cdots x_N; x_1' \cdots x_k' \cdots x_N')$$
$$= \frac{1}{N!} \sum_{P} (-1)^P \rho_D(x_1 \cdots x_N; Px_1' \cdots Px_k' \cdots Px_N'),$$

where

$$(-1)_P = \begin{cases} 1 \text{ for even permutation} \\ -1 \text{ for odd permutation.} \end{cases}$$

Problem: Modify the argument leading to Eq. (2.156) so as to obtain the corresponding result for a noninteracting Fermi gas.

2.9 DENSITY SUBMATRIX

For N particles, the density matrix can be written

$$\rho(x_1, x_2, \ldots, x_N; x_1', x_2', \ldots x_N') = \sum_i w_i \varphi_i(x_1, \ldots, x_N) \varphi_i^*(x_1', \ldots, x_N'). \tag{2.157}$$

Notice that we are not now specializing to the case of statistical mechanics. To compute the expectation value of $P_1^2/2m$, we proceed as follows:

$$\begin{aligned}\left\langle \frac{P_1^2}{2m} \right\rangle &= \frac{1}{\operatorname{Tr}\rho} \operatorname{Tr} \rho \frac{P_1^2}{2m} \\ &= -\frac{1}{\operatorname{Tr}\rho} \frac{\hbar^2}{2m} \int \frac{\partial^2}{\partial x_1^2} \rho(x_1, \ldots, x_N; x_1', \ldots, x_N') \Big|_{x_i = x_i'} dx_1 \cdots dx_N \\ &= -\frac{\hbar^2}{2m} \int dx_1 \frac{\partial^2}{\partial x_1^2} \rho_1(x_1, x_1') \Big|_{x_1 = x_1'} = \operatorname{Tr} \rho_1 \frac{P_1^2}{2m}, \end{aligned} \tag{2.158}$$

where we define

$$\rho_1(x, x') = \frac{\int \rho(x, x_2, \ldots, x_N; x', x_2, \ldots, x_N)\, dx_2 \cdots dx_N}{\operatorname{Tr}[\rho]}. \tag{2.159}$$

ρ_1 is the one-particle density submatrix. Using this matrix we can calculate, for example,

$$\left\langle \sum_k \frac{p_k^2}{2m} \right\rangle = -N \int \left[\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x'^2} \rho_1(x, x') \right]_{x' = x} dx = N \operatorname{Tr} \left[\frac{p^2}{2m} \rho_1 \right] \tag{2.160}$$

(assuming that the particles are essentially identical).

Similarly,

$$\left\langle \sum_k V(x_k) \right\rangle = N \int V(x) \rho_1(x, x)\, dx = N \operatorname{Tr}[V\rho_1]. \tag{2.161}$$

The two-particle density submatrix is defined as

$$\rho_2(x_1 x_2; x_1' x_2') = \frac{\int \rho_2(x_1 x_2 x_3 \cdots x_N; x_1' x_2' x_3 \cdots x_N)\, dx_3 \cdots dx_N}{\operatorname{Tr}[\rho]}. \tag{2.162}$$

From Eq. (2.162), the two-particle potential energy $V(x_1 x_2)$ has expectation

$$\left\langle \sum_{i<j} V(x_i, x_j) \right\rangle = \frac{N(N-1)}{2} \int V(x_1 x_2) \rho_2(x_1 x_2; x_1 x_2)\, dx_1\, dx_2. \tag{2.163}$$

Special cases of density submatrices are the distribution functions. The single-particle distribution function is defined as

$$p_1(x) = \rho_1(x; x). \tag{2.164}$$

The two-particle distribution function is

$$p_2(x, y) = \rho_2(x, y; x, y). \tag{2.165}$$

In a uniform substance, such as a gas or a liquid, $p_1(x)$ is independent of x, and $p_2(x, y)$ depends only on the difference $(x - y)$.

2.10 PERTURBATION EXPANSION OF THE DENSITY MATRIX

Recall that the density matrix in statistical mechanics satisfies

$$\frac{\partial \rho}{\partial \beta} = -H\rho. \tag{2.166}$$

There are very few Hamiltonians for which we can solve Eq. (2.166) exactly, but it may be that H is close to one such Hamiltonian, H_0.

$$H = H_0 + H_1, \tag{2.167}$$

$$\frac{\partial \rho_0}{\partial \beta} = -H_0\rho_0. \tag{2.168}$$

We would like to use ρ_0 to obtain an approximation to ρ, which we expect to be close to $\rho_0 = e^{-\beta H_0}$. Because ρ is close to $e^{-\beta H_0}$, we expect $e^{H_0\beta}\rho$ to vary slowly with β.

$$\frac{\partial}{\partial \beta}(e^{H_0\beta}\rho) = H_0 e^{H_0\beta}\rho + e^{H_0\beta}\frac{\partial \rho}{\partial \beta} = e^{H_0\beta}H_0\rho - e^{H_0\beta}H\rho = -e^{H_0\beta}H_1\rho. \tag{2.169}$$

Integrating Eq. (2.169) from zero to β, and remembering that if $\beta = 0$, we find

$$e^{H_0\beta}\rho = 1,$$

$$e^{H_0\beta}\rho(\beta) - 1 = -\int_0^\beta e^{H_0\beta'}H_1\rho(\beta')\,d\beta'. \tag{2.170}$$

Therefore,

$$\rho(\beta) = \rho_0(\beta) - \int_0^\beta \rho_0(\beta - \beta')H_1\rho(\beta')\,d\beta'. \tag{2.171}$$

The last term in Eq. (2.171) is small if H_1 is small, and it is a correction term to the approximate equation $\rho \approx \rho_0$. If the correction term is small, we can use an approximate $\rho(\beta')$ to obtain a much more accurate approximation to $\rho(\beta)$. For example, if we make the approximation $\rho(\beta') \approx \rho_0(\beta')$, we have

$$\rho(\beta) \approx \rho_0(\beta) - \int_0^\beta \rho_0(\beta - \beta')H_1\rho_0(\beta')\,d\beta'. \tag{2.172}$$

In using Eq. (2.172) as a new approximation to $\rho(\beta')$, we can find a still better approximation to ρ. Continuing in this manner we get

$$\begin{aligned}\rho(\beta) = \rho_0(\beta) &- \int_0^\beta d\beta'[\rho_0(\beta - \beta')H_1\rho_0(\beta')] \\ &+ \int_0^\beta d\beta' \int_0^{\beta'} d\beta''[\rho_0(\beta - \beta')H_1\rho_0(\beta' - \beta'')H_1\rho_0(\beta'')] \\ &- \int_0^\beta d\beta' \int_0^{\beta'} d\beta'' \int_0^{\beta''} d\beta'''[\quad] + \cdots \end{aligned} \tag{2.173}$$

We can easily rewrite Eq. (2.173) in coordinate representation. For example,

$$\begin{aligned}\rho(x, x'; \beta) &= \langle x|\rho(\beta)|x'\rangle \\ &\approx \rho_0(x, x'; \beta) - \int_0^\beta \langle x|\rho_0(\beta - \beta')\left(\int |x''\rangle\langle x''|dx''\right) H_1\rho_0(\beta')|x'\rangle\, d\beta',\end{aligned} \tag{2.174}$$

where we use

$$\int_{-\infty}^{\infty} |x''\rangle\langle x''|\, dx'' = 1.$$

If $H_1 = V(x)$, then

$$\langle x''|H_1\rho_0(\beta')|x'\rangle = V(x'')\rho_0(x'', x'; \beta'),$$

and Eq. (2.174) becomes

$$\begin{aligned}\rho(x, x'; \beta) &= \rho_0(x, x', \beta) \\ &- \int_{-\infty}^{\infty} \int_0^\beta \rho_0(x, x''; \beta - \beta')V(x'')\rho_0(x'', x'; \beta')\, d\beta'\, dx'' + \cdots\end{aligned} \tag{2.175}$$

2.11 PROOF THAT $F \le F_0 + \langle H - H_0 \rangle_0$

Now that we have a perturbation theory, we are in a position to prove as promised in Section 2.3, that $F \le F_0 + \langle H - H_0 \rangle_0$. Let $V = H - H_0$. Since F is determined by the equation $e^{-\beta F} = \mathrm{Tr}\, e^{-\beta(H_0+V)}$, the most obvious approach is to find an approximation to F by using the perturbation expansion of $e^{-\beta(H_0+V)} = \rho$.

$$\begin{aligned}e^{-\beta(H_0+V)} &= e^{-\beta H_0} - \int_0^\beta e^{-(\beta-u)H_0} V e^{-uH_0}\, du \\ &+ \int_0^\beta \int_0^{u_1} du_1\, du_2\, e^{-(\beta-u_1)H_0} V e^{-(u_1-u_2)H_0} V e^{-u_2 H_0} - \cdots.\end{aligned} \tag{2.176}$$

Taking the trace of Eq. (2.176) we obtain

$$e^{-\beta F} = \mathrm{Tr}\, e^{-\beta(H_0+V)} = e^{-\beta F_0} - \int_0^\beta \mathrm{Tr}\,[e^{-(\beta-u)H_0} V e^{-uH_0}]\, du$$
$$+ \int_0^\beta \int_0^{u_1} du_1\, du_2\, \mathrm{Tr}\,[e^{-(\beta-u_1)H_0} V e^{-(u_1-u_2)H_0} V e^{-u_2 H_0}] - \cdots. \tag{2.177}$$

Using the fact that $\mathrm{Tr}\, AB = \mathrm{Tr}\, BA$, we find that

$$e^{-\beta F} = e^{-\beta F_0} - \int_0^\beta du\, \mathrm{Tr}\, e^{-\beta H_0} V$$
$$+ \int_0^\beta \int_0^{u_1} du_1\, du_2\, \mathrm{Tr}\,[e^{-\beta H_0} e^{(u_1-u_2)H_0} V e^{-(u_1-u_2)H_0} V] - \cdots. \tag{2.178}$$

It is possible to simplify the second-order term. Let $w = u_1 - u_2$ and let $x = u_1$. Then the integral becomes

$$\int_A \mathrm{Tr}\,[e^{-\beta H_0} e^{wH_0} V e^{-wH_0} V], \tag{2.179}$$

where A is the shaded region in Fig. 2.4. Now let $w' = \beta - w$ and let $x' = \beta - x$. Then the integral becomes

$$\int_{A'} \mathrm{Tr}\,[e^{-\beta H_0} e^{wH_0} V e^{-wH_0} V]. \tag{2.180}$$

Averaging Eqs. (2.179) and (2.180), we find, from Eq. (2.178),

$$e^{-\beta F} = e^{-\beta F_0} - \beta\, \mathrm{Tr}\,[e^{-\beta H_0} V] + \frac{\beta}{2} \int_0^\beta dw\, \mathrm{Tr}\,[e^{-\beta H_0} e^{wH_0} V e^{-wH_0} V] + \cdots. \tag{2.181}$$

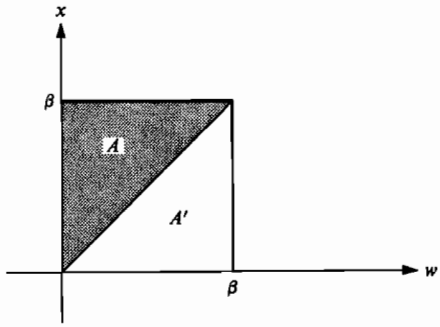

Fig. 2.4 The shaded area A is the region over which Eq. (2.179) is integrated.

Let $|m\rangle$ and $|n\rangle$ be eigenstates of H_0, and let E_m and E_n be their respective eigenvalues. Then

$$\mathrm{Tr}\,[e^{-\beta H_0}V] = \sum_n \langle n|e^{-\beta H_0}V|n\rangle = \sum_n e^{-\beta E_n}V_{nn}. \tag{2.182}$$

$$\begin{aligned}\mathrm{Tr}\,[e^{-\beta H_0}e^{wH_0}Ve^{-wH_0}V] &= \sum_n \langle n|e^{-\beta H_0}e^{wH_0}V \sum_m |m\rangle\langle m|e^{-wH_0}V|n\rangle \\ &= \sum_{nm} e^{-\beta E_n}e^{w(E_n-E_m)}V_{nm}V_{mn},\end{aligned} \tag{2.183}$$

where we have used the fact that $\sum_m |m\rangle\langle m| = 1$. Equation (2.181) becomes

$$e^{-\beta F} = e^{-\beta F_0} - \sum_n \beta e^{-\beta E_n}V_{nn} + \frac{\beta}{2}\sum_{mn}\frac{e^{-\beta E_m} - e^{-\beta E_n}}{E_n - E_m}|V_{mn}|^2 + \cdots. \tag{2.184}$$

Note that if $m = n$,

$$\frac{e^{-\beta E_m} - e^{-\beta E_n}}{E_n - E_m} = e^{-\beta E_n}\frac{e^{\beta(E_n-E_m)} - 1}{E_n - E_m} = \beta e^{-\beta E_n}.$$

In order to get an expansion for F, we write $F = F_0 + F_1 + F_2 + \cdots$, where F_1 is a first-order perturbation and F_2 is a second-order perturbation. Then

$$\begin{aligned}e^{-\beta F} &= e^{-\beta F_0}e^{-\beta(F_1+F_2+\cdots)} \\ &= e^{-\beta F_0}(1 - \beta F_1 - \beta F_2 + \frac{\beta^2}{2}(F_1 + F_2 + \cdots)^2 + \cdots) \\ &= e^{-\beta F_0}\left[1 - \beta F_1 + \left(\frac{\beta^2}{2}F_1^2 - \beta F_2\right) + \cdots\right].\end{aligned} \tag{2.185}$$

up to and including the second order. We have a perturbation expansion for F if we set

$$-e^{-\beta F_0}\beta F_1 = -\sum_n \beta e^{-\beta E_n}V_{nn}$$

and (2.186)

$$e^{-\beta F_0}\left(\frac{\beta^2}{2}F_1^2 - \beta F_2\right) = \frac{\beta}{2}\sum_{mn}\frac{e^{-\beta E_m} - e^{-\beta E_n}}{E_n - E_m}|V_{mn}|^2.$$

Solving these equations we obtain

$$F_1 = \frac{\mathrm{Tr}\,Ve^{-\beta H_0}}{\mathrm{Tr}\,e^{-\beta H_0}} = \langle V\rangle_{H_0} \tag{2.187}$$

and

$$\begin{aligned}F_2 = \frac{\beta}{2}\left[\frac{\sum_n V_{nn}e^{-\beta E_n}}{\sum_n e^{-\beta E_n}}\right]^2 - \frac{1}{2}e^{\beta F_0}\sum_{\substack{mn \\ m\neq n}}\frac{e^{-\beta E_m} - e^{-\beta E_n}}{E_n - E_m}|V_{mn}|^2 \\ -\frac{\beta}{2}\left[\frac{\sum_n |V_{nn}|^2 e^{-\beta E_n}}{\sum_n e^{-\beta E_n}}\right].\end{aligned} \tag{2.188}$$

With the Cauchy-Schwarz inequality,

$$\left|\sum_n a_n b_n\right|^2 \leq \left[\sum_n |a_n|^2\right]\left[\sum_n |b_n|^2\right],$$

it is easy to show that for any set of positive real numbers, w_n,

$$\left[\frac{\sum_n w_n a_n}{\sum_n w_n}\right]^2 \leq \frac{\sum_n w_n |a_n|^2}{\sum_n w_n}.$$

Applying this inequality to Eq. (2.188), we have

$$F_2 \leq -\frac{1}{2} e^{\beta F_0} \sum_{\substack{mn \\ m \neq n}} \frac{e^{-\beta E_m} - e^{-\beta E_n}}{E_n - E_m} |V_{mn}|^2 \leq 0. \tag{2.189}$$

Now, let

$$H(\alpha) = H_0 + \alpha V,$$

and let $F(\alpha)$ be the free energy calculated from $H(\alpha)$. Then $F(0) = F_0$ and $F(1) = F$. From the preceding perturbation analysis,

$$F(\alpha) = F_0 + \alpha F_1 + \alpha^2 F_2 + 0(\alpha^3).$$

So $F'(0) = F_1$. If we can prove that

$$F''(\alpha) \leq 0 \text{ for all } \alpha, \tag{2.190}$$

then $F(\alpha)$ will be concave downward (see Fig. 2.5) and thus it lies below the line $F(0) + \alpha F'(0) = F_0 + \alpha F_1$. From this we can conclude that

$$F \leq F_0 + F_1 = F_0 + \langle V \rangle_{H_0},$$

or

$$F \leq F_0 + \langle H - H_0 \rangle_{H_0}, \tag{2.191}$$

which was to be proved.

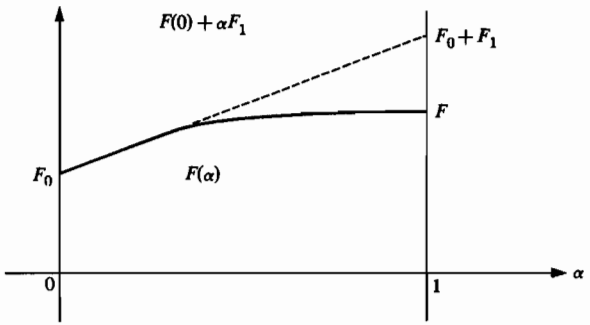

Fig. 2.5 The situation when $F(\alpha)$ is concave downward.

To prove Eq. (2.190) we apply the perturbation analysis to $F(\alpha)$ for each α; that is, we write

$$F(\alpha + \gamma) = F_0(\alpha) + \gamma F_1(\alpha) + \gamma^2 F_2(\alpha) + 0(\gamma^3),$$

where $F_0(\alpha) = F(\alpha)$, and $\gamma F_1(\alpha)$ and $\gamma^2 F_2(\alpha)$ are as in Eqs. (2.187) through (2.189), with H_0 replaced by $H(\alpha)$ and H by $H(\alpha + \gamma) = H(\alpha) + \gamma V$. The result of Eq. (2.189) is then

$$F_2(\alpha) \leq 0,$$

so that

$$F''(\alpha) = \frac{d^2}{d\gamma^2} F(\alpha + \gamma)\bigg|_{\gamma=0} = 2F_2(\alpha) \leq 0.$$

This proves Eq. (2.190) and thus completes the proof of Eq. (2.191).

Applying this same theorem to F_0, we obtain

$$F_0 \leq F + \langle -V \rangle_H. \tag{2.192}$$

Combining Eqs. (2.191) and (2.192), we see that

$$F_0 + \langle V \rangle_H \leq F \leq F_0 + \langle V \rangle_{H_0}. \tag{2.193}$$

A more accurate approximation F is obtained by including F_2:

$$F \approx F_0 + \langle V \rangle_0 + \frac{\beta}{2} \langle V \rangle_0^2 - \frac{1}{2} \left\langle \int_0^\beta e^{wH_0} V e^{-wH_0} V \, dw \right\rangle_0. \tag{2.194}$$

CHAPTER 3

PATH INTEGRALS

3.1 PATH INTEGRAL FORMULATION OF THE DENSITY MATRIX

Earlier it has been shown that the density matrix satisfies the equation

$$\hbar \frac{\partial \rho(u)}{\partial u} = -H\rho(u). \tag{3.1}$$

and has the formal solution

$$\rho(u) = e^{-Hu/\hbar}, \tag{3.2}$$

where it will be noted that u has been redefined to be $\beta\hbar$, which has the dimensions of time. Furthermore, the "time" u can be broken up into divisions of length $\varepsilon(n\varepsilon = u)$ so that ρ can be developed incrementally as follows:

$$\begin{aligned} \rho(u) &= e^{-H\varepsilon/\hbar} e^{-H\varepsilon/\hbar} \cdots e^{-H\varepsilon/\hbar} \\ &= \rho_\varepsilon \rho_\varepsilon \cdots \rho_\varepsilon (n \text{ factors}). \end{aligned} \tag{3.3*}$$

In the coordinate representation for the density matrix $\rho(x, x'; u)$ the solution is represented as

$$\rho(x, x'; u) = \int \cdots \int \rho(x, x_{n-1}; \varepsilon)\rho(x_{n-1}, x_{n-2}; \varepsilon) \cdots \rho(x_1, x'; \varepsilon) dx_1 \cdots dx_{n-1}. \tag{3.4}$$

The representation in Fig. 3.1 is useful in the interpretation of the above expression. The particle travels from x' to x through a series of intermediate steps, $x_1, x_2 \ldots, x_{n-1}$, which define a "path". The total amplitude $\rho(x, x', u)$ for the particle to begin at x' and end up at x is given by a sum over all possible paths, that is, for all possible values of the intermediate positions x_i. As the time increment ε is allowed to approach zero, the number of integrations on the intermediate variables become infinite and Eq. (3.4) can be written symbolically

$$\rho(x, x'; U) = \int\!\!\!\int \Phi[x(u)]\, \mathscr{D}x(u), \tag{3.5}$$

* The reason we want to reduce ε as much as possible is, of course, that the asymptotic behavior of ρ at high temperatures can be easily found (see below).

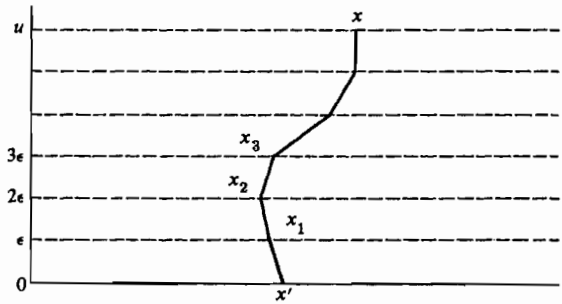

Fig. 3.1 A particle travelling from x' to x along a path x_1, x_2 etc.

where

$$\Phi[x(u)] = \lim_{\substack{\varepsilon \to 0 \\ n\varepsilon = u}} \rho(x, x_{n-1}; \varepsilon)\rho(x_{n-1}, x_{n-2}; \varepsilon) \cdots \rho(x_1, x'; \varepsilon), \tag{3.6}$$

and

$$\mathscr{D}x(u) = \lim_{n \to \infty} dx_1\, dx_2 \cdots dx_{n-1}.$$

Although all possible paths are to be considered in Eq. (3.5), the main contribution comes from the limited set of paths for which none of the $\rho(x, x'; \varepsilon)$ is too small. For example, we know that for low ε,

$$\rho(x, x'; \varepsilon) \approx \sqrt{\frac{m}{2\pi\hbar\varepsilon}}\, e^{-(m/2\hbar\varepsilon)(x-x')^2},$$

which is the density matrix for a free particle. The term $\rho(x, x'; \varepsilon)$ is small if $x - x' > \sqrt{2\hbar/m}\sqrt{\varepsilon}$; so the main contribution comes from reasonably smooth paths. Now let us find Φ more explicitly, first for a free particle and then for one moving subject to a constraint.

Free Particle:

The Hamiltonian is

$$H = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}. \tag{3.7}$$

From previous results we know

$$\rho(x, x'; \varepsilon) = \sqrt{\frac{m}{2\pi\hbar\varepsilon}}\, e^{-(m/2\hbar\varepsilon)(x-x')^2} \tag{3.8}$$

and

$$\rho(x, x'; U) = \lim_{\varepsilon \to 0} \int \cdots \int \exp\left\{-\frac{m\varepsilon}{2\hbar}\left[\left(\frac{x - x_{n-1}}{\varepsilon}\right)^2 + \left(\frac{x_{n-1} - x_{n-2}}{\varepsilon}\right)^2 \cdots \left(\frac{x_1 - x'}{\varepsilon}\right)^2\right]_x\right\}$$
$$\times \frac{dx_1}{\sqrt{2\pi\hbar\varepsilon/m}} \frac{dx_2}{\sqrt{2\pi\hbar\varepsilon/m}} \cdots \frac{dx_{n-1}}{\sqrt{2\pi\hbar\varepsilon/m}}. \tag{3.9}$$

Thus, for a free particle,

$$\Phi[x(u)] = \lim_{\varepsilon \to 0} \exp\left\{-\frac{m\varepsilon}{2\hbar}\left[\left(\frac{x - x_{n-1}}{\varepsilon}\right)^2 + \cdots + \left(\frac{x_1 - x'}{\varepsilon}\right)^2\right]\right\}. \tag{3.10}$$

As the interval ε is made smaller,

$$\frac{x_k - x_{k-1}}{\varepsilon} \to \left.\frac{dx(u)}{du}\right|_{u=k\varepsilon} \equiv \left.\dot{x}(u)\right|_{u=k\varepsilon}$$

and Eq. (3.10) becomes

$$\Phi[x(u)] = \exp\left\{-\frac{1}{\hbar}\int_0^U \frac{m}{2}[\dot{x}(u)]^2\, du\right\}. \tag{3.11}$$

Particle Moving in One Dimension in a Potential $V(x)$:

$$H = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x).$$

The equation for ρ is explicitly

$$-\hbar\frac{\partial \rho}{\partial u} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\rho + V(x)\rho. \tag{3.12}$$

In an infinitesimal "time" interval ε the particle may be considered as moving freely with a small correction due to the presence of the potential $V(x)$. To a very good approximation

$$\delta\rho = -\frac{V(x)}{\hbar}\varepsilon\rho_{\text{free}}. \tag{3.13}$$

This result is demonstrated as follows: From perturbation theory we have

$$\delta\rho(x, x'; \varepsilon) \approx -\int_{-\infty}^{\infty} dx'' \int_0^{\varepsilon} \rho_0(x, x''; \varepsilon - u)V(x'')\rho_0(x'', x'; u)\frac{du}{\hbar}, \tag{3.14}$$

where in our case, $\rho_0 = \rho_{\text{free}}$. For low ε, because ρ_{free} is a very localized Gaussian, most of the contribution in the integral over x'' occurs near $x'' = x_0$, where it can be shown that

$$x_0 = \frac{ux + (\varepsilon - u)x'}{\varepsilon}.$$

So we can, for small ε, write Eq. (3.14) as

$$\begin{aligned}\delta\rho(x, x'; \varepsilon) &\approx -\int_0^\varepsilon \frac{du}{\hbar} V(x_0) \int_{-\infty}^{\infty} dx'' \, \rho_0(x, x''; \varepsilon - u)\rho_0(x'', x'; u) \\ &= -\int_0^\varepsilon \frac{du}{\hbar} V(x_0)\rho_{\text{free}}(x, x'; \varepsilon). \qquad (3.15)\end{aligned}$$

Now, if $x \approx x'$, x_0 is also close to x and $V(x_0)$ is constant over the range of integration. Equation (3.15) goes over to Eq. (3.13):

$$\delta\rho(x, x'; \varepsilon) \approx -\frac{\varepsilon}{\hbar} V(x)\rho_{\text{free}}(x, x'; \varepsilon)$$

under the assumption that $|x - x'|$ and ε are small.

Of course, as $|x - x'|$ increases, our conclusion that $\int_0^\varepsilon V(x_0)\, du \approx \varepsilon V(x)$ becomes less accurate, and it might be thought preferable to write something like

$$\varepsilon\left[\frac{V(x) + V(x')}{2}\right] \quad \text{or} \quad \varepsilon V\left(\frac{x + x'}{2}\right)$$

in place of $\varepsilon V(x)$. But for our purposes, Eq. (3.13) is adequate. So

$$\rho(\varepsilon) \approx \rho_{\text{free}}(\varepsilon)\left[1 - \frac{V(x)}{\hbar}\varepsilon\right] \approx \rho_{\text{free}}(\varepsilon)e^{-V(x)\varepsilon/\hbar}. \qquad (3.16)$$

Thus

$$\rho(x, x'; \varepsilon) \approx \sqrt{\frac{m}{2\pi\hbar\varepsilon}} \exp\left[-\frac{m(x - x')^2}{2\hbar\varepsilon} - \frac{\varepsilon V(x)}{\hbar}\right]. \qquad (3.17)$$

To find $\rho(x, x'; u)$ it is only necessary to apply the correction $e^{-V(x)\varepsilon/\hbar}$ to each element of the "path" of the free particle from x to x'. Thus, as $\varepsilon \to 0$,

$$\begin{aligned}\rho(x, x'; U) &= \int\!\!\!\int \exp\left\{-\frac{1}{\hbar}\int_0^U \left[\frac{m\dot{x}(u)^2}{2} + V(x(u))\right] du\right\} \mathscr{D}x(u) \qquad (3.18) \\ &\equiv \int\!\!\!\int \Phi[x(u)]\, \mathscr{D}x(u),\end{aligned}$$

where $x(0) \equiv x'$ and $x(U) = x$. Thus

$$\Phi[x(u)] = \exp\left\{-\frac{1}{\hbar}\int_0^V \left[\frac{mx'(u)^2}{2} + V(x(u))\right] du\right\}. \tag{3.18a}$$

The free energy is derived from

$$\begin{aligned} e^{-\beta F} &= \int \rho(x, x)\, dx \\ &= \int \left\{ \iint \exp\left\{-\frac{1}{\hbar}\int_0^U \left[\frac{m\dot{x}(u)^2}{2} + V(x(u))\right] du\right\} \mathscr{D}x(u)\right\}_{x(0)=x(U)=x} dx \end{aligned} \tag{3.19}$$

where $U = \beta\hbar$.

Having derived the general expression for Φ the question now is, which "paths" are the most important? If the potential is not very dependent upon position, then the more direct paths contribute the most. This is because longer paths, with higher mean square "velocities," $[\dot{x}(u)]^2$, make the exponent of the weighting functional $\Phi[x(u)]$ more negative. On the other hand, suppose the potential does depend on position. For simplicity, let the potential increase for larger values of x. Then a direct path such as 1 in Fig. 3.2 does not necessarily give the largest contribution, because the integrated value of the potential is higher than over another path such as 2 or 3. Consider a path such as 2 that deviates widely from the direct path. Then $V(x)$ decreases but $\dot{x}$ increases over the path. In this case we would expect that the increased velocity $\dot{x}$ would more than compensate for the decreased potential over the path. Thus, the most important path would be one for which any smaller integrated value of potential energy is more than compensated for by an increase in kinetic energy (symbolically represented by 3 in Fig. 3.2).

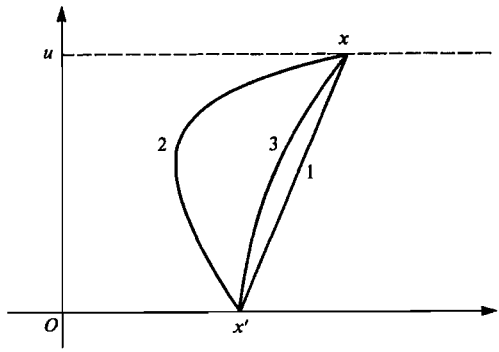

Fig. 3.2 Three paths between x' and x.

In the classical limit, the temperature is high and U is small. We want the classical partition function, and we would like to see how it is related to the quantum-mechanical partition function. From Eq. (3.18)

$$\rho(x, x; U) = \iint_{x(0)=x(U)=x} \exp\left\{-\frac{1}{\hbar}\int_0^U \left[\frac{m\dot{x}(u)^2}{2} + V(x(u))\right] du\right\} \mathscr{D}x(u). \tag{3.20}$$

If we assume that the most important paths are those for which $V(x(u)) \approx V(x)$, Eq. (3.19) can be approximated by

$$\begin{aligned}\rho(x, x; U) &\approx \exp\left[-\frac{U}{\hbar}V(x)\right] \iint_{x(0)=x(U)=x} \exp\left[-\frac{1}{\hbar}\int_0^U \frac{m\dot{x}(u)^2}{2}\right] \mathscr{D}x(u) \\ &= \exp\left[-\frac{U}{\hbar}V(x)\right] \rho_{\text{free}}(x, x; U) \\ &= \sqrt{\frac{m}{2\pi\hbar U}} \exp\left[\frac{-m(x-x')^2}{\hbar U} - \frac{UV(x)}{\hbar}\right]\end{aligned} \tag{3.21}$$

Equation (3.21) is just a specialization of Eq. (3.17). But from the derivation of Eq. (3.21) it is easy to see the direction of the error in the classical limit. We assumed that the most important paths had $x(u) \approx x$, whereas in fact the most important paths are those for which $\int_0^U [m\dot{x}(u)^2/2 + V(x(u))]\, du$ is a minimum, which occurs when $(x(u))$ goes over paths that tend to let $V(x(u))$ be lower than $V(x)$. It follows, then, that

$$\rho_{\text{classical}}(x, x; U) < \rho(x, x; U). \tag{3.22}$$

The classical partition function is

$$e^{-\beta F_{\text{cl}}} = \int_{-\infty}^{\infty} \rho_{\text{cl}}(x, x; U)\, dx = \sqrt{\frac{m}{2\pi\hbar U}} \int_{-\infty}^{\infty} e^{-UV(x)/\hbar} dx. \tag{3.23}$$

From Eq. (3.22) we obtain

$$F_{\text{cl}} \gtrsim F. \tag{3.24}$$

To find the condition for Eq. (3.21) to be a good approximation to the path integral, note that the paths for which

$$\frac{1}{\hbar}\int_0^U \frac{m\dot{x}^2}{2}\, du \gg 1$$

do not contribute much to the path integral. Using the Cauchy-Schwarz inequality, we find

$$\left|\int_a^b fg\right|^2 \leq \int_a^b |f|^2 \int_a^b |g|^2.$$

With $f = \dot{x}$ and $g = 1$, we have for $0 \leq u_1 \leq U$:

$$|x(u_1) - x(0)|^2 = \left|\int_0^{u_1} \dot{x}(u)\right|^2 \leq \left[\int_0^{u_1} \dot{x}^2(u)\right] u_1 \leq \left(\int_0^{u_1} \dot{x}^2\right) U. \tag{3.25}$$

So for the important paths, $(m/2\hbar U)|x(u_1) - x(0)|^2 < 1$. If d is the maximum value of $|x(u_1) - x(0)|$ for $0 < u_1 < U$, then the paths that contribute significantly satisfy

$$\frac{mkTd^2}{2\hbar^2} \leq 1. \tag{3.26}$$

For the hydrogen atom at 1°K Eq. (3.26) yields $d \leq 9$ Å. For oxygen at 1°K, $d \leq 2.5$ Å; for oxygen at 100°K, $d \leq 0.25$ Å.

If the potential varies so slowly that it is approximately unchanged even when the path varies by more than d, then Eq. (3.21) is good. Thus, for heavy particles at normal temperatures, systems may be treated by classical statistical mechanics. However, the classical approximation does not apply to liquid helium, nor does it apply to electrons in a metal, nor even to solids if the temperature is too low.

A few things should be noticed about this formulation of statistical mechanics:

1. We have discussed only the case of one coordinate. However, the extension to N degrees of freedom is obvious; if $x^{(1)}, x^{(2)}, \ldots, x^{(N)}$ are the coordinates, $\mathscr{D}x(u)$ becomes

$$\mathscr{D}x(u) = \lim_{n\to\infty} dx_1^{(1)}\, dx_1^{(2)} \cdots dx_1^{(N)}\, dx_2^{(1)} \cdots dx_2^{(N)} \cdots dx_n^{(N)}. \tag{3.27}$$

2. If "u" is replaced by "iu" in Eq. (3.1), we have Schrödinger's equation. In a similar manner to that of statistical mechanics, quantum mechanics can be formulated in terms of path integrals. For mathematicians, statistical mechanics is easier to deal with because the exponentials in the path integrals are exponentials of real quantities.

3. We have not dealt with relativity (in which particles appear and disappear) and spin (which is a discrete, rather than a continuous, variable) in the path-integral formulation. If such effects exist, we can only get approximations. Actually, spin *can* be put into the formalism, but our interpretation of $\mathscr{D}x(u)$ as a set of paths differing by an infinitesimal amount no longer holds.

3.2 CALCULATION OF PATH INTEGRALS

We will discuss two techniques for the calculation of path integrals:

1. Integrals whose exponents are quadratic in $\dot{x}$ and x will be calculated exactly.

2. Perturbation expansions.

Consider the element

$$f(x_2, x_1, U) = \iint \exp\left\{-\frac{1}{\hbar}\int_0^U \frac{m[\dot{x}(u)]^2}{2}\,du\right\}\mathscr{D}x(u) \tag{3.28}$$

where $x(0) = x_1$ and $x(U) = x_2$. This is the density matrix element for a free particle. Let us expand each path about the straight-line path in the xu-plane. Call the straight line path $\mathbf{x}(u)$. Then

$$x(u) = \mathbf{x}(u) + y(u) \tag{3.29}$$

and

$$f(x_2, x_1, U) = \iint \exp\left[-\frac{1}{\hbar}\frac{m}{2}\int_0^U (\dot{\mathbf{x}}(u) + \dot{y}(u))^2\,du\right]\mathscr{D}y(u). \tag{3.30}$$

Now

$$\frac{d\mathbf{x}(u)}{du} = \frac{x_2 - x_1}{U} = v = \text{velocity} \tag{3.31}$$

so that

$$\int_0^U [\dot{\mathbf{x}} + \dot{y}]^2\,du$$
$$= \int_0^U (v^2 + 2v\dot{y} + \dot{y}^2)\,du = v^2U + 2v[y(U) - y(0)] + \int_0^U \dot{y}^2\,du. \tag{3.32}$$

But $y(U) = y(0) = 0$. Therefore

$$f(x_2, x_1, U) = \exp\left[-\frac{mv^2U}{2\hbar}\right]\iint \exp\left[-\frac{m}{2\hbar}\int_0^U \dot{y}^2\,du\right]\mathscr{D}y. \tag{3.33}$$

Thus we have split the path integral into a factor dependent upon the endpoints and U, and a path integral dependent only upon U. Writing

$$\iint_{y(0)=y(U)=0} \exp\left[-\frac{m}{2\hbar}\int_0^U \dot{y}^2\,du\right]\mathscr{D}y = F(U),$$

we get

$$\rho(x_2, x_1, U) = F(U)\exp\left[-m(x_2 - x_1)^2/2\hbar U\right]. \tag{3.34}$$

We can obtain an equation for $F(U)$ by using

$$\rho(x, y; u_1 + u_2) = \int dx'\,\rho(x, x'; u_2)\rho(x', y; u_1). \tag{3.35}$$

The reader should derive that equation for $F(u)$ and show that its most general continuous solution is

$$F(U) = \sqrt{m/2\pi\hbar U}\, e^{\alpha U}, \tag{3.36}$$

where α is an arbitrary number. Physically, the value of α is immaterial, because a change in the zero of the potential does not change the physics; it does, however, change the value of α. As a matter of fact, since the expectation value of any quantity, A, is given by

$$\langle A \rangle = \frac{\operatorname{Tr} \rho A}{\operatorname{Tr} \rho}, \tag{3.37}$$

$F(U)$ need not be found at all. It cancels out in the division. We know that α is zero, however, from other methods of calculating ρ (see Eq. (3.8)).

Now consider the path integral for the harmonic oscillator:

$$f(x_2, x_1, U) = \iint \exp\left\{-\frac{1}{\hbar}\int_0^U \left[\frac{m\dot{x}^2}{2} + \frac{m\omega^2}{2}x^2\right] du\right\} \mathscr{D}x.$$

Let us expand each path about the path that makes the largest contribution to P. For this path,

$$\int_0^U G(u)\, du = \int_0^U \left[\frac{m\dot{x}^2}{2} + \frac{m\omega^2}{2}x^2\right] du = \text{minimum},$$

where $x(0) = x_1$, $x(U) = x_2$. Thus, our integrand must satisfy the Euler equation,

$$\frac{\partial}{\partial u}\frac{\partial G(u)}{\partial \dot{x}} - \frac{\partial G}{\partial x} = 0 \qquad \text{or} \qquad m\ddot{x} - m\omega^2 x = 0. \tag{3.38}$$

$$x = Ae^{+\omega u} + Be^{-\omega u}.$$

For any path, we write

$$x = x + y$$

so $y(0) = 0$, $y(U) = 0$ and the integral over u becomes

$$\begin{aligned}\int_0^U \left[\frac{m}{2}(\dot{x} + \dot{y})^2 + \frac{m\omega^2}{2}(x + y)^2\right] du \\ &= \int_0^U \frac{m}{2}[\dot{x}^2 + 2\dot{x}\dot{y} + \dot{y}^2 + \omega^2 x^2 + \omega^2 2xy + \omega^2 y^2]\, du \\ &= \int_0^U \frac{m}{2}(\dot{x}^2 + \omega^2 x^2)\, du + \frac{m}{2}\int_0^U (2\dot{x}\dot{y} + 2\omega^2 xy)\, du \\ &\quad + \frac{m}{2}\int_0^U (\dot{y}^2 + \omega^2 y^2)\, du.\end{aligned}$$

Now

$$\int_0^U 2\dot{x}\dot{y}\,du = 2y\dot{x}\Big|_0^U - \int_0^U 2\ddot{x}y\,du.$$

Because $y(0) = 0$ and $y(U) = 0$, we have

$$\int_0^U 2(\dot{x}\dot{y} + \omega^2 xy)\,du = \int_0^U 2y(-\ddot{x} + \omega^2 x)\,du = 0,$$

since x satisfies the Euler equation. Thus, by expanding about the classical path, we have eliminated the integrand linear in the deviation. Thus we may write

$$f(x_2, x_1; U) = \exp\left[-\frac{1}{\hbar}\int_0^U \left(\frac{m}{2}\dot{x}^2 + \frac{m\omega^2}{2}x^2\right)du\right]F(U),$$

where

$$F(U) = \iint \exp\left[-\frac{1}{\hbar}\int_0^U \left(\frac{m}{2}\dot{y}^2 + \frac{m\omega^2}{2}y^2\right)du\right]\mathscr{D}y(u),$$

with $y(0) = 0$ and $y(U) = 0$, is a factor independent of the endpoints x_1 and x_2. Thus expansion about the classical path allows us to separate the path integral into a factor dependent upon U and the endpoints, and another path integral dependent upon U alone. Thus the important dependence on x_1 and x_2 can be found by simply solving the minimizing differential equations (Eq. 3.38) subject to the end conditions $x(0) = x_1$ and $x(U) = x_2$ and calculating the integral

$$\int_0^U \left(\frac{m}{2}\dot{x}^2 + \frac{m\omega^2}{2}x^2\right)du.$$

In fact even this integral can be simplified. We have

$$\int_0^U \frac{m}{2}\dot{x}^2\,du = \frac{m}{2}x\dot{x}\Big|_0^U - \frac{m}{2}\int_0^U x\ddot{x}\,du.$$

Therefore

$$\int_0^U \frac{m}{2}(\dot{x}^2 + \omega^2 x^2)\,du = \frac{m}{2}x\dot{x}\Big|_0^U + \int_0^U \frac{mx}{2}(-\ddot{x} + \omega^2 x)\,du = \frac{m}{2}x\dot{x}\Big|_0^U.*$$

As a new problem, suppose we have two interacting systems, one of which is a harmonic oscillator. Let x describe the position of the oscillator and q that of the other system. A possible Hamiltonian is

$$H = \frac{m\dot{x}^2}{2} + \frac{m\omega^2}{2}x^2 + \frac{M\dot{q}^2}{2} + V(q) - \gamma xq.$$

* You can compare your result with Eq. (2.83).

The partition function is

$$Q = \iint \exp\left[-\frac{1}{\hbar}\int_0^U \left(\frac{m\dot{x}^2}{2} + \frac{m\omega^2 x^2}{2} - \gamma q(u)x(u)\right) du\right]$$
$$\times \exp\left[-\frac{1}{\hbar}\int_0^U \left(\frac{M\dot{q}^2}{2} + V(q)\right) du\right] \mathscr{D}x\, \mathscr{D}q.$$

We cannot, in general, evaluate this path integral, but we can integrate out the x variable. To do so we must find

$$F[q(u)] = \iint \exp\left[-\frac{1}{\hbar}\int_0^U \left(\frac{m\dot{x}^2}{2} + \frac{m\omega^2 x^2}{2} - \gamma q x\right) du\right] \mathscr{D}x.$$

This integral can be evaluated by the same method as that used for a harmonic oscillator. More generally, we can evaluate

$$F_1[f; x, x'] = \iint_{\substack{x(0)=x \\ x(U)=x'}} \exp\left[-\frac{1}{\hbar}\int_0^U \left(\frac{m\dot{x}^2}{2} + \frac{m\omega^2 x^2}{2} + if(u)x(u)\right) du\right] \mathscr{D}x.$$

The result is

$$F_1(f; x, x') = \sqrt{m\omega/2\pi\hbar \sinh \omega U}\; e^{-\Phi/\hbar}, \tag{3.39}$$

where

$$\Phi = \frac{1}{4m\omega}\int_0^U \int_0^U e^{-\omega|u-u'|} f(u)f(u')\, du\, du' + \frac{m\omega}{2 \sinh \omega U}$$
$$\times \left[(x^2 + x'^2)\cosh \omega U - 2xx' + 2A(xe^{\omega U} - x') + 2B(x'e^{\omega U} - x)\right.$$
$$\left. + (A^2 + B^2)e^{\omega U} - 2AB\right], \tag{3.40}$$

where

$$A = \frac{i}{2m\omega}\int_0^U e^{-\omega u} f(u)\, du,$$
$$B = \frac{i}{2m\omega}\int_0^U e^{-\omega(U-u)} f(u)\, du. \tag{3.41}$$

Now let

$$\mathfrak{E}[f] = \left\langle \exp\left[-\frac{i}{\hbar}\int_0^U f(u)x(u)\, du\right]\right\rangle_{\text{harm. osc.}}$$
$$= \frac{\int F_1(f; x, x)\, dx}{\int F_1(0; x, x)\, dx}. \tag{3.42}$$

Eventually one will arrive at the result:

$$\mathfrak{E}[f] = \exp\left[-\frac{1}{4m\omega h}\int_0^U\int_0^U \frac{\cosh\omega(|u-u'|-\frac{1}{2}U)}{\sinh\frac{1}{2}\omega U} f(u)f(u')\,du\,du'\right]. \tag{3.43}$$

If we define f outside the interval $0 \le u \le U$ by letting f be periodic with period U, we may write

$$\mathfrak{E}[f] = \exp\left[-\frac{1}{4m\omega h}\int_{-\infty}^{\infty}\int_0^U e^{-\omega|u-u'|}f(u)f(u')\,du\,du'\right]. \tag{3.43a}$$

An alternative way of finding $F_1(f; x, x)$ is to Fourier analyze $f(u)$ and $x(u)$:

$$x(u) = a_0 + \sum_{n=1}^{\infty}\left(a_n\cos\frac{2\pi nu}{\beta} + b_n\sin\frac{2\pi nu}{\beta}\right),$$

$$f(u) = f_0 + \sum_{n=1}^{\infty}\left(f_n\cos\frac{2\pi nu}{\beta} + g_n\sin\frac{2\pi nu}{\beta}\right).$$

The path integral

$$F_1[f] = \int F_1(f; x, x)\,dx$$

can then be written as

$$F_1[f] = \iiint\cdots\int da_0\,da_1\cdots db_1\,db_2\cdots$$
$$\times\exp\left[-\frac{1}{h}\int_0^U\left(\frac{m\dot{x}^2}{2} + \frac{m\omega^2}{2}x^2 + if(u)x(u)\right)\right]du,$$

and can easily be evaluated.*

---

* The equation

$$\frac{1}{2a^2} + \sum_{n=1}^{\infty}\frac{\cos n\theta}{n^2+a^2} = \frac{\pi}{2a}\frac{\cosh(\pi - |\theta|)a}{\sinh\pi a}$$

is useful. $F_1[f]$ as found by the above method has an unknown multiplicative constant as a consequence of the Jacobian for the change of variables implicit in $\mathscr{D}x(u) \to da_1\cdots db_1\,db_2\cdots$; but Eq. (3.43) can be found unambiguously by the above method.

For a free particle, we can find $F_1[f]$ either by letting $\omega \to 0$ or by direct calculation. The result is

$$\iint \exp\left[-\frac{1}{\hbar}\int_0^U [\tfrac{1}{2}m\dot{x}(u)^2 + if(u)x(u)]\,du\right]\mathscr{D}x(u)$$
$$= \sqrt{\frac{2\pi\hbar m}{U}}\,\delta\left[\int_0^U f(u)\,du\right]\exp\left\{\frac{1}{4m\hbar}\left(\int_0^U\int_0^U |u-u'|f(u)f(u')\,du\,du' + \frac{2}{U}\left[\int_0^U uf(u)\,du\right]^2\right)\right\}. \tag{3.44}$$

We will make use of this last result later.

3.3 PATH INTEGRALS BY PERTURBATION EXPANSION

Suppose that we have found the density matrix element $\rho_0(x_2, x_1, U)$ for some system of potential V_0, and we wish to know $\rho(x_2, x_1, U)$ for another system with potential $V_0 + V'$, where $V' \ll V_0$. Then

$$\rho(x_2, x_1, U) = \iint \exp\left\{-\frac{1}{\hbar}\int_0^U\left[\frac{m\dot{x}^2}{2} + V_0(x) + V'(x)\right]du\right]\mathscr{D}x\Bigg\}$$
$$= \iint\left[\exp\left\{-\frac{1}{\hbar}\int_0^U\left[\frac{m\dot{x}^2}{2} + V_0(x)\right]du\right\} \times \exp\left\{-\frac{1}{\hbar}\int_0^U V'(x)\,du\right\}\right]\mathscr{D}_x$$

Expanding the second exponential, we have

$$= \iint \exp\left\{-\frac{1}{\hbar}\int_0^U\left[\frac{m\dot{x}^2}{2} + V_0(x)\right]du\right\}$$
$$\times\left[1 - \frac{1}{\hbar}\int_0^U V'[x(u)]\,du + \frac{1}{2\hbar^2}\left(\int_0^U V'[x(u)]\,du\right)^2 + \cdots\right]\mathscr{D}x$$
$$= \rho_0 - \frac{1}{\hbar}\iint\int_0^U \exp\left\{-\frac{1}{\hbar}\int_0^u\left[\frac{m\dot{x}^2}{2} + V_0(x)\right]du'\right\}$$
$$\times \exp\left\{-\frac{1}{\hbar}\int_u^U\left[\frac{m\dot{x}^2}{2} + V_0(x)\right]du'\right\} V'[x(u)]\,du\,\mathscr{D}x + \cdots$$
$$= \rho_0 - \frac{1}{\hbar}\int_0^U du\iint \exp\left\{-\frac{1}{\hbar}\int_0^u\left[\frac{m\dot{x}^2}{2} + V_0(x)\right]du'\right\}$$
$$\times V'[x(u)]\exp\left\{-\frac{1}{\hbar}\int_u^U\left[\frac{m\dot{x}^2}{2} + V_0(x)\right]du'\right\} + \ldots$$

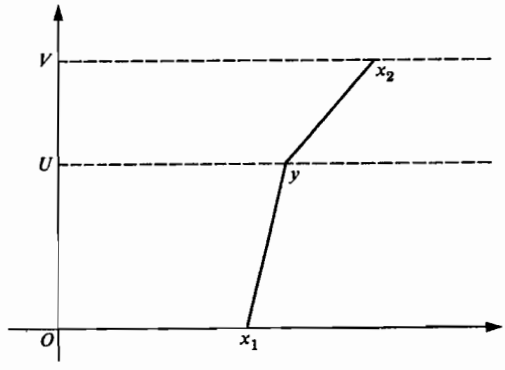

Fig. 3.3 Integrand of first-order term

In order to calculate the path integral, we first integrate over all paths for which $x(u) = y$ (with $x(0) = x_1$, $x(U) = x_2$) and then integrate over y.

$$\rho(x_2, x_1 U) = \rho_0 - \frac{1}{\hbar}\int_0^U \int_{\text{all } y} \rho_0(x_2, y, U - u)V(y)\rho_0(y, x_1, u)\, dy\, du + \cdots$$

Pictorially, the integrand may be represented as shown in Fig. 3.3.

We can think of the zeroth-order term, ρ_0, as being the contribution to the density matrix that results from the particle going from x_1 to x_2 in "time" U without being affected by the potential. Then the first-order term, which has its integrand represented in the Fig. 3.3, may be thought of as the contribution resulting from the particle being essentially free, but being "scattered" once at some "time" u and some position y. Similarly, the second-order term can be considered as the contribution from two "scatterings", and can be written as

$$+\frac{1}{\hbar}\int_0^U du \int_0^u dv \int_{-\infty}^{\infty} dy \int_{-\infty}^{\infty} dz$$
$$\times\ \rho_0(x_2, z, U - u)V(z)\rho_0(z, y, u - v)V(y)\rho_0(y, x_1, v)$$

Pictorially, this integrand may be represented as shown in Fig. 3.4.

Notice that we have not found any new results. We have already derived this perturbation expansion without the use of path integrals.

Problem: Find to what order the nth-order integral expansion of $\rho(x_2, x_1, U)$ solves

$$-\frac{\partial\rho}{\partial u} = -\frac{\hbar^2}{2m}\frac{\partial^2\rho}{\partial x^2} + [V_0(x) + V'(x)]\rho,$$

knowing that

$$-\frac{\partial\rho_0}{\partial u} = -\frac{\hbar^2}{2m}\frac{\partial^2\rho_0}{\partial x^2} + V_0(x)\rho_0.$$

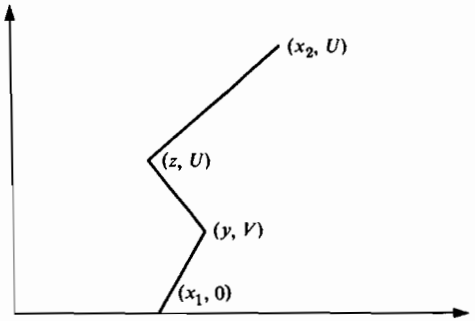

Fig. 3.4 Integrand of second-order term.

3.4 VARIATIONAL PRINCIPLE FOR THE PATH INTEGRAL

For the free energy we know the relation

$$F \leq F_0 + \langle H - H_0 \rangle_0, \tag{3.45a}$$

or

$$F \leq \langle H \rangle_0 - TS_0. \tag{3.45b}$$

This is written in terms of the Hamiltonians H and H_0. We ask ourselves if we can write a corresponding minimum relation using the path integral.

In the path-integral form, the partition function is written as

$$e^{-\beta F} = \int_{x(0)=x(U)} e^{-S[x(u)]} \mathscr{D}x(u). \tag{3.46}$$

Suppose we have another "S", S_0, that is easier to work with. (Here S_0 does not mean entropy). Then Eq. (3.46) may be written as

$$e^{-\beta F} = \frac{\int e^{-(S-S_0)} e^{-S_0} \mathscr{D}x}{\int e^{-S_0} \mathscr{D}x} e^{-\beta F_0}, \tag{3.47}$$

where

$$e^{-\beta F_0} \equiv \int e^{-S_0} \mathscr{D}x. \tag{3.48}$$

The first factor of Eq. (3.47) has the form of an average of exp $[-(S - S_0)]$ with e^{-S_0} as the weight for a certain path $x(u)$. We may write Eq. (3.47) as

$$e^{-\beta F} = \langle e^{-(S-S_0)} \rangle_{S_0} e^{-\beta F_0}. \tag{3.49}$$

Now we suppose that S and S_0 are real and use the inequality

$$\langle e^{-f} \rangle \geq e^{-\langle f \rangle}. \tag{3.50}$$

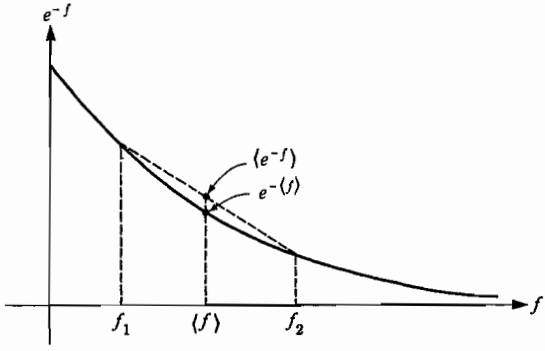

Fig. 3.5 Geometrical interpretation of $\langle e^{-f}\rangle \geq e^{-\langle f\rangle}$.

The geometrical interpretation of this relation is as shown in Fig. 3.5. In Fig. 3.1, $\langle e^{-f}\rangle$ is always above the curve e^{-f}. Note that Eq. (3.50) does not depend on how the f's are distributed. When we apply Eq. (3.50) to Eq. (3.49) we can write

$$e^{-\beta F} \geq e^{-\langle S-S_0\rangle}\, e^{-\beta F_0}, \tag{3.51}$$

where

$$\langle S - S_0\rangle \equiv \frac{\iint (S - S_0)e^{-S_0}\,\mathscr{D}x}{\iint e^{-S_0}\,\mathscr{D}x}. \tag{3.52}$$

Usually Eq. (3.52) is simpler to calculate than the first factor in Eq. (3.47). We thus have a theorem:

$$F \leq F_0 + \frac{1}{\beta}\langle S - S_0\rangle_{S_0}. \tag{3.53}$$

Exercise: When

$$\begin{aligned} S &= \int_0^\beta \left[\frac{m\dot{x}^2(u)}{2} + V(x(u))\right] du, \\ S_0 &= \int_0^\beta \left[\frac{m\dot{x}^2(u)}{2} + V_0(x(u))\right] du, \end{aligned} \tag{3.54}$$

show that

$$\frac{1}{\beta}\langle S - S_0\rangle_0 = \langle V - V_0\rangle_0. \tag{3.55}$$

Thus Eq. (3.53) contains Eq. (3.45) as a special case. (Hint: from Eq. (3.54)),

$$S - S_0 = \int_0^\beta (V - V_0)\, du. \tag{3.56}$$

The numerator of the right-hand side of Eq. (3.52) becomes an integral over u, the integrand of which is independent of u.

3.5 AN APPLICATION OF THE VARIATION THEOREM

Finally in this chapter let us explore an application of our variation theorem, Eq. (3.53).

In a one-dimensional, one-particle case, we have

$$S = \frac{1}{\hbar}\int_0^U \left[\frac{m\dot{x}(u)^2}{2} + V(x(u))\right] du, \tag{3.57}$$

where

$$U \equiv \beta\hbar. \tag{3.58}$$

To avoid unnecessary clutter, we will set $\hbar$ equal to unity in most of the following work. Every once in a while we will restore the $\hbar$'s, so that the reader can see where they belong.

The partition function may be written as

$$e^{-\beta F} = \oint \exp\left\{-\int_0^U \left[\frac{m\dot{x}(u)^2}{2} + V(x(u))\right] du\right\} \mathscr{D}x(u)\, dx(0). \tag{3.59}$$

Here, we first fix $x(0)$ and $x(U) = x(0)$, and then do the path integral over all paths; then we vary $x(0)$. The zero in $\oint$ indicates that we integrate over $x(0)$.

In the classical limit of high temperature, or small $\hbar$, U is small, so that the path does not deviate much from the initial point $x(0)$. Thus the first approximation in Eq. (3.59) is to replace $V(x(u))$ by the initial value $V(x(0))$ to obtain

$$e^{-\beta F} = \sqrt{\frac{mkT}{2\pi\hbar^2}} \int e^{-V(x)/kT}\, dx. \tag{3.60}$$

This is the well-known classical result.

How can we improve on this approximation to take the quantum effect into account?

First we observe that, because $x(0)$ and $x(U)$ are equal, it appears more natural to use an average

$$x \equiv \frac{1}{U}\int_0^U x(u)\, du \tag{3.61}$$

in place of $x(0)$. Second, the deviation of the path from the classical straight line may be taken into account by some average of $V(x)$ over the path rather than by a constant $V(x(0))$. These considerations lead to the following trial S_0:

$$S_0 = \int_0^U \frac{m\dot{x}(u)^2}{2}\, du + Uw(x). \tag{3.62}$$

Here $w(x)$ is a still undetermined function, which is to be varied later to minimize Eq. (3.53). In the first approximation one would expect to choose $w(x) = V(x)$ but we can do much better with our variational principle.

$$e^{-\beta F_0} = \iint_{\text{all closed paths}} \mathscr{D}x \exp\left[-\int_0^U \left[\frac{m\dot{x}^2}{2} + Uw(x)\right] du\right]$$

$$= \int dx \iint_{\substack{\text{all closed paths } x \\ \text{with fixed average } x}} \exp\left[-\int_0^U \left[\frac{m\dot{x}^2}{2}\, du + Uw(x)\right] du\right] \mathscr{D}x.$$

Now let $y = x - x(0)$. Since $\dot{y}(u) = \dot{x}(u)$, we obtain

$$e^{-\beta F_0} = \int dx\, e^{-Uw(x)} \iint_{\substack{\text{all closed paths} \\ y \text{ with } y(0) = 0}} \exp\left[-\int_0^U \frac{m\dot{y}^2}{2}\right] \mathscr{D}y.$$

In the path integral above, there is no restriction on y because $y = x - x(0)$, and we integrate over $x(0)$. The reasons for integrating over the paths chosen can be made clearer by drawing diagrams; the reader should do this if he needs clarification.

$$e^{-\beta F_0} = \int dx\, e^{-Uw(x)} \rho_{\text{free}}(0, 0; U) = \sqrt{\frac{m}{2\pi\hbar U}} \int_{-\infty}^{\infty} e^{-Uw(y)}\, dy. \tag{3.63}$$

It should be noted here that in the classical case Eq. (3.60), $w(y)$ is replaced by $V(y)$.

Next, we calculate $\langle S - S_0 \rangle$. Using Eqs. (3.57) and (3.62), we see by definition

$$\begin{aligned}
&\frac{1}{\beta}\langle S - S_0 \rangle \\
&\quad = \frac{\iint 1/\beta(S - S_0) \exp\left[-\int m\dot{x}^2/2\, du\right] \exp\left[-Uw(x)\right] \mathscr{D}x}{\iint \exp\left[-\int m\dot{x}^2/2\, du\right] \exp\left[-Uw(x)\right] \mathscr{D}x} \\
&\quad = \frac{\iint \{1/U \int_0^U V(x(u'))\, du' - w(x)\} \exp\left[-\int m\dot{x}^2/2\, du\right] \exp\left[-Uw(x)\right] \mathscr{D}x}{\iint \exp\left[-\int m\dot{x}^2/2\, du\right] \exp\left[-Uw(x)\right] \mathscr{D}x}.
\end{aligned} \tag{3.64}$$

Here x is defined in Eq. (3.61) and is functional of $x(u)$. The first term in the numerator is simplified as follows:

$$\begin{aligned}
&\iint \frac{1}{U} \int_0^U V(x(u'))\, du' \exp\left[-\int_0^U \frac{m\dot{x}(u)^2}{2}\, du\right] \exp\left[-Uw(x)\right] \mathscr{D}x \\
&\quad = \int_0^U \frac{du'}{U} \iint V(x(u')) \exp\left[-\int_0^U \frac{m\dot{x}(u)^2}{2}\, du\right] \exp\left[-Uw(x)\right] \mathscr{D}x \\
&\quad = \int_0^U \frac{du'}{U} \int dx'\, V(x') \iint_{\substack{x(0)=x(U) \\ x(u')=x'}} \exp\left[-\int_{u'}^U \frac{m\dot{x}^2(u)}{2}\, du \int_0^{u'} \frac{m\dot{x}^2(u)}{2}\, du\right] \\
&\qquad \times \exp\left[-Uw(x)\right] \mathscr{D}x(u).
\end{aligned}$$

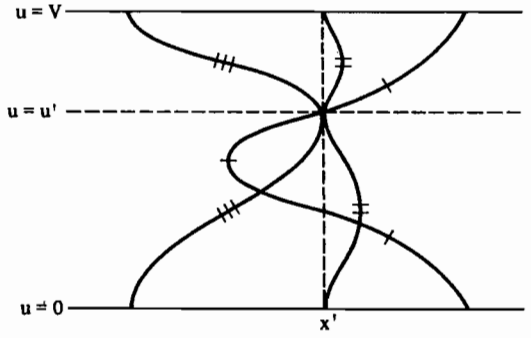

Fig. 3.6a Paths of integration.

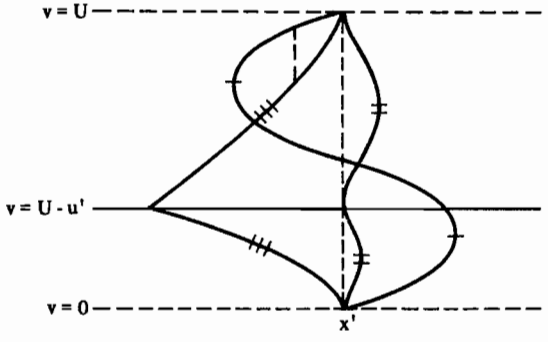

Fig. 3.6b

We are integrating over paths like those in Fig. 3.6a. By making the change of variables

$$V = \begin{cases} u + U - u' & 0 \leq u \leq u' \\ u - u' & u' \leq u \leq U \end{cases}$$

we integrate over paths like those in Fig. 3.6b. Because of the condition that $x(u = 0) = x(u = U)$, we do not get a discontinuity in x at $V = U - u'$; therefore

$$\int_0^U \frac{m\dot{x}^2(u)}{2}\, du = \int_0^U \frac{m\dot{x}^2(V)}{2}\, dV.$$

Also,

$$x = \frac{1}{U}\int_0^U x(u)\, du = \frac{1}{U}\int_0^U x(V)\, dV.$$

The first term in the numerator of Eq. (3.64) becomes

$$\int_0^U \frac{du'}{U} \int dx' V(x') \iint_{x(0)=x(U)=x'} \exp\left[-\int_0^U \frac{m\dot{x}^2(V)}{2}\, dV\right] \exp\left[-Uw(x)\right]$$

$$= \iint_{x(0)=x(U)} V(x(0)) \exp\left[-\int_0^U \frac{m\dot{x}^2(u)}{2}\, du\right] \exp\left[-Uw(x)\right] \mathscr{D}x(u). \tag{3.65}$$

Thus Eq. (3.64) may be written as

$$\frac{1}{\beta}\langle S - S_0\rangle_{S_0} = \langle V(x(0))\rangle - \langle w(x)\rangle. \tag{3.66}$$

To transform further we use the following result (proved previously):

$$\iint \exp\left[-\int_0^U \frac{m\dot{x}(u)^2}{2}\, du\right] \exp\left[-i\int_0^U f(u)x(u)\, du\right] \mathscr{D}x(u)$$

$$= \sqrt{\frac{2\pi m}{U}}\, \delta\left(\int_0^U f(u)\, du\right) \exp\left\{\frac{1}{4m}\left[\int_0^U \int_0^U |u - u'| f(u) f(u')\, du\, du' + \frac{2}{U}\left(\int_0^U u f(u)\, du\right)^2\right]\right\} \tag{3.67}$$

In evaluating Eq. (3.65) we first do the path integral keeping x fixed; then Eq. (3.65) is written

$$\iint V(x(0)) \exp\left[-\int \frac{m\dot{x}(u)^2}{2}\, du\right] \exp\left[-Uw(x)\right] \mathscr{D}x = \int K(y) \exp\left[-Uw(y)\right] dy \tag{3.68}$$

where

$$K(y) = \iint V(x(0)) \exp\left[-\int_0^U \frac{m\dot{x}(u)^2}{2}\, du\right] \mathscr{D}x \tag{3.69}$$

such that $x = y$. Equation (3.69) may be written

$$K(y) = \iint V(x(0)) \exp\left[-\int_0^U \frac{m\dot{x}(u)^2}{2}\, du\right] \delta(x - y)\, \mathscr{D}x(u). \tag{3.70}$$

For this we use the Fourier transform:

$$V(x) \equiv \int v(q) e^{iqx}\, dq \tag{3.71}$$

to write Eq. (3.70) as

$$K(y) = \int v(q)\, dq \int dk \iint \exp\,[iqx(0)] \exp\,[ik(x - y)]$$
$$\times \exp\left[-\int_0^U \frac{m\dot{x}^2}{2}\, du\right] \mathscr{D}x. \tag{3.72}$$

Here $\delta(x - y)$ of Eq. (3.70) was also Fourier transformed, and we are ignoring factors such as 2π. Use Eq. (3.61) in Eq. (3.72) to write

$$K(y) = \int v(q)\, dq \int dk \exp\,[-iky] \iint \exp\,[iqx(0)] \exp\left[i\frac{k}{U}\int_0^U x(u)\, du\right]$$
$$\times \exp\left[-\int_0^U \frac{m\dot{x}^2}{2}\, du\right] \mathscr{D}x. \tag{3.73}$$

Now we see that the path-integral part of this expression can be brought into the form of Eq. (3.67) if we define

$$f(u) \equiv q\delta(u - 0) + \frac{k}{U} \tag{3.74}$$

because

$$\int_0^U f(u)x(u)\, du = qx(0) + \frac{k}{U}\int_0^U x(u)\, du. \tag{3.75}$$

When we use Eq. (3.74) in Eq. (3.67) we see that

$$\int_0^U f(u)\, du = q + k, \tag{3.76}$$

$$\int_0^U uf(u)\, du = \frac{kU}{2}, \tag{3.77}$$

$$\int_0^U \int_0^U |u - u'| f(u)f(u')\, du\, du' = 2\int_0^U du \int_0^u (u - u')f(u')\, du' f(u)$$
$$= 2\int_0^U du \left[\int_0^u (u - u')\frac{k}{U}\, du' + uq\right] f(u)$$
$$= 2\int_0^U du \left[\frac{ku^2}{2U} + uq\right] f(u) = \frac{k^2U}{3} + kqU. \tag{3.78}$$

Use Eqs. (3.76), (3.77), and (3.78) in Eq. (3.67) and also in the path-integral part of Eq. (3.73) to write the latter as

$$K(y) = \int dq v(q) \int dk \exp[-iky]\delta(q + k) \exp\left[\frac{U}{4m}\left(\frac{k^2}{3} + kq + \frac{k^2}{2}\right)\right]$$

$$= \int dq v(q) \int dk \exp[-iky]\delta(q + k) \exp\left[-\frac{k^2 U}{24m}\right]$$

$$= \int dk v(-k) \exp[-iky] \exp\left[-\frac{k^2 U}{24m}\right]. \tag{3.79}$$

Using the inverse Fourier transform, we find that Eq. (3.79) becomes

$$K(y) = \int dk \exp[-iky] \int V(z) \exp[ikz]\, dz \exp\left[-\frac{k^2 U}{24m}\right]$$

$$= \int dz V(z) \exp\left[-\frac{6m}{U}(y - z)^2\right]. \tag{3.80}$$

When the correct factors are taken into account, the final form is

$$K(y) = \sqrt{\frac{6mkT}{\pi\hbar^2}} \int dz V(z) \exp\left[-\frac{6mkT(y - z)^2}{\hbar^2}\right]. \tag{3.81}$$

It should be noticed that $K(y)$ is $V(z)$ averaged over a Gaussian. The root mean square of the Gaussian spread is

$$\frac{\hbar}{\sqrt{12\, mkT}}. \tag{3.82}$$

This spread is about 1 Å for a helium atom at 2°K, and is a very narrow spread at room temperatures. At the limit of infinite temperature, the Gaussian becomes a delta function, and $K(y) \to V(y)$.

Summarizing the results obtained so far, we have

$$F \leq F_0 + \frac{1}{\beta}\langle S - S_0\rangle_{S_0}, \tag{3.83a}$$

$$e^{-\beta F_0} = \int \exp[-\beta w(y)]\, dy \sqrt{\frac{m}{2\pi\beta\hbar^2}}, \tag{3.83b}$$

$$\frac{1}{\beta}\langle S - S_0\rangle_{S_0} = \frac{\int (K(y) - w(y))e^{-\beta w(y)}\, dy}{\int e^{-\beta w(y)}\, dy}. \tag{3.83c}$$

Equation (3.83b) is Eq. (3.63). Equation (3.83c) follows from Eqs. (3.64), (3.65), and (3.68). Further, $K(y)$ in Eq. (3.83c) is given in Eq. (3.81).

Now we ask what is the best choice of $w(y)$? Changing

$$w(y) \to w(y) + \eta(y) \tag{3.84}$$

where $\eta(y)$ is small, we see from Eqs. (3.83b) and (c)

$$\delta F_0 = \frac{\int \eta e^{-\beta w}\, dy}{\int e^{-\beta w}\, dy} \tag{3.85a}$$

$$\delta \frac{1}{\beta} \langle S - S_0 \rangle = \frac{\int e^{-\beta w}[-\beta\eta(K - w) - \eta\, dy]}{\int e^{-\beta w}\, dy} + \frac{\int e^{-\beta w}(K - w)\, dy \int \beta\eta e^{-\beta w}\, dy}{(\int e^{-\beta w}\, dy)^2}. \tag{3.85b}$$

Thus

$$\delta\left(F_0 + \frac{1}{\beta}\langle S - S_0\rangle_{S_0}\right) = 0 \tag{3.86}$$

leads to

$$w(y) = K(y). \tag{3.87}$$

This is the best choice of $w(y)$.

In this case Eq. (3.83c) yields

$$\langle S - S_0\rangle_{S_0} = 0, \tag{3.88}$$

so that, from Eq. (3.83a),

$$F \leq F_{\mathrm{cl}\,K} \tag{3.89}$$

where $F_{\mathrm{cl}\,K}$ means the classical free energy with the potential $V(y)$ replaced by $K(y)$. Alternatively,

$$e^{-\beta F_{\mathrm{cl}\,K}} = \sqrt{\frac{m}{2\pi\hbar U}} \int e^{-\beta K(y)}\, dy \tag{3.90}$$

compare Eq. (3.60) . $K(y)$ is defined in Eq. (3.81). $F_{\mathrm{cl}\,K}$ is a better approximation than the ordinary classical form F_{cl} in Eq. (3.60).

Example: To find out how good $F_{\mathrm{cl}\,K}$ is, let us consider the harmonic oscillator. In this case the potential is

$$V(x) = m\frac{\omega^2 x^2}{2}. \tag{3.91}$$

From Eq. (3.81)

$$K(y) = \sqrt{\frac{6mkT}{\pi\hbar^2}}\,\frac{m\omega^2}{2}\int z^2 \exp\left[-\frac{6mkT}{\hbar^2}(y - z)^2\right] dz = \frac{m\omega^2}{2}\left(y^2 + \frac{\hbar^2\beta}{12m}\right). \tag{3.92}$$

Using this result in Eq. (3.90), we obtain

$$e^{-\beta F_{\mathrm{cl}\,K}} = \sqrt{\frac{m}{2\pi\hbar^2\beta}} \int \exp\left[-\beta \frac{m\omega^2}{2}\left(y^2 + \frac{\hbar^2\beta}{12m}\right)\right] dy$$

$$= (1/\hbar\beta\omega)e^{-\hbar^2\beta^2\omega^2/24}. \tag{3.93}$$

Thus, we have the following results:

$$F_{\mathrm{true}} = \frac{1}{\beta} \ln\left[2 \sinh\left(\frac{\hbar\omega\beta}{2}\right)\right], \tag{3.94a}$$

$$F_{\mathrm{cl}\,K} = \frac{1}{\beta}\left[\ln(\hbar\omega\beta) + \frac{\hbar^2\beta^2\omega^2}{24}\right], \tag{3.94b}$$

$$F_{\mathrm{cl}} = \frac{1}{\beta} \ln(\hbar\omega\beta). \tag{3.94c}$$

When we write

$$G \equiv \frac{2F}{\hbar\omega} \quad \text{and} \quad f \equiv \frac{\hbar\omega\beta}{2}. \tag{3.95}$$

Eq. (3.94) becomes

$$G_{\mathrm{true}} = \frac{1}{f} \ln(2 \sinh f), \tag{3.95a}$$

$$G_{\mathrm{cl}\,K} = \frac{1}{f} \ln(2f) + \frac{f^2}{6}, \tag{3.95b}$$

$$G_{\mathrm{cl}} = \frac{1}{f} \ln(2f). \tag{3.95c}$$

Table 3.1 is a numerical comparison of G_{true}, $G_{\mathrm{cl}\,K}$ and G_{cl}. This table shows the remarkable improvement $G_{\mathrm{Cl}\,K}$ has over G_{cl}. Remember that $f = \frac{1}{2}$ corresponds to the temperature $kT = \hbar\omega$ where the quantum effect is large.

Table 3.1

	$f = 1/2$	$f = 1$	$f = 2$
G_{true}	0.08263	0.8544	0.9908
$G_{\mathrm{cl}\,K}$	0.08333	0.8598	1.0264
G_{cl}	0	0.6931	0.6931

The expression $F_{\mathrm{cl}\,K}$ in Eq. (3.90) combined with Eq. (3.81) is better than the series expansion of $K(y)$ in terms of $V(y)$, $d^2V/dy^2, \ldots$.

Nevertheless, $F_{\mathrm{cl}\,K}$ is not as useful as Table 3.1 might indicate. First, it cannot be used when quantum-mechanical exchange effects exist. Second, it fails in its present form when the potential $V(y)$ has a very large derivative as in the case of hard-sphere interatomic potential.